ATMOSPHERIC AND OCEANIC FLUID DYNAMICS

FUNDAMENTALS AND LARGE-SCALE CIRCULATION

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Contents

An asterisk indicates more advanced, but usually uncontroversial, material that may be omitted on a first reading. A dagger indicates material that is still a topic of research or that is not settled. *Caveat emptor.*

We must be ignorant of much, if we would know anything.
Cardinal John Newman (1801–1890).

Preface

THIS IS A BOOK on the fluid dynamics of the atmosphere and ocean, with an emphasis on the fundamentals and on the large-scale circulation, the latter meaning flows from a hundred or so kilometers to the global scale. The book is primarily a textbook; it is designed to be accessible to students and could be used as a text for graduate courses. It may be also useful as an introduction to the field for scientists in other areas and as a reference for researchers in the field, and some aspects of the book have the flavour of a research monograph.

Atmospheric and oceanic fluid dynamics (AOFD) is fascinating field, and simultaneously both pure and applied. It is a pure field because it is intimately tied to some of the most fundamental and unsolved problems in fluid dynamics — problems in turbulence and wave–mean flow interaction, problems in chaos and predictability, and problems in the general circulation itself. Yet it is applied because the climate and weather so profoundly affect the human condition, and so a great deal of effort goes into making predictions — indeed the practice of weather forecasting is a remarkable example of a successful applied science, in spite of the natural limitations to predictability that we now understand because of basic research. The field is also very broad, encompassing such diverse topics as the general circulation, gyres, boundary layers, waves, convection and turbulence. My goal in this book is present a coherent selection of these topics, concentrating on the foundations but without shying away from the boundaries of active areas of research — for a book that limits itself to what is absolutely settled would, I think, be rather dry, a quality best reserved for martinis and humour.

The book is an outgrowth of various courses that I have taught over the years, mainly at Princeton University but also at the University of California and at summer schools or similar in Boulder and Kyoto. There are four parts to the book — fundamentals of geophysical fluid dynamics (GFD); instabilities, wave–mean flow interaction and turbulence; atmospheric circulation; and ocean circulation. Each corresponds, very roughly, to a one-term graduate course, although sections of Part I could also be used for advanced undergraduates. Limitations enforced both by the need to keep the book coherent and focussed, and my own expertize or (especially) lack thereof, naturally limit the choice of topics. In particular the chapters on the circulation focus on the steady and statistically steady large-scale circulation and perforce a number of important topics are omitted — equatorial dynamics, the spin-up of the ocean circulation, the quasi-biennial oscillation and so on.

I have tried to keep the overall treatment of topics as straightforward and as clear as I know how, and in practice this means that the level should be appropriate for graduate students. There is a certain amount of repetition in this book, which serves both to emphasize the important things, and to keep chapters and sections reasonably self-contained. Obviously the chapters are intellectually linked — for example, heat transport in the atmosphere depends on baroclinic instability, but hopefully the reader already familiar with the latter will be able to read about the former without too much cross-referencing. The treatment generally is fairly physical and phenomenological, and rigour in the mathematical sense is absent — we treat the derivatives of integrals and of infinitesimal quantities rather informally, for example.

An asterisk, *, next to a section means that it may be omitted on first reading; it may be more advanced and is not essential for most of the subsequent material. A dagger, †, next to section means that the section discusses topics of research. Very roughly speaking, one might interpret as asterisk as indicating there is advanced *manipulation* of the equations, whereas a dagger might indicate there is *approximation* of the equations. These sections may be regarded as providing an introduction to the literature, rather than a compete or finished treatment. If the asterisk or dagger is applied to a section it applies to all the subsections within, and if a dagger or asterisk appears within a section that is already so-marked, the warning is even more emphatic. Problems marked with black diamonds may be difficult, and I do not know the solutions to all of them. Good answers to some of them are probably publishable and I would appreciate hearing about any such work. Qui docet discit.

Finally, I should say that this book owes its existence in part to my own hubris and selfishness: hubris to think that others might wish to read what I have written, and selfishness because the enjoyable task of writing such a book masquerades as work.

Summary of Contents

The chapters within each part, and the sections within each chapter, form *logical* units, and do not necessarily directly correspond to a single lecture or number of lectures.

Part I. Fundamentals of geophysical fluid dynamics

Chapter 1 is a brief introduction to fluid dynamics and the basic equations of motion, assuming no prior knowledge of the field. Readers who have a background in fluid dynamics might skim it lightly, concentrating on those aspects unique to the atmosphere or ocean.

Chapter 2 introduces the effects of stratification and rotation, these being the two main effects that most differentiate AOFD from other branches of fluid dynamics. Fundamental topics such as the primitive equations, the Boussinesq equations, and Ekman layers are introduced here, and these form the foundation for the rest of the book.

Chapter 3 focusses on the shallow water equations. Many of the principles of geophysical fluid dynamics have their simplest expression in the shallow water equations, because the effects of stratification are either eliminated or much simplified. The equations thus provide a relatively gentle introduction to the field.

Chapter 4 discusses vorticity and potential vorticity. Potential vorticity plays an especially important role in rotating, stratified flows, especially on the large scale, and its conservation provides the basis for the equation sets of chapter 5.

Chapter 5 derives simplified equation sets for large-scale flows, in particular the quasi-geostrophic and planetary geostrophic equation sets, and introduces a simple application, Rossby waves. Much of our theoretical understanding of the large-scale circulation has arisen through the use of these equations.

Part II. Instabilities, wave–mean flow interaction and turbulence

Chapter 6 covers barotropic and baroclinic instability, the latter being the instability that gives rise to weather — and therefore being, perhaps, the form of hydrodynamic instability that most affects the human condition.

Chapter 7 provides an introduction to the important topic, albeit one that is regarded as difficult, of wave–mean flow interaction. That is, how do the waves and instabilities affect the mean flow in which they propagate?

Chapter 8 and 9 are on the statistical theory of turbulence. Chapter 8 introduces the basic concepts of two- and three-dimensional turbulence, and chapter 9 applies similar ideas to geostrophic turbulence.

Chapter 10 discusses turbulent diffusion — a hoary subject, both used and abused, yet one that plays a central role in our thinking about the transport properties of eddies in the atmosphere and ocean.

Part III. Large-scale atmospheric circulation

Chapter 11 is mostly concerned with the dynamics of the Hadley Cell, and, rather descriptively, with the Ferrel Cell.

Chapter 12 addresses the mid-latitude circulation. The goal of this chapter is to provide the basis of an understanding of such topics as the surface westerly winds, the dynamics of the Ferrel Cell, the stratification of the atmosphere and the height of the tropopause. These are still active topics of research, so we cannot always be definitive, although many of the underlying principles are now established and our discussion emphasizes the fundamentals.

Chapter 13 discusses stationary waves in the atmosphere, mainly produced by the interaction of the zonal wind with mountains and land-sea temperature contrasts. It also discusses the vertical propagation of Rossby waves, and how these induce a stratospheric circulation.

Part IV. Large-scale oceanic circulation

Chapter 14 discusses the wind-driven circulation, in particular the ocean gyres, either ignoring buoyancy effects or assuming that bouyancy forcing acts primarily to set up a stratification that can be taken as a given.

Chaper 15 discusses the buoyancy-driven circulation, largely neglecting the effects of wind forcing.

Chapter 16 addresses the combined effects of wind and buoyancy forcing in setting up the stratification and in driving both the predominantly horizontal gyral circulation and the overturning circulation. As with Part III, many of these topics are still being actively researched, although, again, many of the underlying principles have, I think, now been established and have permanent value.

Acknowledgements

My first exposure to the field came as a graduate student in the marvellous atmosphere of the Atmospheric Physics Group at Imperial College in the late 1970's and early 1980's. I'd like to thank everyone who was there at that time, without exception. You know who you are.

This book would not have been possible without the input, criticism, encouragement and advice of a large number of people, from graduate students to senior scientists. Students at the Princeton University, New York University, Columbia University, California Institute of Technology and the University of Toronto have used earlier versions of the text in various courses, and I am grateful for the feedback received from the instructors and students about what works and what doesn't, as well as for numerous detailed comments.

The following individuals have provided comments or corrections on the text, figures, notes, or conversational input. Brian Arbic, Pavel Berloff, Agatha deBoer, Thomas Birner, Paola Cessi, Sorin Codoban, Roland deSzoeke, Andreas Dornbräck, Janusz Eluszkiewicz, Gavin Esler, Raffaele Ferrari, Baylor Fox-Kemper, Dargan Frierson, Steve Garner, Ed Gerber, Steve Griffies, Brian Hoskins, Chris Hughes, Laura Jackson, Martin Juckes, Allan Kaufman, Samar Khatiwhala, Paul Kushner, Joe LaCasce, Ants Leetmaa, Trevor McDougall, Michael McIntyre, Tim Palmer, David Pearson, Anders Persson, Jeff Polton, Lorenzo Polvani, Cathy Raphael, Roger Samelson, Adam Scaife, John Scinocca, Rob Scott, Sabrina Speich, Robbie Toggweiler, Yue-Kin Tsang, Ross Tulloch, Eli Tziperman, Jacques Vanneste, Jeff Varanyuk, Chris Walker, Darryn Waugh, Ric Williams, John Wilson, Rong Zhang, Rongrong Zhao, Michal Ziemiánski. I would like to thank them all. I am particularly grateful to Jurgen Theiss, Tapio Schneider, Adam Sobel, Andy White and Paul O'Gorman who read and gave detailed comments on many chapters of text, to Isaac Held for many conversations over the years, especially on the atmospheric general circulation, to Shafer Smith for both comments and code, to Bill Young for very useful input on thermodynamics and the Boussinesq equations. Needless to say, I take full responsibility for the errors that undoubtedly remain.

The micro-biographies at the end of each chapter draw from a variety of sources, including http://www.maths.tcd.ie/pub/HistMath/, as well as the references cited. The book was written in LaTeX using the AlphaX editor. The Lucida family of fonts is used for both text and mathematics. Most of the figures were prepared using Matlab, Ferret, Adobe Illustrator, and/or Apple Keynote, which, apart from Ferret (a NOAA product), are all registered trademarks. I would also like to thank NSF and NOAA for providing financial support over the years, and Bess for support of a greater kind.

NOTATION

The list below contains only the more important variables, or instances of non-obvious notation. Distinct meanings are separated with a semi-colon. Variables are normally set in italics, constants (e.g, π) in roman (i.e., upright), differential operators in roman, vectors in bold, and tensors in sans serif. Thus, vector variables are in bold italics, vector constants (e.g., unit vectors) in bold roman, and tensor variables are in slanting sans serif. Physical units are set in roman. A subscript denotes a derivative only if the subscript is a coordinate, such as x, y or z; a subscript 0 generally denotes a constant reference value (e.g., ρ_0). The components of a vector are denoted by superscripts.

Variable	Description
b	Buoyancy, $-g\delta\rho/\rho_0$ or $g\delta\theta/\widetilde{\theta}$.
$\boldsymbol{c}_g$	Group velocity, (c_g^x, c_g^y, c_g^z).
c_p	Phase speed; heat capacity at constant pressure.
c_v	Heat capacity constant volume.
c_s	Sound speed.
f, f_0	Coriolis parameter, and its reference value.
$\boldsymbol{g}, g$	Vector acceleration due to gravity, magnitude of $\boldsymbol{g}$.
h	Layer thickness (in shallow water equations).
$\mathbf{i}, \mathbf{j}, \mathbf{k}$	Unit vectors in (x, y, z) directions.
i	An integer index.
i	Square root of -1.
$\boldsymbol{k}$	Wave vector, with components (k, l, m) or (k^x, k^y, k^z).
k_d	Wave number corresponding to deformation radius.
L_d	Deformation radius.
L, H	Horizontal length scale, vertical (height) scale.
m	Angular momentum about the earth's axis of rotation.
M	Montgomery function, $M = c_pT + \Phi$.
N	Buoyancy, or Brunt-Väisälä, frequency.
p	Pressure.
Pr	Prandtl ratio, f_0/N.
q	Quasi-geostrophic potential vorticity.
Q	Potential vorticity (in particular Ertel PV).
$\dot{Q}$	Rate of heating.
Ra	Rayleigh number.
Re	Real part of expression.
Re	Reynolds number, UL/ν.
Ro	Rossby number, U/fL.
S	Salinity; source term on right-hand side of evolution equation.
T	Temperature.
t	Time.
$\boldsymbol{u}$	Two-dimensional, horizontal velocity, (u, v).
$\boldsymbol{v}$	Three-dimensional velocity, (u, v, z).
x, y, z	Cartesian coordinates, usually in zonal, meridional and vertical directions.
Z	Log-pressure, $-H \log p/p_R$. We often use $H = 7.5\,\mathrm{km}$ and $p_R = 10^5\,\mathrm{Pa}$.

Variable	Description
$\mathcal{A}$	Wave activity.
α	Inverse density, or specific volume; aspect ratio.
β	Rate of change of f with latitude, $\partial f/\partial y$.
β_T, β_S	Coefficient of expansion with respect to temperature, salinity.
ϵ	Generic small parameter (epsilon).
ε	Cascade or dissipation rate of energy (varepsilon).
η	Specific entropy; perturbation height; enstrophy cascade or dissipation rate.
$\boldsymbol{\mathcal{F}}$	Eliassen Palm flux, $(\mathcal{F}^y, \mathcal{F}^z)$.
γ	Vorticity gradient, $\beta - u_{yy}$; the ratio c_p/c_v.
Γ	Lapse rate.
κ	Diffusivity; the ratio R/c_p.
$\mathcal{K}$	Kolmogorov or Kolomogorov-like constant.
Λ	Shear, e.g., $\partial U/\partial z$.
ν	Kinematic viscosity.
v	Meridional component of velocity.
ϕ	Pressure divided by density, p/ρ; passive tracer.
Φ	Geopotential, usually gz.
Π	Exner function, $\Pi = c_p T/\theta = c_p(p/p_R)^{R/c_p}$.
$\boldsymbol{\omega}$	Vorticity.
$\Omega, \boldsymbol{\Omega}$	Rotation rate of earth and associated vector.
ψ	Streamfunction.
ρ	Density.
ρ_θ	Potential density.
σ	Layer thickness, $\partial z/\partial\theta$; Prandtl number ν/κ; measure of density, $\rho - 1000$.
$\boldsymbol{\tau}$	Wind stress.
τ	Zonal component or magnitude of wind stress; eddy turnover time.
θ	Potential temperature.
ϑ, λ	Latitude, longitude.
ζ	Vertical component of vorticity.
$\left(\dfrac{\partial a}{\partial b}\right)_c$	Derivative of a with respect to b at constant c.
$\left.\dfrac{\partial a}{\partial b}\right\rvert_{a=c}$	Derivative of a with respect to b evaluated at $a = c$.
∇_a	Gradient operator at constant value of coordinate a, e.g., $\nabla_z = \mathbf{i}\,\partial_x + \mathbf{j}\,\partial_y$.
$\nabla_a\cdot$	Divergence operator at constant value of coordinate a, e.g., $\nabla_z\cdot = (\mathbf{i}\,\partial_x + \mathbf{j}\,\partial_y)\cdot$.
$\nabla^\perp$	Perpendicular gradient, $\nabla^\perp\phi \equiv \mathbf{k}\times\nabla\phi$.
curl_z	Vertical component of $\nabla\times$ operator, $\mathrm{curl}_z\boldsymbol{A} = \mathbf{k}\cdot\nabla\times\boldsymbol{A} = \partial_x A^y - \partial_y A^x$.
$\dfrac{\mathrm{D}}{\mathrm{D}t}$	Material derivative (generic).
$\dfrac{\mathrm{D}_3}{\mathrm{D}t}, \dfrac{\mathrm{D}_2}{\mathrm{D}t}$	Material derivative in three dimensions and in two dimensions, for example $\partial/\partial t + \boldsymbol{v}\cdot\nabla$ and $\partial/\partial t + \boldsymbol{u}\cdot\nabla$ respectively.
$\dfrac{\mathrm{D}_g}{\mathrm{D}t}$	Material derivative using geostrophic velocity, for example $\partial/\partial t + \boldsymbol{u}_g\cdot\nabla$.

Part I

FUNDAMENTALS OF GEOPHYSICAL FLUID DYNAMICS

Part II

INSTABILITY, WAVE–MEAN FLOW INTERACTION AND TURBULENCE

Part III

LARGE-SCALE ATMOSPHERIC CIRCULATION

Part IV

LARGE-SCALE OCEANIC CIRCULATION

To increase our knowledge the subject [oceanography] must be made attractive to men who do not mind facing up to the difficulties of fluid mechanics.

From *Nature*, 25 December 1954.

CHAPTER FOURTEEN

Wind-Driven Gyres

UNDERSTANDING THE CIRCULATION OF THE OCEAN involves a combination of observations, comprehensive numerical modelling, and more conceptual modelling, or 'theory'. All have become essential, but in this chapter, and the ones following, our emphasis is on the last of the tripos. Its (continuing) role is not to explain every feature of the observed ocean circulation, nor to necessarily describe details best left to numerical simulations. It is to provide a conceptual and theoretical framework for understanding the circulation of the ocean, for interpreting observations and suggesting how new observations may best be made, and to aid the development and interpretation of experiments with numerical models.

The main currents of the worlds oceans are sketched in Fig. 14.1.[1] (Over most of the ocean, the vertically averaged currents have a similar sense as the surface currents, one exception being at the equator where the surface currents are mainly westwards but the vertical integral is dominated by the eastwards undercurrent.) Two seemingly dichotomous aspects stand out: (i) the complexity of the currents as they interact with topography and the geography of the continents; (ii) the simplicity and commonality of the large-scale structures in the major ocean basins, and in particular the ubiquity of subtropical and subpolar gyres. Indeed these gyres, sweeping across the great oceans carrying vast quantities of water and heat, are perhaps the single most conspicuous feature of the circulation. The subtropical gyres are anticyclonic, extending polewards to about 45°, and the subpolar gyres are cyclonic and polewards of this, primarily in the Northern Hemisphere. The existence of gyres, and that they are strongest in the west, has been known for centuries; this *western intensification* leads to such well-known currents as the Gulf Stream in the Atlantic (charted by Benjamin Franklin), the Kuroshio in the Pacific, and the Brazil Current in the South Atlantic. Although even today we barely have sufficient observations to produce a

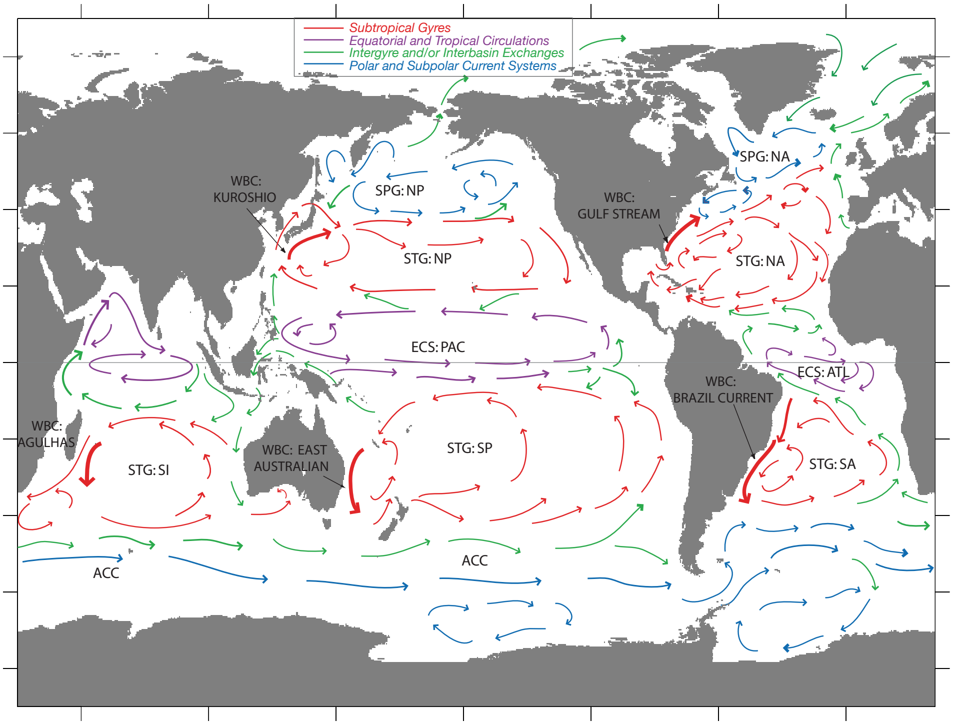


Fig. 14.1 A schema of the main currents of the global ocean. Key: STG – Sub-Tropical Gyre; SPG – Sub-Polar Gyre; WBC – Western Boundary Current; ECS – Equatorial Current System; NA – North Atlantic; SA – South Atlantic; NP – North Pacific; SP – South Pacific; SI – South Indian; ACC – Antarctic Circumpolar Current; ATL – Atlantic; PAC – Pacific. The figure is a qualitative, and not quantitative, representation of the actual flow.

synoptic map of the ocean currents, except at the surface, the large-scale mean currents are fairly well mapped and Fig. 14.2 illustrates the average current pattern of the North Atlantic using a combination of observations and a numerical model, and the Gulf Stream is clearly visible.[2]

For much of this chapter we consider a model, and variations about it, that explains the large-scale features of ocean gyres and that lies at the core of ocean circulation theory — the steady, forced-dissipative, homogeneous model of the ocean circulation first formulated by Stommel.[4] Such models explain many of the zeroth-order features of the ocean, in particular the existence of gyres and the appearance of intense western boundary currents. In the later part of the chapter we examine the vertical structure of the wind-driven circulation, taking the stratification as given. In the two chapters following we consider the maintenance of that stratification.

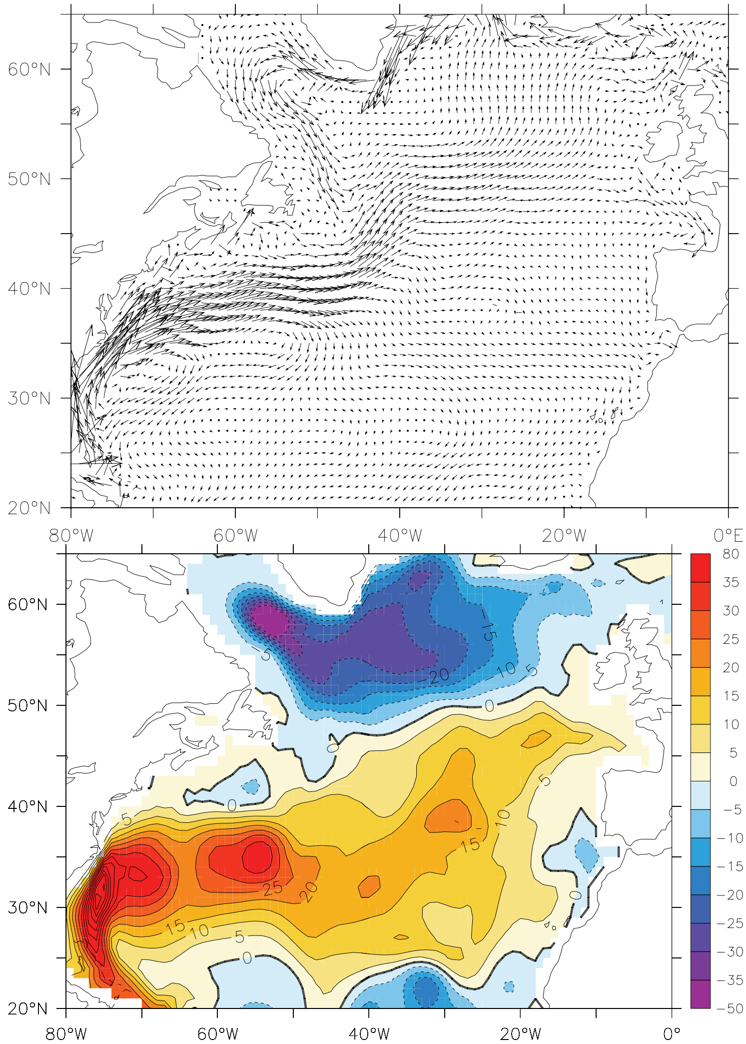


Fig. 14.2 Top: The time-averaged velocity field at a depth of 75 m in the North Atlantic, obtained by constraining a numerical model to observations. Bottom: The streamfunction of the vertically integrated flow, in Sverdrups ($1\,\mathrm{Sv} = 10^9\,\mathrm{kg\,s^{-1}}$). Note the presence of an anticyclonic subtropical gyre, a cyclonic subpolar gyre, and intense western boundary currents.[3]

14.1 THE DEPTH INTEGRATED WIND-DRIVEN CIRCULATION

The equations that govern the large-scale flow in the oceans are the planetary-geostrophic equations. Greatly simplified as these are compared to the Navier-Stokes equations, or even the hydrostatic Boussinesq equations, they are still quite daunting: a prognostic equation for buoyancy is coupled to the advecting velocity via hydrostatic and geostrophic balance, and the resulting problem is formidably nonlinear. However, it turns out that thermodynamic effects can effectively be eliminated by the simple device of vertical integration; the resulting equations are linear, and the only external forcing is that due to the wind-stress. The resulting model then, at the price of some comprehensiveness, gives a useful picture of the *wind-driven* circulation of the ocean.

14.1.1 The Stommel Model

The planetary geostrophic equations for a Boussinesq fluid are:

$$\frac{\mathrm{D}b}{\mathrm{D}t} = \dot{b}, \qquad \nabla_3 \cdot \boldsymbol{v} = 0, \tag{14.1a,b}$$

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi + \frac{1}{\rho_0}\frac{\partial \boldsymbol{\tau}}{\partial z}, \qquad \frac{\partial \phi}{\partial z} = b. \tag{14.2a,b}$$

These equations are, respectively, the thermodynamic equation (14.1a), the mass continuity equation (14.1b), the horizontal momentum equation (14.2a), (i.e., geostrophic balance, plus a stress term), and the vertical momentum equation (14.2b), (i.e, hydrostatic balance). These equations are derived more fully in chapter 5, but they are essentially the Boussinesq primitive equations with the advection terms omitted in the horizontal momentum equation, on the basis of small Rossby number. In this chapter we will absorb the factor of ρ_0 into the $\boldsymbol{\tau}$, so that $\boldsymbol{\tau}$ is the kinematic stress, and the gradient operator will be two dimensional, in the x-y plane, unless noted.

Take the curl of (14.2a) (that is, cross differentiate its x and y components) and integrate over the depth of the ocean to give

$$\int \boldsymbol{f}\nabla \cdot \boldsymbol{u}\,\mathrm{d}z + \frac{\partial f}{\partial y}\int v\,\mathrm{d}z = \mathrm{curl}_z(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B), \tag{14.3}$$

where the operator curl_z is defined by $\mathrm{curl}_z\boldsymbol{A} \equiv \partial A^y/\partial x - \partial A^x/\partial y = \mathbf{k}\cdot\nabla\times\boldsymbol{A}$, and the subscripts T and B are for top and bottom. The divergence term vanishes if the vertical velocity is zero at the top and bottom of the ocean. Strictly, at the top of the ocean the vertical velocity is given by the material derivative of height of the ocean's surface, Dh/Dt, but on the large-scales this has a negligible effect and we may make the rigid-lid approximation and set it to zero. At the bottom of the ocean the vertical velocity is only zero if the ocean in flat-bottomed; otherwise it is $\boldsymbol{u}\cdot\nabla\eta_B$, where η_B is the orographic height at the ocean floor. The neglect of this topographic term is probably the most restrictive single approximation in the model. Given this neglect, (14.3) becomes

$$\beta\overline{v} = \mathrm{curl}_z(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B) \tag{14.4}$$

where henceforth, in this section, quantities with an overbar are understood to be the vertical integral over the depth of the ocean. If the stresses depend only on the velocity fields then thermodynamic fields do not affect the vertically integrated flow.

At the top of the ocean, the stress is given by the wind. At the bottom, in the absence of topography we assume that the stress may be parameterized by a linear drag, or Rayleigh friction, as might be generated by an Ekman layer; it is this assumption that particularly characterizes this model as that due to Stommel. (Note that we parameterize the friction by a drag acting on the vertically integrated velocity. Using the velocity at the bottom of the ocean would be more realistic, but this wrinkle is beyond the scope of vertically integrated models.) Eq. (14.4) then becomes

$$\beta\overline{v} = -r\overline{\zeta} + W_E(x,y) \tag{14.5}$$

Formulation of the Stommel/Munk model

- ⋆ Vertically integrated planetary-geostrophic equations, or a homogeneous fluid with nonlinearity neglected.
- ⋆ Friction parameterized by a linear drag (Stommel model) or a harmonic Newtonian viscosity (Munk model) or both (Stommel-Munk model).
- ⋆ Flat bottomed ocean.

Variations on the theme include allowing nonlinearity in the vorticity equation, posing the problem in domains of various shapes, and allowing bottom topography (in particular sloping sidewalls). The solutions are normally calculated in the boundary-layer approximation, but some exact solutions exist.

where W_E is proportional to the wind-stress curl at the top of the ocean and is a known function. Because the velocity is divergence-free, we can define a streamfunction ψ such that $\overline{u} = -\partial\psi/\partial y$ and $\overline{v} = \partial\psi/\partial x$. Eq. (14.5) then becomes

$$r\nabla^2\psi + \beta\frac{\partial\psi}{\partial x} = W_E(x, y). \tag{14.6}$$

This equation is often referred to as the *Stommel problem* or *Stommel model,* and may be posed in a variety of two dimensional domains.

14.1.2 Alternative formulations

A number of alternative formulations leading to (14.6) are possible. None are perhaps as well justified as the derivation via the planetary-geostrophic equations but the differences in the specific assumptions made both give some indication of the robustness of the derivation, and show how the model might be extended to include topographic or nonlinear effects.

I. A Homogeneous Model

Rather than vertically integrating, we may suppose that the ocean is a homogeneous fluid obeying the shallow water equations (chapter 3). The potential vorticity equation [c.f., (3.80) on page 138] is:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{\zeta + f}{h}\right) = \frac{F}{h}, \tag{14.7}$$

where F represents friction and forcing [c.f., (3.80)]. In an ocean with a rigid-lid and flat bottom then we obtain the barotropic vorticity equation,

$$\frac{\mathrm{D}\zeta}{\mathrm{D}t} + \beta v = F. \tag{14.8}$$

The forcing term F we again represent as a wind-stress curl and a linear drag. Further, since the horizontal velocity is divergence-free (because of the flat-bottom and rigid-lid assumptions) we may represent it as a streamfunction, whence we obtain the closed equation

$$\frac{\mathrm{D}}{\mathrm{D}t}\nabla^2\psi + \beta\frac{\partial\psi}{\partial x} = W_E(x,y) - r\nabla^2\psi. \tag{14.9}$$

where W_E represents the wind forcing. This equation is the 'time-dependent nonlinear Stommel problem'. The steady nonlinear problem is sometimes of interest, too, and this is just

$$J(\psi,\nabla^2\psi) + \beta\frac{\partial\psi}{\partial x} = W_E(x,y) - r\nabla^2\psi. \tag{14.10}$$

To obtain the original Stommel model we just ignore the advective derivative, which will be valid if $|\zeta| \sim Z \ll \beta L$ where $Z = U/L$ is a representative value of vorticity. This condition is equivalent to

$$R_\beta \equiv \frac{U}{\beta L^2} \ll 1. \tag{14.11}$$

R_β is called the *beta-Rossby number.* On sufficiently large scales, $\beta \sim f/L$ and (14.11) is similar to a small Rossby number assumption.

II. Quasi-geostrophic formulation

In the PG formulation, the horizontal velocity is divergence-free only because we have vertically integrated. If the scales of motion are not too large the horizontal flow at every level is divergence-free for another reason: because it is in geostrophic balance. In reality, over a single oceanic gyre (say from 15° to 40° latitude), variations in Coriolis parameter are not large, and this prompts us to formulate the model in terms of the quasi-geostrophic equations. Formally, such a model would then be restricted to length-scales L no more than $\mathcal{O}(Ro^{-1})$ larger than the deformation radius, and for gyre scales this criterion is marginally satisfied if $L_d = 100\,\mathrm{km}$. An advantage of the quasi-geostrophic equations is that it readily allows for the inclusion of both nonlinearity and stratification. [For an informal summmary of quasi-geostrophy, see the box on page 224.] The quasi-geostrophic vorticity equation for a Boussinesq system is:

$$\frac{\mathrm{D}\zeta}{\mathrm{D}t} + \beta v = f_0\frac{\partial w}{\partial z} + \mathrm{curl}_z\frac{\partial \boldsymbol{\tau}}{\partial z} \tag{14.12}$$

If we neglect the advective derivative, and vertically integrate, we obtain

$$\beta\int v\,\mathrm{d}z = f_0[w]_B^T + \mathrm{curl}_z[\boldsymbol{\tau}]_B^T. \tag{14.13}$$

where T denotes the ocean top and B the bottom. We now make one of two virtually equivalent choices.

(i) We suppose that the integration is over the entire depth of the ocean, in which case the term $[w]_B^T$ vanishes given a rigid lid and a flat bottom. If the stress at the top of the ocean is given by the wind stress, and at the bottom of the ocean it is parameterized by a linear drag, then we obtain

$$\beta\overline{v} = W_E(x,y) - r\overline{\zeta} \tag{14.14}$$

just as in (14.5), and where an overbar denotes a vertical integral. Writing $\overline{v} = \partial\psi/\partial x$ and $\overline{\zeta} = \nabla^2\psi$ then gives (14.6).

(ii) We suppose that the integration is between two thin Ekman layers at the top and bottom of the ocean. The stress is zero at the interior edge of these layers, but the vertical velocity is not. At the base of the upper Ekman layer it is given by:

$$w(x, y, -\delta_u) = \mathrm{curl}_z(\boldsymbol{\tau}_T/f_0) \tag{14.15}$$

where the top of the ocean is at $z = 0$ and δ_u is the thickness of the upper Ekman layer. Similarly, at the top of the lower Ekman layer, the vertical velocity is:

$$w(x, y, -H + \delta_B) = \delta_B\zeta \tag{14.16}$$

where $z = -H$ at the ocean bottom and δ_B is the thickness of the bottom Ekman layer. Neglecting the advective derivative, and integrating over the ocean between the two Ekman layers, (14.12) becomes

$$\beta\overline{v} = \mathrm{curl}_z\boldsymbol{\tau}_T - f_0\delta_B\overline{\zeta}. \tag{14.17}$$

Defining the drag coefficient r by $r = f_0\delta_B$, and introduncing a streamfunction gives, as before,

$$r\nabla^2\psi + \beta\frac{\partial\psi}{\partial x} = \mathrm{curl}_z\boldsymbol{\tau}_T. \tag{14.18}$$

14.1.3 Approximate solution of Stommel model

Sverdrup balance

Eq. (14.6) is linear and it is possible to obtain an exact, analytic solution. However, it is more insightful to approach the problem perturbatively, by supposing that the frictional term is small, meaning there is an approximate balance between wind-stress and the β-effect.[5] Friction is small if $|r\zeta| \ll |\beta v|$ or

$$\frac{r}{L} = \frac{f\delta}{HL} \ll \beta \tag{14.19}$$

using $r = fd/H$ where d is the thickness of the Ekman layer and L is the horizontal scale of the motion, and generally speaking this inequality is well satisfied for large-scale flow. The vorticity equation becomes

$$\beta\overline{v} \approx \mathrm{curl}_z\boldsymbol{\tau}_T, \tag{14.20}$$

or *Sverdrup balance.*[6] The observational support for Sverdrup balance is rather mixed, discrepancies arising not so much from failure of (14.19), but from the presence of small-scale eddying motion with concomitantly large nonlinear terms, and the presence of non-negligible vertical velocities induced by the interaction with bottom topography.[7] Nevertheless, Sverdrup balance provides a useful, if not impregnable, foundation on which to build. (Sometimes 'Sverdrup balance' is taken to mean the linear geostrophic vorticity balance $\beta v = f\partial w/\partial z$, but we will restrict its use to a balance between the beta effect and wind stress curl.)

Boundary-layer solution

For simplicity, consider a square domain of side a and rescale the variables by setting

$$x = a\hat{x}, \qquad y = a\hat{y}, \qquad \tau = \tau_0\hat{\tau}, \qquad \psi = \hat{\psi}\frac{\tau_0}{\beta} \tag{14.21}$$

where τ_0 is the amplitude of the windstress. The hatted variables are nondimensional and, assuming our scaling to be sensible, these are $\mathcal{O}(1)$ quantities in the interior. Eq. (14.18) becomes

$$\frac{\partial\hat{\psi}}{\partial\hat{x}} + \epsilon_S\nabla^2\hat{\psi} = \mathrm{curl}_z\hat{\boldsymbol{\tau}}_T, \tag{14.22}$$

where $\epsilon_S = (r/a\beta) \ll 1$, in accord with (14.19). For the rest of this section we will drop the hats over nondimensional quantities. Over the interior of the domain, away from boundaries, the frictional term in (14.22) is small. We can take advantage of this by writing

$$\psi(x,y) = \psi_I(x,y) + \phi(x,y) \tag{14.23}$$

where ψ_I is the interior streamfunction and ϕ is a boundary layer correction. Away from boundaries ψ_I is presumed to dominate the flow, and this satisfies

$$\frac{\partial\psi_I}{\partial x} = \mathrm{curl}_z\boldsymbol{\tau}_T. \tag{14.24}$$

The solution of this equation (called the 'Sverdrup interior') is

$$\psi_I(x,y) = \int_0^x \mathrm{curl}_z\boldsymbol{\tau}(x',y)\,\mathrm{d}x' + g(y), \tag{14.25}$$

where $g(y)$ is an arbitrary function of integration, and which gives rise to an arbitrary zonal flow. The corresponding velocities are

$$v_I = \mathrm{curl}_z\boldsymbol{\tau}, \qquad u_I = -\frac{\partial}{\partial y}\int_0^x \mathrm{curl}_z\boldsymbol{\tau}(x',y)\,\mathrm{d}x' - \frac{\partial g(y)}{\partial y}. \tag{14.26}$$

The dynamics is most clearly illustrated if we now restrict attention to a windstress curl that is zonally uniform, and that vanishes at two latitudes, $y = 0$ and $y = 1$. An example is

$$\tau_T^y = 0, \qquad \tau_T^x = -\cos(\pi y), \tag{14.27}$$

for which $\mathrm{curl}_z\boldsymbol{\tau}_T = -\pi\sin(\pi y)$. The Sverdrup (interior) flow may then be written as

$$\psi_I(x,y) = [x - C(y)]\mathrm{curl}_z\boldsymbol{\tau}_T = \pi[C(y) - x]\sin\pi y, \tag{14.28}$$

where $C(y)$ is the arbitrary function of integration. If we choose C to be a constant, the zonal flow associated with it is $C\,\mathrm{curl}_z\boldsymbol{\tau}_T$. We can then satisfy $\psi = 0$ at *either* $x = 0$ (if $C = 0$) *or* $x = 1$ (if $C = 1$). These solutions are illustrated in Fig. 14.3 for the particular stress (14.27).

Regardless of our choice of C we cannot satisfy $\psi = 0$ at both zonal boundaries. We must choose one, and then construct a *boundary layer* solution (i.e., determine ϕ) to satisfy the other condition. Which choice do we make? On intuitive grounds it seems that we should choose the solution that satisfies $\psi = 0$ at $x = 1$ (the solution on the left in Fig. 14.3), for then the interior flow then goes round in the same

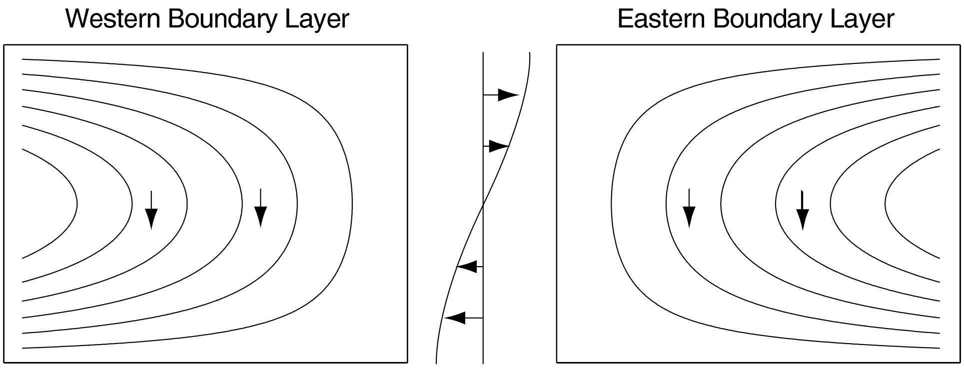


Fig. 14.3 Two possible Sverdrup flows, ψ_I, for the wind stress shown in the center. Each solution satisfies the no-flow condition at either the eastern or western boundary, and a boundary layer is therefore required at the other boundary. Both flows have the same, equatorward, meridional flow in the interior. Only the flow with the western boundary current is physically realizable, however, because only then can friction produce a torque that opposes that of the wind-stress, so allowing the flow to equilibrate.

direction as the wind: the wind is supplying a clockwise torque, and to achieve an angular momentum balance anticlockwise angular momentum must be removed by friction. We can imagine that this would be provided by the frictional forces at the western boundary layer if the interior flow is clockwise, but not by friction at an eastern boundary layer when the interior flow is anticlockwise. Note that this argument is not dependent on the sign of the windstress-curl: if the wind blew the other way a similar argument still implies that a western boundary layer is needed. We will now see if and how the mathematics reflects this intuitive but non-rigorous argument.

Asymptotic Matching

Near the walls of the domain the boundary layer correction $\phi(x, y)$ must become become important in order that the boundary conditions may be satisfied, and the flow, and in particular $\phi(x, y)$, will vary rapidly with x. To reflect this, let us *stretch* the x-coordinate near this point of failure (i.e., at $x = 0$ or $x = 1$, but we don't know which yet) and let

$$x = \epsilon\alpha \qquad \text{or} \qquad x - 1 = \epsilon\alpha \tag{14.29a,b}$$

Here, α is the stretched-coordinate, which has values $\mathcal{O}(1)$ in the boundary layer, and ϵ is a small parameter, as yet undetermined. We then suppose that $\phi = \phi(\alpha, y)$, and using (14.23) in (14.22), we obtain

$$\epsilon_S(\nabla^2\psi_I + \nabla^2\phi) + \frac{\partial \psi_I}{\partial x} + \frac{1}{\epsilon}\frac{\partial \phi}{\partial \alpha} = \text{curl}_z \boldsymbol{\tau}_T, \tag{14.30}$$

where $\phi = \phi(\alpha, y)$ and $\nabla^2\phi = \epsilon^{-2}\partial^2\phi/\partial\alpha^2 + \partial^2\phi/\partial y^2$. Now, by choice, ψ_I exactly satisfies Sverdrup balance, and so (14.30) becomes

$$\epsilon_S\left(\nabla^2\psi_I + \frac{1}{\epsilon^2}\frac{\partial^2\phi}{\partial\alpha^2} + \frac{\partial^2\phi}{\partial y^2}\right) + \frac{1}{\epsilon}\frac{\partial\phi}{\partial\alpha} = 0. \tag{14.31}$$

We now choose ϵ to obtain a physically meaningful solution. The obvious choice is $\epsilon = \epsilon_S$, for then the leading-order balance in (14.31) is

$$\frac{\partial^2 \phi}{\partial \alpha^2} + \frac{\partial \phi}{\partial \alpha} = 0, \tag{14.32}$$

the solution of which

$$\phi = A(y) + B(y)\,\mathrm{e}^{-\alpha}. \tag{14.33}$$

Evidently, ϕ grows exponentially in the negative α direction. If this were allowed, it would violate our assumption that solutions are small in the interior, and we must eliminate this possibility by allowing α to take only positive values in the interior of the domain, and by setting $A(y) = 0$. We therefore choose $x = \epsilon\alpha$ so that $\alpha > 0$ for $x > 0$; the boundary layer is then at $x = 0$, that is, it is a *western boundary,* and it decays eastward in the direction of increasing α — that is, into the ocean interior. We now choose $C = 1$ in (14.28) to make $\psi_I = 0$ at $x = 1$ in (14.28) and then, for the windstress (14.27), the interior solution is given by

$$\psi_I = \pi(1 - x)\sin \pi y. \tag{14.34}$$

This alone satisfies the boundary condition at the eastern boundary. The function $B(y)$ is chosen to satisfy the additional condition that

$$\psi = \psi_I + \phi = 0 \qquad \text{at} \quad x = 0, \tag{14.35}$$

and using (14.34) this gives

$$\pi \sin \pi y + B(y) = 0. \tag{14.36}$$

Using this in (14.33), with $A(y) = 0$, then gives the boundary layer solution

$$\phi = -\pi \sin \pi y\, \mathrm{e}^{-x/\epsilon_S}. \tag{14.37}$$

The composite (boundary layer plus interior) solution is the sum of (14.34) and (14.37), namely

$$\boxed{\psi = (1 - x - \mathrm{e}^{-x/\epsilon_S})\pi \sin \pi y}. \tag{14.38}$$

With dimensional variables this is

$$\psi = \frac{\tau_0 \pi}{\beta}\left(1 - \frac{x}{a} - \mathrm{e}^{-x/(a\epsilon_S)}\right) \sin \frac{\pi y}{a}. \tag{14.39}$$

This is a 'single gyre' solution. Two or more gyres can be obtained with a different wind forcing, such as $\tau^x = -\tau_0 \cos(2\pi y)$, as in Fig. 14.4.

It is a relatively straightforward matter to generalize to other wind stresses, provided these also vanish at the two latitudes between which the solution is desired. It is left as a problem to show that in general

$$\psi_I = \int_{x_E}^{x} \mathrm{curl}_z \boldsymbol{\tau}(x', y)\, \mathrm{d}x', \tag{14.40}$$

and that the composite solution is

$$\psi = \psi_I - \psi_I(0, y)\,\mathrm{e}^{-x/(x_E \epsilon_S)}. \tag{14.41}$$

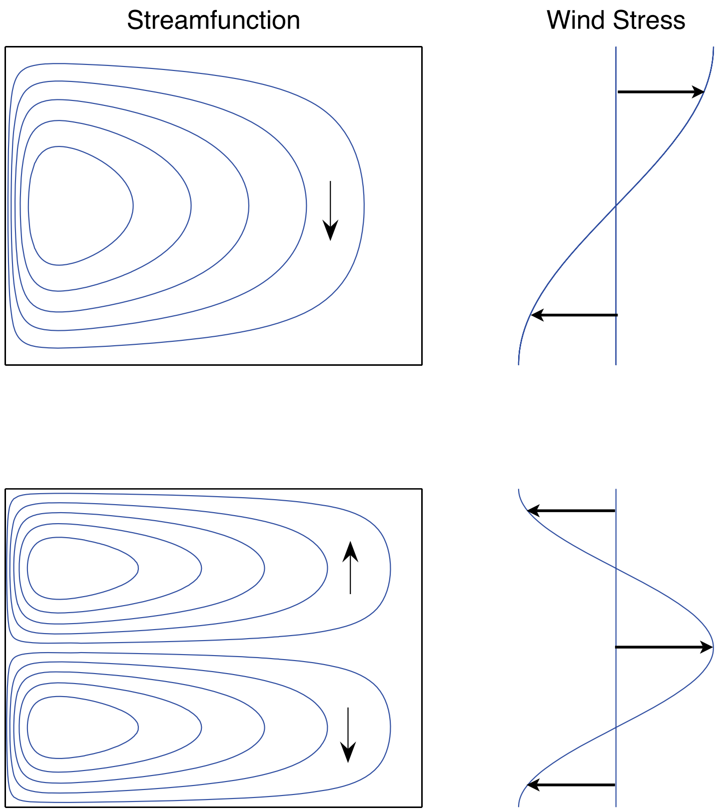


Fig. 14.4 Two solutions of the Stommel model. Upper panel shows the streamfunction of a single-gyre solution, with a windstress proportional to $-\cos(\pi y/a)$ (in a domain of side a), and the lower panel shows a two-gyre solution, with windstress proportional to $\cos(2\pi y/a)$. In both cases $\epsilon_S = 0.04$.

14.2 USING VISCOSITY INSTEAD OF DRAG

A natural variation on on the Stommel problem is to use a harmonic viscosity, $\nu\nabla^2\zeta$, in place of the drag term $-r\zeta$ in the vorticity equation, the argument being that the wind-driven circulation does not reach all the way to the ocean bottom so that an Ekman drag is not appropriate. This variation is called the 'Munk problem' or 'Munk model',[8] and if both drag and viscosity are present we have the 'Stommel-Munk' model. The particular form of the lateral friction used in the Munk problem is still somewhat hard to justify because it relies on an ill-founded eddy diffusion of relative vorticity (chapter 10). Our treatment of this problem is relatively brief, concentrating only on those areas where the problem differs from the Stommel problem. The problem is to find and understand the solution to the (dimensional) equation

$$\beta\frac{\partial\psi}{\partial x} = \mathrm{curl}_z\boldsymbol{\tau}_T + \nu\nabla^2\zeta = \mathrm{curl}_z\boldsymbol{\tau}_T + \nu\nabla^4\psi \tag{14.42}$$

in a given domain, for example a square of side a. We need two boundary conditions at each wall to solve the problem uniquely, and as before for one of them we choose $\psi =$

Some Properties of the Stommel Model

(i) The transport in the Sverdrup interior is equatorwards for an anti-cyclonic wind-stress curl. This transport is exactly balanced by the polewards transport in the boundary layer.

(ii) There must be a boundary layer to satisfy mass conservation, and this must be a western boundary layer if the friction acts to provide a torque of opposite sign to the motion itself. As there is a balance between friction and the β-effect, it is a 'frictional boundary layer'. The western location does not depend on the sign of the wind-stress, nor on the sign of the Coriolis parameter, but it does depend on the sign of β, and so on the direction of rotation of the earth.

(iii) The boundary layer width arises by noting that the terms $r\nabla^2\psi$ and $\beta\partial\psi/\partial x$ are in approximate balance in the western boundary layer, implying a length-scale of $L_W = (r/\beta)$. If r, the inverse frictional time, is $20\,\mathrm{days}^{-1}$ then $L_W \approx 60\,\mathrm{km}$, similar to the width of the Gulf Stream.

0 to satisfy the no-normal-flow condition. For the other condition, two possibilities present themselves:

(i) Zero vorticity, or $\zeta = 0$. Since $\psi = 0$ along the boundary, this possibility is equivalent to $\partial^2\psi/\partial n^2 = 0$ where $\partial/\partial n$ denotes a derivative normal to the boundary. This is known as a 'free-slip' condition. At $x = 0$, for example, the condition becomes $\partial v/\partial x = 0$; that is, there is no horizontal shear at the boundary.

(ii) No flow along the boundary (the 'no slip' condition). That is $\psi_n = 0$ where the subscript denotes the normal derivative of the streamfunction. At $x = 0$ we have $v = 0$.

There is little a priori justification for choosing either of these. The second choice would be demanded if ν were a molecular viscosity, but then we would have to resolve a molecular boundary layer which perhaps would be a few millimeters thick. Instead, ν must be interpreted as some form of eddy viscosity, as we discuss in chapter 10. In that case, one might argue that the free-slip condition should be preferred, but in the absence of a proper theoretical basis for such eddy viscosities there is no truly rational way to make the choice. We will solve the no-slip problem; see exercise 14.3 for the free-slip problem.

Let the wind stress be the canonical $\tau^x = -\cos(\pi y/a)$. Then the interior (Sverdrup) flow is given by (14.34), as for the Stommel problem. This satisfies the free-slip boundary conditions at $y = 0, 1$, namely $\partial_x^2\psi = 0$, automatically. However, we will need boundary layers at both western and eastern boundaries, because the interior solution cannot satisfy all four boundary conditions required by (14.42). The eastern boundary layer will be relatively weak, and needed only to satisfy the no-slip or free-slip condition but, as in the Stommel problem, there will be a strong western boundary layer, needed to satisfy the no-normal flow condition. How thick will this be? Inspection of (14.42) suggests that the frictional term and the β term will balance in a boundary layer of thickness of order L_M where

$$L_M = \left(\frac{\nu}{\beta}\right)^{1/3}. \tag{14.43}$$

Nondimensionalizing (14.42) in a similar way as was done for the Stommel problem yields

$$-\epsilon_M \nabla^4 \hat{\psi} + \frac{\partial \hat{\psi}}{\partial \hat{x}} = \operatorname{curl}_z \hat{\boldsymbol{\tau}}_T \tag{14.44}$$

where $\epsilon_M = (\nu/\beta a^3)$. Considering only the western boundary layer correction, we let the solution be the sum of an interior (Sverdrup) streamfunction plus a boundary layer correction:

$$\hat{\psi} = \psi_I + \phi_W(\alpha, \hat{y}) \tag{14.45}$$

where α is a stretched coordinate such that $\hat{x} = \epsilon\alpha$ where ϵ is some small parameter. Substituting (14.45) into (14.44) and subtracting the Sverdrup balance gives

$$-\epsilon_M \left(\nabla^4 \psi_I + \frac{1}{\epsilon^4}\frac{\partial^4 \phi_W}{\partial \alpha^4}\right) + \frac{1}{\epsilon}\frac{\partial \phi_W}{\partial \alpha} = 0. \tag{14.46}$$

A nontrivial balance is obtained when $\epsilon = \epsilon_M^{1/3}$, implying a dimensional western boundary thickness of $\epsilon a = (\nu/\beta)^{1/3}$, consistent with (14.43). Eq. (14.46) then becomes, at leading order,

$$-\frac{\partial^4 \phi_W}{\partial \alpha^4} + \frac{\partial \phi_W}{\partial \alpha} = 0 \tag{14.47}$$

The boundary conditions on (14.47) are that:

(i) $\phi_W \to 0$ as $\alpha \to \infty$. This states that the perturbation decays as it extends into the interior.

(ii) $\phi_W = -\psi_I$ at $\hat{x} = \alpha = 0$. This is the no-normal-flow condition on the meridional boundary.

(iii) $\partial \phi_W/\partial \hat{x} = -\partial \psi_I/\partial \hat{x}$ at $\hat{x} = \alpha = 0$. This is the no-slip condition.

In addition, zonal boundary layers must exist at $\hat{y} = 0, 1$ and another meridional boundary layer must exist at $\hat{x} = 1$, in order to satisfy the no slip condition. Solving the full problem is a straightforward albeit non-trivial algebraic exercise and, omitting the weak zonal boundary layers at $\hat{y} = 0, 1$ but including the eastern boundary layer correction, we eventually find

$$\hat{\psi} = \psi_I - \psi_I(0, \hat{y})\, \mathrm{e}^{-\hat{x}/(2\epsilon)} \left[\cos\left(\frac{\sqrt{3}\hat{x}}{2\epsilon}\right) + \frac{1}{\sqrt{3}} \sin\left(\frac{\sqrt{3}\hat{x}}{2\epsilon}\right)\right] - \epsilon\, \mathrm{e}^{(\hat{x}-1)/\epsilon} \frac{\partial}{\partial \hat{x}} \psi_I(1, \hat{y}). \tag{14.48}$$

where $\epsilon = (\nu/\beta a^3)^{1/3}$. With the canonical windstress (14.27) this solution becomes

$$\hat{\psi} = \pi \sin(\pi \hat{y}) \left\{1 - \hat{x} - \mathrm{e}^{-\hat{x}/(2\epsilon)} \left[\cos\left(\frac{\sqrt{3}\hat{x}}{2\epsilon}\right) + \frac{1}{\sqrt{3}} \sin\left(\frac{\sqrt{3}\hat{x}}{2\epsilon}\right)\right] + \epsilon\, \mathrm{e}^{(\hat{x}-1)/\epsilon}\right\} \tag{14.49}$$

The solutions of this are plotted in Fig. 14.5. Note how the Munk layers bring the tangential as well as the normal velocity to zero. The eastern boundary layer has

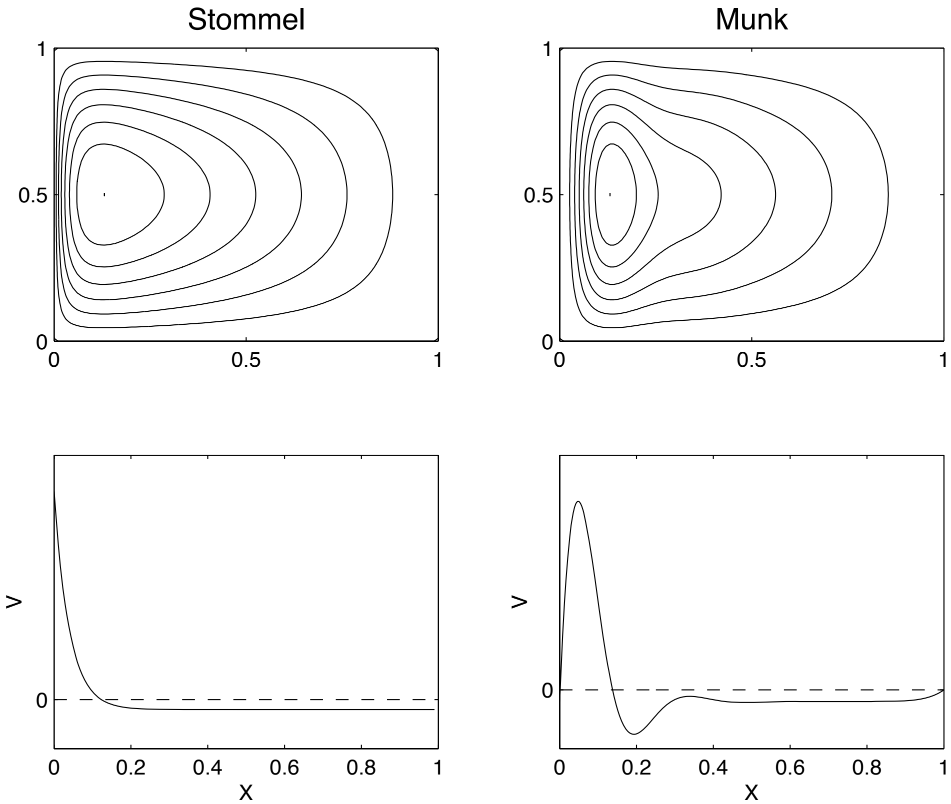


Fig. 14.5 The Stommel and Munk solutions, (14.49) with $\epsilon_S = \epsilon_M^{1/3} = 0.04$, with the windstress $\tau = -\cos \pi y$, for $x, y \in (0, 1)$. Upper panels are contours of streamfunction in the x-y plane, and the flow is clockwise. The lower panels are plots of meridional velocity as a function of x, in the center of the domain ($y = 0.5$). The Munk solution can satisfy both no-normal flow and one other boundary condition at each wall, here chosen to be no-slip.

similar thickness to the western boundary layer, but is not as dynamically important since its *raison d'etre* is to enable the no-slip condition to be satisfied, a relatively weak frictional constraint that manifests itself by a boundary layer in which the flow parallel to the boundary is slowed down. On the other hand the western boundary layer exists in order that the no-normal flow condition can be satisfied, which causes a qualitative change in the flow pattern. It should be emphasized that neither the Stommel nor Munk models are quantitative descriptors of the real ocean, but taken together the similarities of their solutions are a powerful argument for the relative insensitivity of the qualitative form of the solution to the detailed form of the friction. The models do in fact produce qualitatively realistic patterns of large-scale flow in the major basins of the world, as illustrated in Fig. 14.6.

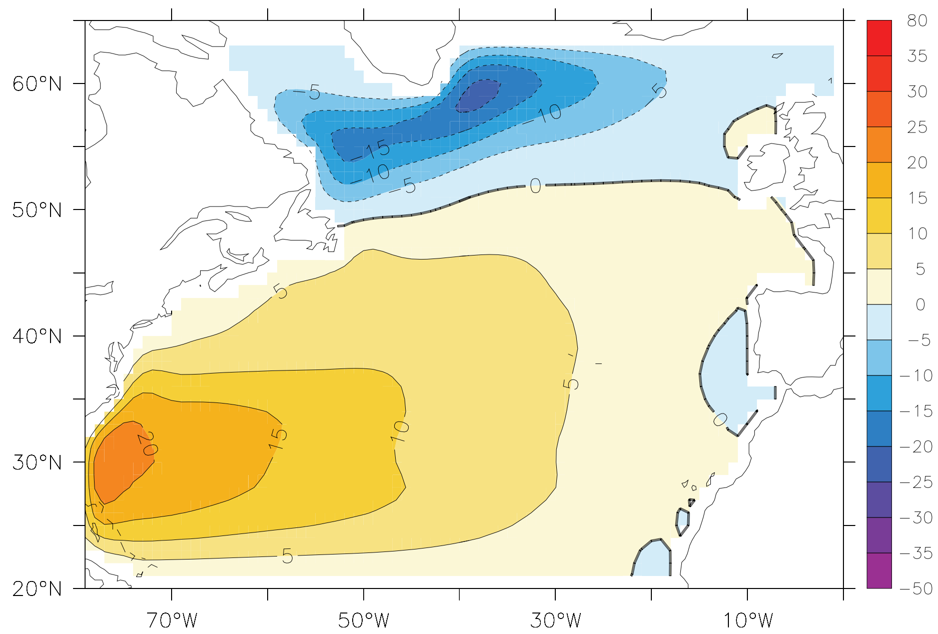


Fig. 14.6 The solution (streamfunction, in Sverdrups) to the Stommel-Munk problem numerically calculated for the North Atlantic, using the observed wind field. The model ocean has realistic geometry, but is flat-bottomed. The calculation qualitatively reproduces the large-scale patterns, including the subtropical and subpolar gyre and the western intensification of both, although the separation of the Gulf Stream from the coast is a little too far North. Compare with Fig. 14.2.[9]

14.3 ZONAL BOUNDARY LAYERS

The canonical wind stress $[\tau^x = -\tau_0 \cos(\pi y/a)]$ is special because its curl vanishes at $y = 0, a$, and so the interior solution satisfies $\psi_I = 0$ at $y = 0,\ 1$. We cannot expect the real wind to be so accomodating. Consider, then, the problem forced by the linear wind profile,

$$\tau^x = \frac{\tau_0}{a}\left(y - \frac{a}{2}\right). \tag{14.50}$$

Scaling the variables in the usual way leads to the nondimensional problem

$$\epsilon_S \nabla^2 \widehat{\psi} + \frac{\partial \widehat{\psi}}{\partial \widehat{x}} = -1. \tag{14.51}$$

The interior flow obeys $\partial \widehat{\psi}/\partial \widehat{x} = -1$ everywhere, and evidently does not satisfy $\widehat{\psi} = 0$ at either $\widehat{y} = 0$ or $\widehat{y} = 1$; thus, boundary layers are needed at the meridional and zonal boundaries, and we let the solution be the sum of five parts,

$$\widehat{\psi} = \psi_I + \phi_W + \phi_E + \phi_N + \phi_S, \tag{14.52}$$

with self-explanatory notation. The interior solution, ψ_I, is the solution of $\partial \psi_I/\partial \widehat{x} = -1$, so that a solution satisfying $\psi_I = 0$ at $\widehat{x} = 1$ is $\psi_I = (1 - \widehat{x})$ and there is no need

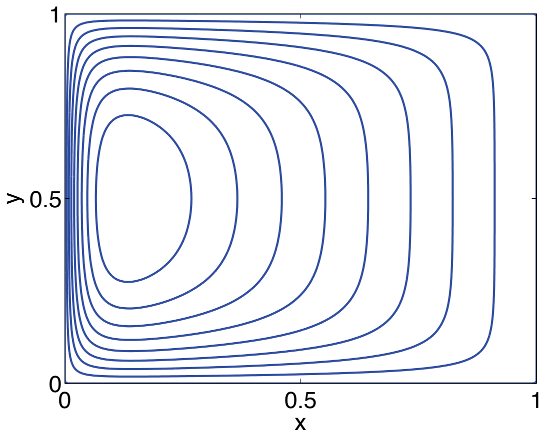


Figure 14.7 Solutions to the Stommel problem with a windstress that increases linearly from $y = 0$ to $y = 1$, as in (14.50). The interior solution is $\psi_I = (1-x)$, or $v_I = -1$, necessitating zonal boundary layers at $y = 0$ and $y = 1$, as well as a western boundary layer at $x = 0$.

for an eastern boundary layer. By the same methods we used in section 14.1.3 the western boundary layer correction is easily found to be

$$\phi_W = -\mathrm{e}^{-\hat{x}/\epsilon_S}. \tag{14.53}$$

It remains to find ϕ_N and ϕ_S, the boundary layer corrections at the northern and southern boundaries.

Consider the boundary layer correction at $y = 1$, and introduce the stretched coordinate α where $\epsilon'\alpha = y - 1$ and where ϵ' is a small parameter. Thus, we let $\phi_N = \phi_N(x, \alpha)$, and on substituting (14.52) into (14.51) we find

$$\epsilon_S \left(\nabla^2 \psi_I + \frac{\partial^2 \phi_N}{\partial \hat{x}^2} + \frac{1}{\epsilon'^2} \frac{\partial^2 \phi_N}{\partial \alpha^2} \right) + \frac{\partial \psi_I}{\partial \hat{x}} + \frac{\partial \phi_N}{\partial \hat{x}} = -1, \tag{14.54}$$

having neglected the small contributions from the other boundary layer streamfunctions (such as ϕ_S). To obtain a nontrivial balance we choose $\epsilon'^2 = \epsilon_S$ and obtain the dominant balance

$$\frac{\partial^2 \phi_N}{\partial \alpha^2} + \frac{\partial \phi_N}{\partial \hat{x}} = 0. \tag{14.55}$$

The boundary conditions necessary to complete the solution are:

(i) The total streamfunction (interior plus boundary correction) must vanish at the northern boundary. That is, $\phi_N(\hat{x}, \alpha = 0) = -\psi_I(\hat{x}, \hat{y} = 1) = -(1 - \hat{x})$.

(ii) The boundary solution should match to the interior streamfunction far from the boundary. That is $\phi_N(\hat{x}, \alpha \to \infty) = 0$.

(iii) At the eastern boundary ϕ_N should also vanish, else it would provide a velocity into the eastern wall. Thus, $\phi_N(1, \alpha) = 0$.

(Actually obtaining the solution is quite algebraically tedious.) The solution at the southern boundary is obtained in an analogous way, and the complete solution is illustrated in Fig. 14.7. The nondimensional thickness of the zonal (northern and southern) boundary layers is $\epsilon_S^{1/2}$, which, because it scales like the half power of a small number, is much thicker than the western boundary current. The thickness arises from the dimensional equations that dominate at the boundary, namely

$$r \frac{\partial^2 \psi}{\partial y^2} + \beta \frac{\partial \psi}{\partial x} = 0. \tag{14.56}$$

This follows from (14.18) by noting that the Laplacian operator must be dominated by the derivatives in the y-direction, and the β term has an (interior) component that anihilates the wind-stress-curl, plus a boundary layer correction to balance the Laplacian. Inspection of (14.56) yields a dimensional thickness $L_Z \sim \sqrt{ra/\beta} = \epsilon_S^{1/2} a$, where a is the length-scale in the x-direction.

14.4 THE NONLINEAR PROBLEM

In the nonlinear problem we seek solutions to

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = -r\nabla^2\psi + \mathrm{curl}_z \boldsymbol{\tau}_T + \nu \nabla^2 \zeta, \tag{14.57}$$

which we have written in dimensional form. In the Stommel problem we set $\nu = 0$ and in the Munk problem we set $r = 0$. In general solutions will be time-dependent and turbulent and this will create motion on small scales, so that ν cannot be neglected. The 'steady nonlinear Stommel-Munk problem' is

$$J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = -r\nabla^2\psi + \mathrm{curl}_z \boldsymbol{\tau}_T + \nu \nabla^2 \zeta. \tag{14.58}$$

We can scale this by first supposing that the leading order balance is Sverdrupian (i.e., $\beta\partial\psi/\partial x \sim \mathrm{curl}_z \boldsymbol{\tau}_T$), from which we obtain the scales $\Psi = |\tau|/\beta$ and $U = |\tau|/(\beta L)$. Equation (14.58) may then be nondimensionalized to yield

$$R_\beta J(\widehat{\psi}, \widehat{\zeta}) + \frac{\partial \widehat{\psi}}{\partial \widehat{x}} = -\epsilon_S \nabla^2 \widehat{\psi} + \mathrm{curl}_z \widehat{\boldsymbol{\tau}}_T + \epsilon_M \nabla^2 \widehat{\zeta} \tag{14.59}$$

where $R_\beta = U/\beta L^2 = |\tau|/(\beta^2 L^3)$, the β-Rossby number for this problem, is a measure of the nonlinearity. Evidently, the nonlinear term increases in importance with increasing wind-stress and for a smaller domain (see also problem 14.8).

14.4.1 A perturbative approach

A direct attack on the full nonlinear problem (14.58) is is possible only through numerical methods, so first we shall explore the problem perturbatively, assuming the nonlinear term to be small; the analysis is straightforward, albeit messy. We begin with the Stommel problem, (14.59) with $\epsilon_M = 0$, and expand the streamfunction in terms of R_β,

$$\widehat{\psi} = \psi_0 + R_\beta \psi_1 + \cdots . \tag{14.60}$$

Now substitute this into (14.59) and equate powers of R_β. The lowest order problem is simply

$$\epsilon_S \nabla^2 \psi_0 + \frac{\partial \psi_0}{\partial \widehat{x}} = \mathrm{curl}_z \widehat{\boldsymbol{\tau}} \tag{14.61}$$

which is the Stommel problem we have already solved. At next order,

$$\epsilon_S \nabla^2 \psi_1 + \frac{\partial \psi_1}{\partial \widehat{x}} = J(\psi_0, \zeta_0) \tag{14.62}$$

This equation has precisely the same form as the Stommel problem, with the known nonlinear term on the right playing the part of the wind-stress. The algebra to obtain the solution is rather tedious, because the right-hand side varies with both $\widehat{x}$ and

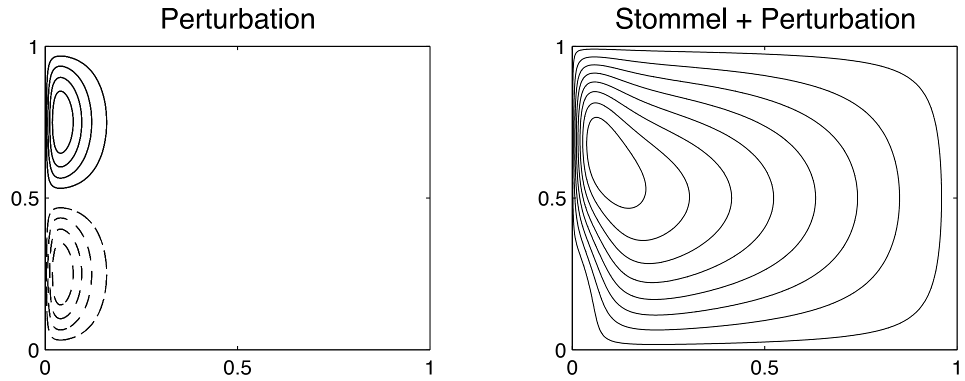


Fig. 14.8 The nonlinear perturbation solution of the Stommel problem, calculated according to (14.63). On the left is the perturbation, $-R_\beta \pi^3/(2\epsilon_S^2)\sin(2\pi y)x e^{-x/\epsilon_S}$, and on the right is the reconstituted solution, using $R_\beta = 10^{-4}$ and $\epsilon = 0.04$. Dashed contours are negative.

$\hat{y}$, but this is much ameleriated by the use of computer algebraic manipulation languages. For the canonical wind stress $\tau^x = -\tau_0\cos(\pi\hat{y})$ the corrected solution, in the boundary layer approximation and ignoring any corrections at the zonal boundaries, is found to be:[10]

$$\hat{\psi} \approx \sin(\pi\hat{y})(1 - \hat{x} - \mathrm{e}^{-\hat{x}/\epsilon_S}) - \frac{R_\beta \pi^3}{2\epsilon_S^3}\sin(2\pi\hat{y})\hat{x}\,\mathrm{e}^{-\hat{x}/\epsilon_S}. \tag{14.63}$$

The solution is illustrated in Fig. 14.8. The perturbation is antisymmetric about $\hat{y} = 1/2$, being positive for $\hat{y} > 1/2$ and negative for $\hat{y} < 1/2$. This tends to move the center of the gyre polewards, narrowing and intensifying the flow in the polewards half of the western boundary current, whereas the western boundary current equatorwards of $\hat{y} = 1/2$ is broadened and weakened. The net effect is that the center of the gyre is pushed polewards — essentially because the western boundary current is advecting the vorticity of the gyre polewards. In the perturbation solution the advection is both by and of the linear Stommel solution; thus, negative vorticity is advected polewards, intensifying the gyre in its polewards half, weakening it in its equatorward half. The solution illustrated in Fig. 14.8 has $\epsilon = 0.04$ and $R_\beta = 10^{-4}$; for larger values of R_β the perturbation itself starts to dominate.

The problem with this perturbative approach is that boundary layer solution to the Stommel problem does not calculate derivatives accurately, so that the nonlinear term $J(\psi, \nabla^2\psi)$ is poorly approximated in the western boundary layer; however, in the interior where the errors are small the perturbative correction is negligible. A more accurate perturbative approach begins with the *exact* solution to the Stommel problem, and then proceeds in the same way. However, the analytic effort is considerable, and the intuitive sense of the way nonlinearities affect the solution is not apparent. (See also problem 14.6.)

14.4.2 A numerical approach

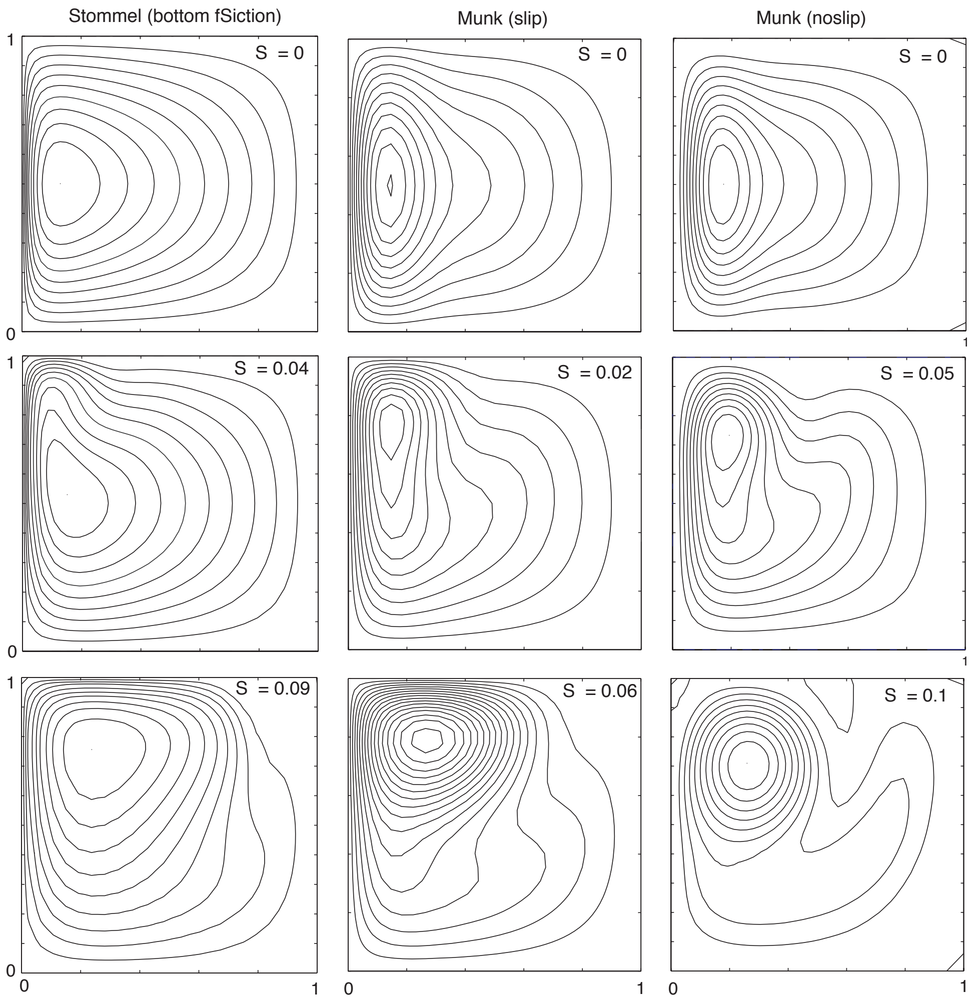


Fig. 14.9 Streamfunctions in solutions of the nonlinear Stommel and Munk problems, obtained numerically with a Newton's method, for various values of the nonlinearity parameter $S = R_\beta^{1/2}$. As in the perturbation solution, for small values of nonlinearity the center of the gyre moves polewards, strengthening the boundary current in the north-western quadrant (for a northern-hemisphere solution). As nonlinearity increases, the recirculation of the gyre dominates, and the solutions become increasingly inertial.

Fully nonlinear solutions show qualitatively similar effects to those seen in the perturbative solutions, as we see in Fig. 14.9, where the solution to (14.59) for the Stommel and Munk problems are obtained numerically by a Newton's method.[11] Just as with the perturbative procedure, for both the Stommel and Munk problems small values of nonlinearity lead to the poleward advection of the gyre's anticyclonic vorticity in the western boundary current, strengthening and intensifying the boundary current in the northwest corner. A higher level of nonlinearity results in a strong recirculating regime in the upper westward quadrant, and ultimately much of the gyre's transport is confined to this regime. The western boundary current itself becomes less noticeable as nonlinearity increases, the more nonlinear solutions have a much greater degree of E-W symmetry than the linear ones, just as the fully nonlinear Fofonoff solutions (section 14.5.3).

Such qualitative effects do not depend on the precise formulation of the model, but the boundary conditions do play an important role in the detailed solution. For example, for a given value of R_β, nonlinearity has a stronger effect in the Munk problem with slip boundary conditions than with no-slip, because in the latter the velocity is reduced to zero at the boundary with a corresponding reduction in the advection term. However, these solutions themselves are unlikely to be relevant for larger values of nonlinearity, because then the flow becomes hydrodynamically unstable.

14.5 * INERTIAL SOLUTIONS

In this section we further explore inertial effects and ask: might purely inertial effects be sufficient to satisfy boundary conditions at the western boundary? Can we envision a purely inertial gyre circulation? Our question is motivated by the steady wind-driven, Rayleigh damped quasi-geostrophic equation, namely

$$J(\psi, \nabla^2\psi) + \beta\frac{\partial\psi}{\partial x} = W_E - r\nabla^2\psi \tag{14.64}$$

$$\frac{U^2}{L^2} \qquad \beta U \quad |W_E| \qquad r\frac{U}{L}, \tag{14.65}$$

where W_E is the wind forcing and the second line indicates the magnitude of the various terms. Since the inertial (advective) terms are higher order than the linear terms, indeed they are higher order than the Rayleigh drag, it is natural to wonder if they themselves might serve to satisfy the no-normal flow condition on the western boundary, without recourse to rather ill-defined frictional terms. The answer is no, as we see below, but nevertheless nonlinear effects may be important in the western boundary layer even if they are small in the interior.

14.5.1 Roles of friction and inertia

The need for friction

Consider the steady barotropic flow satisfying

$$\boldsymbol{u}\cdot\nabla q = \mathrm{curl}_z\boldsymbol{\tau}_T + Fr \tag{14.66}$$

where $q = \nabla^2\psi + \beta y$ and Fr represents frictional effects. Noting that $\boldsymbol{u}$ is divergence-free and integrating the left-hand side of (14.66) over the area, A, between two closed streamlines (ψ_1 and ψ_2, say) and using the divergence theorem we find

$$\int_A \nabla\cdot(\boldsymbol{u}q)\,\mathrm{d}A = \oint_{\psi_1}\boldsymbol{u}q\cdot\boldsymbol{n}\,\mathrm{d}l - \oint_{\psi_2}\boldsymbol{u}q\cdot\boldsymbol{n}\,\mathrm{d}l = 0. \tag{14.67}$$

Here, $\boldsymbol{n}$ is the unit vector normal to the streamline so that $\boldsymbol{u}\cdot\boldsymbol{n} = 0$. The integral of the wind-stress curl over the same area will not in general be zero. Now, we can take these two streamlines as close together as we wish; thus, a balance in (14.66) can only be achieved if *every* closed contour passes through a region where frictional effects are non-zero. This does not mean that nonlinear terms may not locally dominate the friction, just that friction must be somewhere important. In the Stommel and Munk problems, it means that every streamline must pass through the frictional western boundary layer.

Frictional and inertial scales

As we have noted, the ratio of the size of the nonlinear terms to the linear terms is given by the beta Rossby number, $R_\beta = U/(\beta L^2)$. In the western boundary layer the length scales can be expected to be much smaller than the basin scale, and if the balance in the western boundary layer were between the nonlinear and beta term, as in

$$u\frac{\partial}{\partial x}\nabla^2\psi \sim \beta\frac{\partial \psi}{\partial x}, \tag{14.68}$$

then the *inertial boundary layer thickness*, δ_I, is given by

$$\delta_I = \left(\frac{U}{\beta}\right)^{1/2}. \tag{14.69}$$

This, of course, gives $R_\beta = 1$ if $L = \delta_I$. The more energetic the flow, the wider the region where nonlinearity is important, and the corresponding scale is sometimes called the Charney thickness.[12]

The linearized Stommel equation has a boundary layer of dimensional thickness of order $\delta_S = (r/\beta)$, obtained by equating $\beta\partial\psi/\partial x$ and $r\nabla^2\psi$. This thickness is equal to the inertial boundary layer thickness when $U = (r^2/\beta)$. If $\delta_I > \delta_S$, i.e., if $U > (r^2/\beta)$, then nonlinearity *must* be important in the western boundary layer, because the nonlinear terms are at least as important as the beta term in (14.64). On the other hand, if the Stommel boundary layer is wider than the inertial boundary layer, there is no obvious need for nonlinearity to be important, since the Stommel boundary layer generates no length scales smaller than δ_I and R_β remains small.

14.5.2 Attempting an inertial western boundary solution

Although friction must be important, it is nevertheless instructive to try to find a purely inertial solution for the western boundary layer, and to see if and how the attempt fails. Let us suppose that the interior solution has purely zonal flow, with flow either towards or away from the western boundary current, and consider the dimensional equation of motion

$$J(\psi, \nabla^2\psi + \beta y) = 0. \tag{14.70}$$

This has the general solution

$$\nabla^2\psi + \beta y = G(\psi) \tag{14.71}$$

where G is an arbitrary function of its argument. Consider first the case with flow entering the boundary layer with a local velocity $-U$ (i.e., westward) — that is, $\psi = Uy$. The potential vorticity in this region (just outside our putative boundary layer) is $Q_I = \beta y$. Thus, the relation between potential vorticity and streamfunction is $Q_I = \beta\psi_I/U$ and so

$$G(\psi) = \frac{\beta\psi}{U}. \tag{14.72}$$

Because we are assuming the flow is inviscid, the fluid will preserve this relationship between potential vorticity and streamfunction *even as it moves through the western boundary layer.* Thus, using (14.71) and (14.72), the flow in the interior *and* in the boundary layer are given by solutions of

$$\nabla^2\psi - \frac{\beta\psi}{U} = -\beta y. \tag{14.73}$$

with $\psi = 0$ on the boundary. To obtain a solution, we let $\psi = \psi_I + \phi$, where $\psi_I = Uy$ is the particular solution to (14.73) (and, of course, the interior flow). The boundary layer correction then obeys

$$\nabla^2\phi - \frac{\beta\phi}{U} = 0. \tag{14.74}$$

with $\phi = -\psi_I$ at $x = 0$. Now, in the boundary layer, length-scales in the x-direction are much smaller than length-scales in the y-direction, and so (14.74) becomes approximately

$$\frac{\partial^2\phi}{\partial x^2} - \frac{\beta\phi}{U} = 0. \tag{14.75}$$

(We could formalize this procedure by nondimensionalizing and introducing a stretched coordinate, as we did for the Stommel problem.) Solutions of (14.75) are $\phi = -\psi_I\,\mathrm{e}^{-x/\delta_I}$, where $\delta_I = (U/\beta)^{1/2}$, and so the full solution is

$$\psi = \psi_I(1 - \mathrm{e}^{-x/\delta_I}). \tag{14.76}$$

Clearly, this solution smoothly transitions into the interior solution for large x.

What about solutions exiting the boundary layer? We might attempt a similar procedure, but now the interior boundary condition that we must match is that of westward flow, namely $\psi_I = -Uy$. Thus, analogously (14.73), to we seek solutions to the problem,

$$\nabla^2\psi + \frac{\beta\psi}{U} = -\beta y \tag{14.77}$$

for which the homogeneous (boundary layer) equation is

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\beta\phi}{U} = 0. \tag{14.78}$$

Solutions of (14.78) are [c.f., (14.76)]

$$\psi = \psi_I(1 - \mathrm{e}^{-x/\delta_I^*}) \tag{14.79}$$

where $\delta_I^* = \mathrm{i}\,\delta_I = \mathrm{i}\,(U/\beta)^{1/2}$. The solution is therefore wavelike, and does not transition smoothly to the interior flow. This suggests that friction must be important in allowing the boundary layer to smoothly connect to the interior for reasons connected to the conservation of potential vorticity, as discussed in the next subsection.[13]

In addition to this transition problem, we note that (14.76) and (14.79) together do not constitute a globally acceptable inviscid solution, for the simple reason that

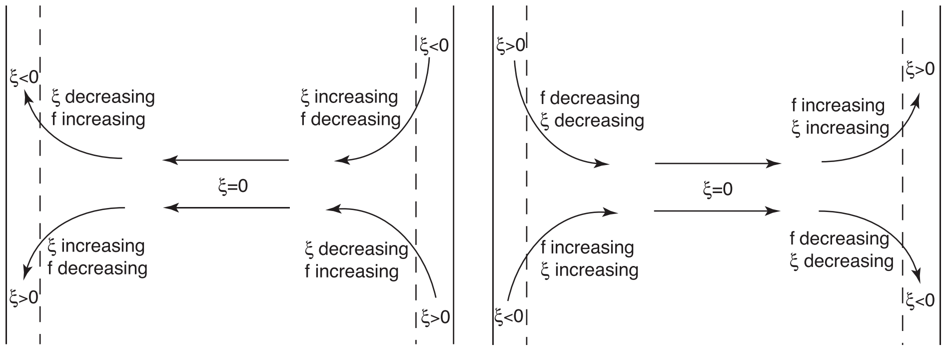


Fig. 14.10 Putative inertial boundary layers connected to a westward flowing internal flow (left panel) or eastward flowing internal flow (right panel), in the Northern Hemisphere. Westward flow into the western boundary layer, or flow emerging from an eastern boundary layer, is able to conserve its potential vorticity through a balance between changes in relative vorticity and Coriolis parameter. But flow cannot emerge smoothly from a western boundary layer into an eastward flowing interior and still conserve its potential vorticity. The right panel thus has inconsistent dynamics.

the flow in the westward flowing interior region has a different potential vorticity–streamfunction (q-ψ) relationship than does the eastward flowing interior. If these two regions are connected by a boundary layer, the flow in that boundary layer must be viscous or unsteady since the value of potential vorticity on the streamlines has changed, whereas inviscid flow conserves potential vorticity.

The connection between the boundary layer and the interior

We have seen that solutions with a boundary layer character that blend smoothly to a flow interior exist only for westward interior flow ($u_I = -U < 0$): an inertial western boundary layer on a beta-plane evidently cannot easily release fluid into the interior. The underlying reason for this stems from the conservation of potential vorticity, as we now explain (and see Fig. 14.10).

Conservation of potential vorticity demands that, in barotropic flow, $v_x - u_y + \beta y$ is a constant on streamlines. In the interior, relative vorticity, ζ, is zero, and in the meridional boundary layers it is effectively v_x. Consider first the case with westward flow in the interior (left panel of Fig. 14.10). Fluid from the irrotational interior approaches the western boundary where it is deflected eiher polewards or equatorwards. In the former case its relative vorticity falls, so allowing potential vorticity to be conserved because the reduction of ζ can be balanced by an increase in the planetary vorticity, f. Similarly, flow deflected southwards produces positive relative vorticity, which is compensated for by the reduced value of f. In the eastern boundary layer, southwards (northwards) moving flow has negative (positive) relative vorticity. As it emerges into the interior its relative vorticity increases (decreases), this being balanced by a fall (rise) in the value of f. Thus, we see that the solution with a westward flowing interior can indeed conserve potential vorticity, at both eastern and western boundaries.

On the other hand suppose the interior flow were eastward (right panel of Fig.

14.10). Flow moving polewards in the western boundary layer has negative relative vorticity. It cannot be freely released into an irrotational interior because f and ζ would then both need to increase, violating potential vorticity conservation. Flow moving southwards with positive relative vorticity similarly is trapped within the western boundary current, unless it meets a zonal boundary which allows a eastward moving boundary current with positive relative vorticity. Similar arguments show that an eastern boundary current cannot entrain fluid from an eastward flowing irrotational interior.

One might ask, why cannot we simply reverse the trajectory of all the fluid parcels in an inviscid flow and thereby obtain a solution with an eastward-flowing interior? The answer is that such a flow will only be a solution if we also reverse the direction of the earth's rotation, and so reverse the sign of β, and hence the location of the frictional boundary current.

14.5.3 A fully inertial approach: the Fofonoff model

Rather than attempt to match an inertial boundary layer with an interior Sverdrup flow, we may look for a purely inertial solution that holds basinwide, and such a construction is known as the *Fofonoff model.*[14] That is, we seek global solutions to the inviscid, unforced problem,

$$J(\psi, \nabla^2\psi + \beta y) = 0 \tag{14.80}$$

We should not regard this problem as representing even a very idealised wind-driven ocean; rather, we may hope to learn about the properties of purely inertial solutions, which might, in turn, tell us something about the ocean circulation.

The general solution to (14.80) is

$$\nabla^2\psi + \beta y = Q(\psi) \tag{14.81}$$

where $Q(\psi)$ is an arbitrary function of its argument. For simplicity we choose the linear form,

$$Q(\psi) = A\psi + B, \tag{14.82}$$

where $A = \beta/U$ and $B = \beta y_0$, where U and y_0 are arbitrary constants. Thus, (14.81) becomes

$$\left(\nabla^2 - \frac{\beta}{U}\right)\psi = \beta(y_0 - y). \tag{14.83}$$

We will further choose $\beta/U > 0$, which we anticipate will provide a westward flowing interior flow, but which (from our experience in the previous section) is more likely to provide a meaningful solution than an eastward interior, and we will use boundary-layer methods to find a solution. A natural scaling for ψ is UL, where L is the domain size, and with this the nondimensional problem is

$$(\epsilon_F\nabla^2 - 1)\widehat{\psi} = \widehat{y}_0 - \widehat{y}, \tag{14.84}$$

where $\epsilon_F = U/(\beta L^2)$, and if we take this to be small (note that $\epsilon_F = R_\beta$) we can find a solution by boundary-layer methods. As usual we write $\widehat{\psi} = \psi_I + \phi$ where

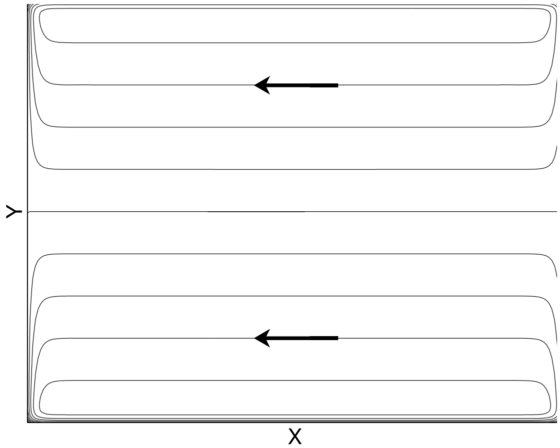


Figure 14.11 The Fofonoff solution. Plotted are contours (streamlines) of (14.88) in the plane $0 < x < x_E$, $0 < y < y_N$ with $U = 1$, $y_N = 1$, $x_E = y_N = 1$, $y_0 = 0.5$ and $\delta_I = 0.05$. The interior flow is westward everywhere, and $\psi = 0$ at $y = y_0$. In addition, boundary layers of thickness $\delta_I = \sqrt{U/\beta}$ bring the solution to zero at $x = (0, x_E)$ and $y = (0, y_N)$, excepting small regions at the corners.

$\psi_I = \hat{y} - \hat{y}_0$ [dimensionally, $U(y - y_0)$] and ϕ is the boundary layer correction, to calculated separately for each boundary using

$$\epsilon_F \nabla^2 \phi - \phi = 0. \tag{14.85}$$

For example, at the northern boundary, $\hat{y} = \hat{y}_N$, the y-derivatives will dominate and (14.85) may be approximated by

$$\epsilon_F \frac{\partial^2 \phi}{\partial \hat{y}^2} - \phi = 0, \tag{14.86}$$

with solution

$$\phi = B\, \mathrm{e}^{-(\hat{y}_N - \hat{y})/\epsilon_F}, \tag{14.87}$$

where $B = \hat{y}_0 - \hat{y}_N$, so satisfying the boundary condition that $\phi(\hat{x}, \hat{y}_N) = -\psi_I(\hat{x}, \hat{y}_N)$. We follow a similar procedure at the other boundaries to obtain the full solution, and in dimensional form this is

$$\psi = U(y - y_0)\left(1 - \mathrm{e}^{-x/\delta_I} - \mathrm{e}^{-(x_E - x)/\delta_I}\right) + U(y_0 - y_N)\, \mathrm{e}^{-(y_N - y)/\delta_I} + U y_0\, \mathrm{e}^{-y/\delta_I}. \tag{14.88}$$

where $\delta_I = (U/\beta)^{1/2}$.

A typical solution is illustrated in Fig. 14.11. On approaching the western boundary layer, the interior flow bifurcates at $y = y_0$. The western boundary layer, of width δ_I, accelerates away from this point, being constantly fed by the interior flow. The westward return flow occurs in zonal boundary layers at the northern and southern edges, also of width δ_I. Flow along the eastern boundary layers is constantly being decelerated, because it is feeding the interior. If one of the zonal boundaries corresponds to y_0 (e.g., if $y_N = y_0$) there would be no boundary layer along it, since ψ is already zero at $y = y_0$. Rather, there would be westward flow along it, just as in the interior. Indeed, a slippery wall placed at $y = 0.5$ would have no effect on the solution illustrated in Fig. 14.11.

14.6 TOPOGRAPHIC EFFECTS ON WESTERN BOUNDARY CURRENTS

The above sections have emphasized the role of friction in satisfying the boundary conditions in the west. However, we should not think of friction as being the *cause* of the western boundary layer and in this section we shall show that if there are sloping sidewalls the role of friction is significantly different, and the western bouundary current may even be largely invisicid![15] The key point is that the flow may be inviscid if it is able to follow potential vorticity contours. In a flat-bottomed western boundary layer the flow is moving to larger values of f (and not as a direct response to the wind) and so the flow *must* be frictional. However, if the sidewalls are sloping, then the flow may preserve its potential vorticity (essentially its value of f/h) if it moves offshore as it moves polewards. We will focus on homogeneous fluids, noting that the interaction of topography and stratification is a subtle and rather complex problem.

14.6.1 Homogeneous model

The potential vorticity evolution equation for a homogeneous model with topography and a rigid lid [c.f., (14.7)] may be written as:

$$\frac{\mathrm{D}q}{\mathrm{D}t} = \frac{F}{h}. \tag{14.89}$$

where $q = (\zeta + f)/h$ where $h = h(x, y)$ is the time-independent depth of the fluid, and F represents vorticity forcing and frictional terms. The advecting velocity is determined by noting that the mass conservation equation is just $\nabla \cdot [\boldsymbol{u}h(x, y)] = 0$, which allows us define the mass-transport streamfunction ψ such that

$$u = -\frac{1}{h}\frac{\partial \psi}{\partial y}, \qquad v = \frac{1}{h}\frac{\partial \psi}{\partial x}. \tag{14.90}$$

The streamfunction is obtained from the vorticity by solving the elliptic equation,

$$\nabla \cdot \left(\frac{1}{h}\nabla\psi\right) = \zeta = qh - f. \tag{14.91}$$

The system — equations (14.89), (14.91) and (14.90) — is then closed. Unlike quasi-geostrophy, neither the Rossby number nor the topography need be small for the validity of the model. Including finite size topography but not stratification is not especially realistic *vis a vis* the real ocean, but nevertheless this model is physically realizable and a useful tool.

The 'topographic Stommel problem' is obtained by neglecting relative vorticity in the potential vorticity in (14.89) (i.e., let $q = f/h$) and using a linear drag acting on the vertically integrated fields for friction, and a windstress curl forcing. Multiplying (14.89) by h, expanding the advective term and omitting time dependence gives

$$J\left(\psi, \frac{f}{h}\right) = -r\nabla^2\psi + \mathrm{curl}_z(\boldsymbol{\tau}_W/h), \tag{14.92}$$

where the boundary conditions are $\psi = 0$ at the domain edges and $\boldsymbol{\tau}_W$ represents the wind-stress.

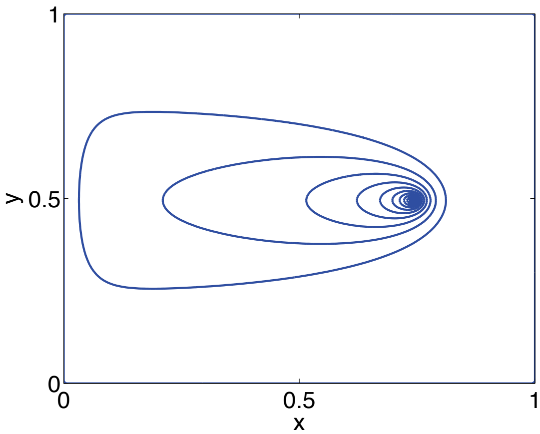


Figure 14.12 The β-plume, namely the Green's function for the Stommel problem. Specifically we plot the solution of (14.94) with $\psi = 0$ at the walls, and a delta-function source at $x = 0.75$, $y = 0.5$. The streamfunction trails westward from the source, as if it were a tracer being diffused while being advected westward along lines of constant f.

14.6.2 Advective dynamics

An illuminating way to begin to study the problem is to write (14.92) in the form[16]

$$J(\Psi, \psi) = +r\nabla^2\psi - \mathrm{curl}_z(\boldsymbol{\tau}_W/h), \tag{14.93}$$

where $\Psi \equiv f/h$. Thus, noting that $J(\Psi, \psi) = \boldsymbol{U}\cdot\nabla\psi$ where $\boldsymbol{U} = \mathbf{k}\times\nabla(f/h)$, we regard ψ as being advected by the pseudo-velocity $\boldsymbol{U}$. This advection is along f/h contours and is quasi-westward, meaning that high values of potential vorticity lie to the right. Eq. (14.93) is then an advection-diffusion equation for the tracer ψ, with the advection occuring along f/h contours. The 'source' of ψ is the wind-stress curl, and ψ is diffused by the first term on the right-hand side of (14.93). This same interpretation applies to the original Stommel problem, of course, where the pseudo-velocity, $-\beta\mathbf{i}$, is purely westward, and it is useful to first revisit this problem.

Consider, then, a flat-bottomed ocean, where the wind stress curl is just a point source at $\boldsymbol{x}_0$. With $\Psi = f$, (14.93) becomes

$$r\nabla^2\psi + \beta\frac{\partial\psi}{\partial x} = \delta(\boldsymbol{x} - \boldsymbol{x}_0). \tag{14.94}$$

This can be transformed to a Helmholtz equation by writing $\phi = \psi\exp(\beta x/2r)$, giving $\nabla^2\phi - [\beta/(2r)]^2\phi = \delta(\boldsymbol{x} - \boldsymbol{x}_0)$. This may then be solved exactly, and the solution (for ψ) is illustrated in Fig. 14.12 — this is the Greens function for the Stommel problem. The tracer ψ is 'advected' westwards along f/h contours — lines of latitude in this case — spreading diffusively as it goes; the resulting structure is called a *β-plume*. The western boundary layer results as a consequence of the f/h contours colliding with the western boundary, along with the need to satisfy the boundary condition $\psi = 0$. If there were no diffusion at all, ψ would just propagate westwards from the source, and if in addition the source were spatially distributed, for example as $\sin \pi y$, the solution streamfunction would represent Sverdrup interior flow.

Now consider the case with a sloping sidewall on the western boundary. The f/h contours (the dashed lines in Fig. 14.13b) tend to converge at the southwest corner of the domain, and only where f/h contours intersect the boundary is a diffusive boundary layer required. The solution to (14.94) with $r = 0$ with a conventional

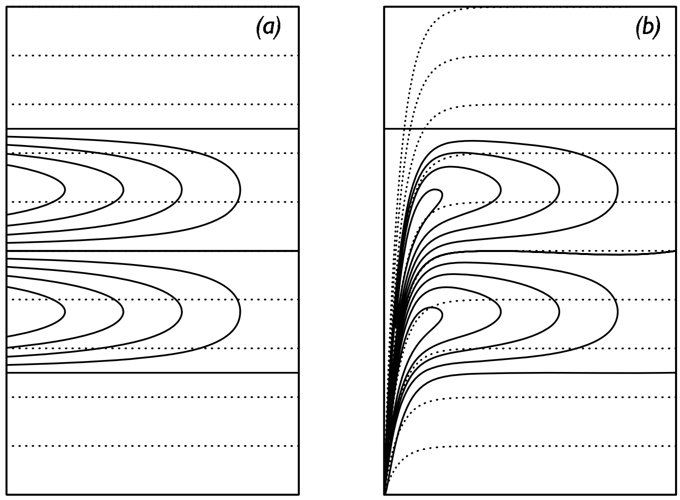


Figure 14.13 The two-gyre Sverdrup flow (solid contours) for (a) a flat-bottomed domain and (b) a domain with sloping sidewalls. The f/h contours are dashed.[17]

'two-gyre' wind-stress are also illustrated in that figure; this is the Sverdrup flow for this problem and it evidently satisfies $\psi = 0$ on the western boundary. In terms of the interpretation above, wind-stress provides a source for the streamfunction ψ and the latter is advected *pseudo-westward* — i.e., along potential vorticity contours, with higher values to the right. The source in this case is distributed over the entire domain, but the contours all converge in the southwest corner. The (numerically obtained) solution to the associated Stommel problem is illustrated in Fig. 14.14, and the western boundary current in this case is no longer a frictional boundary layer. Friction is necessarily important where the flow crosses f/h contours; linear theory suggests that this will occur at the southwest corner. It also occurs on the western boundary where the f/h contours are densely packed and the vorticity in the topographic Sverdrup flow is large, and the friction enables the flow to move across the f/h contours. (In the flat bottomed case the f/h contours are zonal and friction allows the flow to move meridionally.)

14.6.3 Bottom pressure torques and form drag

In the homogeneous problem f and h appear only in the combination f/h, and we may solve the problem entirely without considering pressure effects. It is nevertheless informative to think about how the pressure interacts with the topography to produce meridional flow, and as way toward addressing the effects of stratification. The geostrophic momentum equation is

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi + \boldsymbol{F} \tag{14.95}$$

where $\boldsymbol{F}$ represents both wind forcing and frictional terms. Integrating this over the depth of the ocean (with $z = 0$ at the top), and using Leibnitz's rule ($\nabla \int_{\eta_B}^0 \phi \, dz = \int_{\eta_B}^0 \nabla\phi \, dz - \phi_B \nabla\eta_B$) to evaluate the pressure term, gives

$$\boldsymbol{f} \times \overline{\boldsymbol{u}} = -\nabla\overline{\phi} - \phi_B \nabla\eta_B + \overline{\boldsymbol{F}}. \tag{14.96}$$

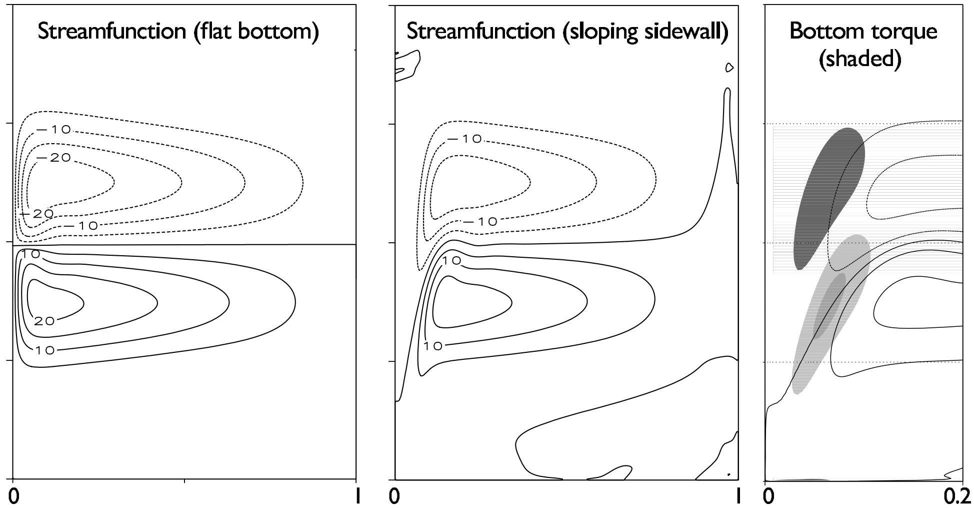


Fig. 14.14 The numerically obtained steady solution to the homogenous problem with a two-gyre forcing and friction, for a flat-bottomed doman and a domain with sloping western sidewall. The shaded regions in the right panel show the regions where bottom pressure torque is important, in the meridional flow of the western boundary currents.[18]

Here, the overbar denotes a vertical integral (e.g., $\overline{\boldsymbol{u}} = \int_{\eta_B}^{0} \boldsymbol{u}\,\mathrm{d}z$), ϕ_B is the pressure at $z = \eta_B$ and η_B is the z-coordinate of the bottom topography. We take the top of the ocean at $z = 0$, and note that $\nabla\eta_B = -\nabla h$ where h is the fluid thickness. Taking the curl of (14.96) gives

$$\beta\overline{v} = -J(\phi_B, \eta_B) + \mathrm{curl}_z\overline{\boldsymbol{F}} \tag{14.97}$$

This equation holds for both a stratified and homogeneous fluid. The first term on the right-hand side is the *bottom pressure torque,* discussed more below.

Equation (14.97) is similar to (14.92), in that both arise from (14.96). To derive a equation with the same form as (14.92) but valid for a stratified fluid, we write the vertical integral of the pressure as

$$\int_{-h}^{0} \phi\,\mathrm{d}z = \int_{-h}^{0} [\mathrm{d}(\phi z) - z(\partial\phi/\partial z)\,\mathrm{d}z] = \int_{-h}^{0} [\mathrm{d}(\phi z) - zb\,\mathrm{d}z] = h\phi_B + \Gamma, \tag{14.98}$$

using hydrostatic balance, and where $\Gamma \equiv -\int_{-h}^{0} zb\,\mathrm{d}z$. Using (14.98) in (14.96) we obtain:

$$\boldsymbol{f} \times \overline{\boldsymbol{u}} = -h\nabla\phi_B + \nabla\Gamma + \overline{\boldsymbol{F}}. \tag{14.99}$$

The curl of this equation just gives back (14.97), but if we divide by h before taking the curl we obtain

$$J(\psi, f/h) + J(h^{-1}, \Gamma) = \mathrm{curl}_z(\overline{\boldsymbol{F}}/h). \tag{14.100}$$

If the stratification vanishes (i.e., b is constant) then Γ is function of h alone and $J(h^{-1}, \Gamma) = 0$, and (14.100) reprises (14.92), given an appropriate choice of $\boldsymbol{F}$. The second term in (14.100) is known as the JEBAR term — joint effect of baroclinicity and relief — and it couples the depth integrated flow with the baroclinic flow.[19] Thus, in the bottom is not flat, the Stommel-Munk models are not solutions for the vertically integrated flow.

Bottom pressure torque in a homogeneous gyre

The first term on the right-hand side of (14.97) is the *bottom pressure torque,* T_B. It be written as the curl of a form stress, $\boldsymbol{\tau}_{form} = -\phi_B \nabla \eta_B$; that is, $T_B \mathbf{k} = \nabla_3 \times \boldsymbol{\tau}_{form} = -\nabla \phi_B \times \nabla \eta_B$, and so is non-zero whenever the pressure gradient has a component parallel to the topographic contour. Note that for unforced and inviscid flow we have both $J(\psi, f/h) = 0$ and, from (14.97), $\beta v = T_B$. That is, flow along f/h contours is equivalent to meridional flow balanced by a bottom pressure torque. On the other hand, if the domain is flat-bottomed, all meridional flow is forced or viscid.

However, unlike viscosity, the bottom pressure torque cannot balance the effects of the wind-stress curl when integrated over the whole domain, or indeed when integrated over an area bounded by a line of constant ϕ_B or constant η_B, because its integral over such an area vanishes and (14.92) cannot be balanced if $r = 0$. Locally, however, bottom pressure torque can be as or more important than either the wind forcing or friction. In the numerical simulations shown in Fig. 14.14, it is the bottom pressure torque term that largely balances the polewards flow (βv) in the vorticity equation in parts of western boundary current, with friction acting in the opposite sense. That is, over some regions where the flow is crossing f/h contours we have the balance

$$[\beta v] \approx [\text{Bottom pressure torque}] - [\text{Friction}], \tag{14.101}$$

where the terms in square brackets are positive, and friction is small. In contrast, in the flat-bottomed case in the western boundary layer we have the classical balance $[\beta v] \approx +[\text{Friction}]$, with both terms positive.

Now consider the balance of momentum, integrated zonally across the domain. We write the vorticity equation (14.97) in the form

$$\nabla \cdot (f\overline{\boldsymbol{v}}) = -\mathrm{curl}_z(\phi_B \nabla \eta_B) + \mathrm{curl}_z \boldsymbol{\tau}_T - \mathrm{curl}_z \boldsymbol{\tau}_B, \tag{14.102}$$

where $\boldsymbol{\tau}_T$ is the wind stress at the top and $\boldsymbol{\tau}_B$ the frictional stress at the bottom. Integrate (14.102) over the area of a zonal strip bounded by two nearby lines of latitude, y_1 and y_2, and the coastlines at either end. The term on the left-hand side vanishes by mass conservation and using Stokes's theorem we obtain:

$$\int_{y_1} \phi_B \frac{\partial \eta_B}{\partial x}\,\mathrm{d}x - \int_{y_2} \phi_B \frac{\partial \eta_B}{\partial x}\,\mathrm{d}x = \int_{y_1} (\tau_T^x - \tau_F^x)\,\mathrm{d}x - \int_{y_2} (\tau_T^x - \tau_F^x)\,\mathrm{d}x. \tag{14.103}$$

If the topography is non-zero, there is nothing in this equation to prevent the wind-stress being balanced by the form stress terms, with the friction being a negligible contribution. If, for example, friction were to be confined to the southwest corner, then bottom pressure torque is the proximate driver of fluid polewards in the western boundary current. This may hold only if the scale of the sloping sidewall is greater than the thickness of the Stommel layer; if the converse holds then the sidewalls appear to be essentially vertical to the flow. If the sidewalls are truly vertical, then the form stress is confined to delta-functions at the walls. Friction must then be important even in the zonal balance, because if we restrict the integral in (14.103) to a strip that does not quite reach the sidewalls, the left-hand side vanishes identically and the wind-stress can only be balanced by friction.

To conclude this discussion, we note that the effects of topography are likely greater in homogeneous fluids than in stratified fluids, because the stratification may partially shield the wind-driven upper ocean from feeling the topography, but we leave the exploration of that topic for another day.

14.7 * VERTICAL STRUCTURE OF THE WIND-DRIVEN CIRCULATION

We now examine the vertical structure of the wind-driven flow.[20] We pose the problem using the quasi-geostrophic equations, taking the overall stratification of the ocean as a given. Of course the production of the stratification is itself one of the most important problems of physical oceanography, but we leave that to the next chapters.

14.7.1 A Two-Layer Quasi-Geostrophic Model

Scales of motion

We will concern ourselves with scales that are sufficiently larger than the deformation radius that we can ignore relative vorticity relative to vortex stretching and the β-effect. We must be careful in so doing, because quasi-geostrophic scaling applies only to scales that are not significantly larger than the deformation scale; thus, our analysis will be formally valid under the following set of inequalities:

$$\begin{aligned}
\beta L &\ll f_0 && \text{(small variations in Coriolis parameter)}\\
\beta L &> U/L && \text{(to ignore relative vorticity compared to planetary vorticity)}\\
L^2 &> L_d^2 && \text{(to ignore relative vorticity compared to vortex stretching)}\\
Ro\,L^2 &\ll L_d^2 && \text{(to keep the variations in stratification small),}
\end{aligned}$$

where L_d is the deformation radius and L the scale of the motion. Since in the mid-latitude ocean $L_d \approx 10^5\,\mathrm{m}$, then the above inequalities are reasonably well satisfied for $L \approx 10^6\,\mathrm{m}$ and $U \approx 0.1\,\mathrm{m\,s^{-1}}$ with $\beta = 10^{-11}\,\mathrm{m^{-1}\,s^{-1}}$ and $f_0 = 10^{-4}\,\mathrm{s^{-1}}$. The first and last of the inequalities are standard quasi-geostrophic requirements, with the '$\gg$' symbol denoting the asympotic ordering. The middle two inequalities are taken within the quasi-geostrophic dynamics, and are needed in order to ignore relative vorticity and give a balance between the β-effect and vortex stretching. The simultaneous satisfaction of all these conditions may seem restrictive, but the plangent dynamics contained within the quasi-geostrophic equations and the generality of the method employed below will suggest that the principle results obtained may transcend the particular limitations of the equations used.

Constructing the model

We now make the following simplifications for our model ocean.

(i) We use the two-layer quasi-geostrophic equations, with layers of equal thickness.

(ii) We seek only statistically-steady solutions.

(iii) We include a frictional term derived by assuming potential vorticity is fluxed downgradient. Given the neglect of relative vorticity, this is equivalent to an interfacial drag.

(iv) We will neglect the western boundary layer.

Because of the equal-layer-thickness assumption, which we make primarily for algebraic simplicity, it is best considered as model for the upper ocean above a level where the vertical velocity is approximately zero. The equations of motion are then

$$J(\psi_1, q_1) = \frac{1}{H_0}\mathrm{curl}_z \boldsymbol{\tau}_T - \nabla\cdot\boldsymbol{T}_1, \qquad J(\psi_2, q_2) = -\nabla\cdot\boldsymbol{T}_2 \tag{14.104a,b}$$

where

$$q_1 = \beta y + F(\psi_2 - \psi_1), \qquad q_2 = \beta y + F(\psi_1 - \psi_2). \tag{14.105a,b}$$

Here, $F = f_0^2/(g'H_0) = 1/L_d^2$ is a measure of the stratification, where H_0 is the thickness of either layer, and the $\nabla\cdot\boldsymbol{T}$ terms are represent interfacial eddy stresses, which we parameterize by by a downgradient flux of potential vorticity:

$$\boldsymbol{T}_1 = -\kappa\nabla q_1 = -R\nabla(\psi_2 - \psi_1), \qquad \boldsymbol{T}_2 = -\kappa\nabla q_2 = -R\nabla(\psi_1 - \psi_2), \tag{14.106}$$

where R and κ are constants, and $R = \kappa F$. We will be interested in the limit of small R, or more specifically $UFL/R \gg 1$, which is a large Péclet number condition. (The Péclet number is similar to a Reynolds number, but with the diffusivity replacing the kinematic viscosity.) So first consider the case when R is identically zero. An *exact* solution to (14.104) has $\psi_2 = 0$, so that (14.104a) becomes $\beta\partial\psi_1/\partial x = H_0^{-1}\mathrm{curl}_z\boldsymbol{\tau}_T$, with solution

$$\psi_1 = -\frac{1}{H_0\beta}\int_x^{x_E} \mathrm{curl}_z\boldsymbol{\tau}_T \,\mathrm{d}x. \tag{14.107}$$

That is, there is no flow in the lower layer, and the upper layer solution is given by Sverdrup balance. The solution satisfies $\psi_1 = 0$ at $x = x_E$ and, because $\psi_2 = 0$, the nonlinear term on the left-hand side of (14.104a) vanishes identically.

General Solution

We now construct the solution without assuming $\psi_2 = 0$. Although the equations are nonlinear, we obtain a linear equation for the lower-layer streamfunction by adding (14.104a) and (14.104b), giving

$$J(\psi_1, \beta y + F(\psi_2 - \psi_1)) + J(\psi_2, \beta y + F(\psi_1 - \psi_2)) = \frac{1}{H_0}\mathrm{curl}_z\boldsymbol{\tau}_T. \tag{14.108}$$

The nonlinear terms cancel leaving

$$J(\overline{\psi}, \beta y) = \frac{1}{H_0}\mathrm{curl}_z\boldsymbol{\tau}_T, \qquad \text{where} \quad \overline{\psi} = \psi_1 + \psi_2, \tag{14.109a,b}$$

with solution [c.f., (14.107)]

$$\overline{\psi} = -\frac{1}{H_0\beta}\int_x^{x_E} \mathrm{curl}_z\boldsymbol{\tau}_T\,\mathrm{d}x'. \tag{14.110}$$

This simply says that the vertically integrated flow obeys Sverdrup balance. For the canonical windstress

$$\boldsymbol{\tau}_T = -\tau_0\cos\pi y\,\mathbf{i} \tag{14.111}$$

we obtain $\overline{\psi} = (\pi\tau_0/\beta H_0)(x_E - x)\sin\pi y$. It is useful to define

$$\overline{q} \equiv (\beta y + F\overline{\psi}), \tag{14.112}$$

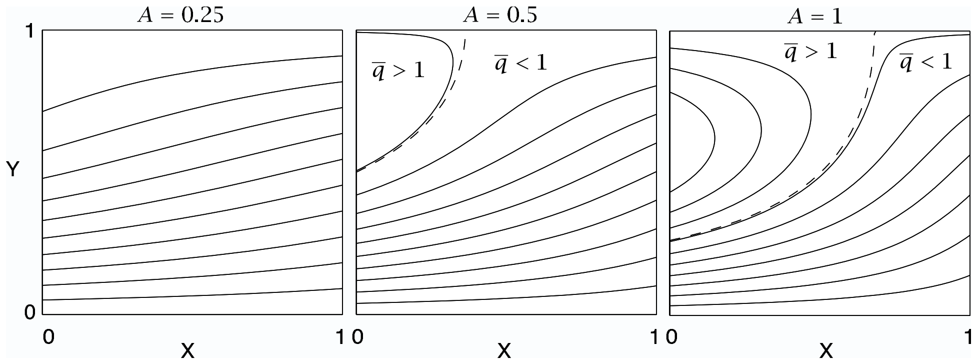


Fig. 14.15 Contours of $\overline{q} = \beta y + A \sin \pi y (1 - x)$, with $\beta = 1$. The dashed line is $\overline{q} = 1$, which separates the blocked region to the east ($\overline{q} < 1$) from the closed region to the west ($\overline{q} > 1$). See Fig. 14.16 for plots of the other fields.

and then $\overline{q} = \beta[y + A(1 - x) \sin \pi y]$, where $A = \pi\tau_0/(\beta H_0)$ parameterizes the wind strength, and this is plotted in Fig. 14.15. For $\overline{q} < 1$ (below and to the right of the dashed line) all the geostrophic contours intersect the eastern boundary and the flow is 'blocked'. For $\overline{q} > 1$ the flow is 'closed'.

Lower layer

Although the full equations are nonlinear, using (14.112) we can obtain a linear equation for the lower layer. Because the Jacobian of a field with itself vanishes, (14.104b) and (14.105b) imply that

$$J(\psi_2, \overline{q}) = -\nabla \cdot \boldsymbol{T}_2, \tag{14.113}$$

and this is useful because $\overline{q}$ is a function of the wind, using (14.112). If $\nabla \cdot \boldsymbol{T}_2 = 0$ then

$$J(\psi_2, \overline{q}) = 0. \tag{14.114}$$

As well as the possibility that $\psi_2 = 0$ we now have the more general solution

$$\psi_2 = G(\overline{q}) \tag{14.115}$$

where G is an *arbitrary* function of its argument. Isolines of ψ_2 and $\overline{q}$ are then coincident. (Contours that are isolines of both streamfunction and potential vorticity are generally known as geostrophic contours.)

Consider a blocked isoline of $\overline{q}$; that is, one that intersects the eastern boundary (see again Fig. 14.15). The ψ_2 contour coincident with this has value zero at the eastern boundary (by the no-normal flow condition). Thus $\psi_2 = 0$ *everywhere* in the blocked region, and $q_2 = \overline{q}$. In this region the Sverdrup transport is carried everywhere by the upper layer, and the lower layer is at rest. This region is called a 'shadow zone', for the fluid is in the shadow of the eastern boundary, and it will re-appear in a model of the thermocline in chapter 16. In the region of closed contours, ψ_2 cannot be given by this argument. But if R is small, we can expect (14.114) to approximately hold, and that the presence of a small amount of dissipation will determine the functional relationship between ψ_2 and $\overline{q}$. Thus, in summary, there are two regions of flow:

(i) The blocked region in which $\psi_1 \approx \overline{\psi} \gg \psi_2$ and ψ_1 is approximately given by (14.107).

(ii) A closed region in which $\psi_2 = G(\overline{q}) + \mathcal{O}(R)$.

14.7.2 The form of $G(\overline{q})$

A general argument

In chapter 10 we showed that, within a region of closed contours, the values of a tracer that is materially conserved except for the effects of a small diffusion would become *homogeneous*. In the case at hand, potential vorticity is that tracer so that within potential vorticity contours or closed streamlines potential vorticity will become homogenized. If we can determine the value of q_2 within the region of closed contours, then from (14.105) ψ_2 is given by

$$\psi_2 = (1/2F)(\overline{q} - q_2) \tag{14.116}$$

and the solution would be complete. Now, outside the closed region $\psi_2 \ll \psi_1$, so that the outermost contour of the closed region must be characterized by $q_2 \approx \overline{q}$, for this makes ψ_2 continuous between closed and blocked regions. Thus, the value of q_2 within the closed homogeneous region is that of $\overline{q}$ (i.e., $\beta y + F\overline{\psi}$) on its boundary. Since this contour intersects the poleward edge of the domain, where $\overline{\psi}$ is zero, the value of this contour is just βy at $y = L$; that is, βL. Thus, within the closed region,

$$q_2 = \beta L. \tag{14.117}$$

A specific calculation

Now consider the steady, lower-layer potential vorticity equation (14.104b); since $J(\psi_2, F(\psi_1 - \psi_2)) = J(\psi_2, F(\psi_1 + \psi_2))$, (14.104b) may be written as

$$J(\psi_2, \overline{q}) = -\nabla \cdot \boldsymbol{T}_2 \tag{14.118}$$

Integrating around a closed contour of $\overline{q}$ the left-hand side vanishes and

$$R \int (\nabla\psi_1 - \nabla\psi_2) \cdot \boldsymbol{n}\, \mathrm{d}l = 0 \tag{14.119}$$

or

$$\oint \boldsymbol{u}_1 \cdot \mathrm{d}\boldsymbol{l} = \oint \boldsymbol{u}_2 \cdot \mathrm{d}\boldsymbol{l} \tag{14.120}$$

thus, the deep circulation around a mean geostrophic contour (i.e., isoline of $\overline{q}$) is equal to the upper level circulation.

Now, previously we argued that

$$\psi_2 = G(\overline{q}) = G(\beta y + F(\psi_1 + \psi_2)). \tag{14.121}$$

where G is an arbitrary function of its argument. In order to satisfy (14.120) (a linear relation between $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$), G must be a linear function, and so we write

$$\psi_2 = C\left[\frac{\beta y}{F} + (\psi_1 + \psi_2)\right] + B, \tag{14.122}$$

Summary of Wind-driven, Two-layer Solution

The vertically integrated flow in a wind-driven two-layer quasi-geostrophic model is determined by Sverdrup balance. The effects of eddies may be crudely parameterized by a downgradient diffusion of potential vorticity. If this is identically zero, then the lower layer flow is identically zero and the upper layer flow carries all the transport. If the diffusion is small but non-zero, the lower layer streamfunction approximately satisfies $J(\psi_2, \overline{q})$, where $\overline{q}$ is given by (14.112), and therefore ψ_2 is a function of $\overline{q}$ — that is, $\psi_2 \approx G(\overline{q})$. For a typical subtropical wind, contours of $\overline{q}$, and therefore contours of ψ_2, are naturally divided into two regions (Fig. 14.15):

(i) A blocked region (the shadow zone), in which contours of $\overline{q}$ intersect the eastern boundary, the lower layer flow is zero and the upper layer carries all the Sverdrup transport.

(ii) A closed region in which (if we envision a nearly inviscid western boundary current) the flow recirculates. In this region we posit that the lower layer potential vorticity becomes homogeneous, with a value determined by the value at the region's boundary, and this in turn is determined by tracing $\overline{q}$ back to the domain boundary.

To satisfy a circulation constraint the function $G(\overline{q})$ must be a linear function, and given this the entire solution may be determined. If, for example, the wind is zonal and a function of y only, and $\mathrm{curl}_z \boldsymbol{\tau}_T = g(y)$ then, in both regions:

$$\overline{\psi} \equiv \psi_1 + \psi_2 = -\frac{1}{\beta H_0} g(y)(x_E - x), \qquad \overline{q} \equiv \beta y + F\overline{\psi} \tag{OC.1a,b}$$

In the blocked region:

$$\psi_2 = 0, \qquad \psi_1 = -\frac{1}{\beta H_0}(x_E - x)g(y), \tag{OC.2a}$$

$$q_1 = \beta y + F(\psi_1 - \psi_2), \qquad q_2 = \beta y + F(\psi_2 - \psi_1). \tag{OC.2b}$$

In the closed region:

$$q_2 = \beta L \quad \text{(by homogenization)}, \tag{OC.3a}$$

$$\psi_2 = \frac{1}{2F}(\overline{q} - \beta L), \qquad \psi_1 = \overline{\psi} - \psi_2, \tag{OC.3b}$$

$$q_1 = \beta y + F(\psi_2 - \psi_1) = 2\beta y - \beta L. \tag{OC.3c}$$

For $g(y) = -\sin \pi y$ these solutions are illustrated in Fig. 14.15 and Fig. 14.16.

This approach provides a solution to the conundrum of what drives the lower ocean ocean, for if there are no eddy effects at all [i.e., $\boldsymbol{T}_1 = \boldsymbol{T}_2 = 0$ in (14.106)], then the lower layer flow is stationary. This seems unrealistic, for the upper layer flow could be made quite shallow. Another solution to this issue is provided in section 16.4, wherein it is assumed that the lower layers may outcrop and so feel the wind directly.

where C and B are constants. This may be re-arranged to give

$$\psi_1 = -C\frac{\beta y}{F} + (\psi_1 + \psi_2)(1 - C) - B. \tag{14.123}$$

This is consistent with (14.120) if $C = 1/2$. With this, (14.122) gives

$$\overline{q} = 2F(\psi_2 - B) \tag{14.124}$$

and the potential vorticity in the closed contour region of the lower layer is

$$q_2 = \beta y + F\overline{\psi} - 2F\psi_2 = -2FB. \tag{14.125}$$

That is, it is constant. Outside the closed contours $\psi_2 \ll \psi_1$ so that $q_2 \approx \overline{q} = \beta y + F\psi_1$. If we trace this contour to the edge of the domain where $\psi_1 = 0$ and $y = L$ then we see that the value of $\overline{q}$ on the contour, and hence q_2 in the closed region, is βL, as in (14.117), and $B = -\beta L/(2F)$. Using (14.124) then gives

$$\psi_2 = (2F)^{-1}(\overline{q} - \beta L). \tag{14.126}$$

Given ψ_2 and q_2 [from (14.117)], we obtain q_1 and ψ_1 using (14.105) and (14.109b), giving

$$q_1 = 2\beta y - \beta L, \qquad \psi_1 = \overline{\psi} - \psi_2. \tag{14.127}$$

All these fields are illustrated in Fig. 14.16, and see the shaded box on the previous page for a summary.

14.8 * A MODEL WITH CONTINUOUS STRATIFICATION

We now look the dynamics of the continuously stratified circulation, largely by way of an extension of our two-layer procedure. Let us first consider how deep the wind's influence is.

14.8.1 Depth of the Wind's Influence

The thermal wind-relationship in the form $f\partial u/\partial z = \partial b/\partial y$ implies a vertical scale H given by

$$H = \frac{fUL}{\Delta b} \tag{14.128}$$

where Δb is a typical magnitude of the horizontal variation of the buoyancy. We can relate this to the Ekman pumping velocity using the linear geostrophic vorticity equation, $\beta v = f\partial w/\partial z$, which, (assuming that the horizontal components of velocity are roughly similar, i.e., $V = U$), implies that

$$U = \frac{fW_E}{\beta H}, \tag{14.129}$$

Equations (14.128) and (14.129) may be combined to give an estimate of the depth of the wind-driven circulation, namely

$$H = \left(\frac{f^2 W_E L}{\beta\,\Delta b}\right)^{1/2}, \tag{14.130}$$

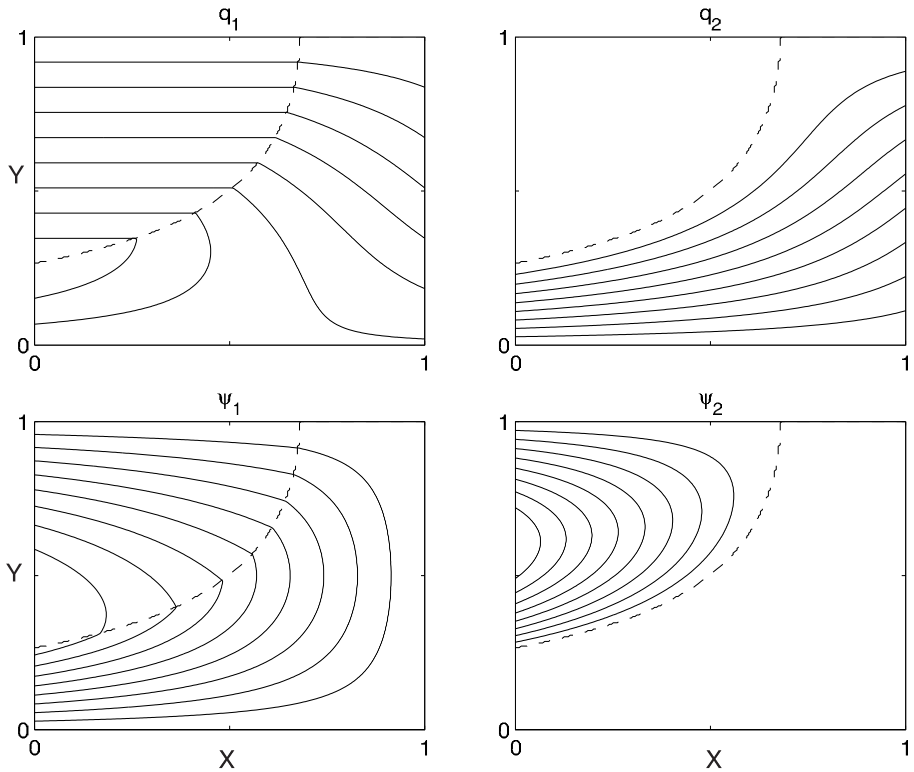


Fig. 14.16 Upper and lower level potential vorticity and streamfunction for the canonical windstress (14.111). The field of $\overline{q}$ is that of Fig. 14.15 with $A = 1$. The dashed line divides the blocked region from the closed region. The lower layer streamfunction ψ_2 is non-zero only in the closed region, and here $q_2 = \beta L$) and $q_1 = 2\beta y - \beta L$. In the blocked region the upper layer carries all of the Sverdrup transport. Both streamfunction and potential vorticity are continuous at the divide: $\psi_2 = 0$ and $q_2 = \overline{q} = \beta L$.

where L may be interpreted as the gyre scale. We now use quasi-geostrophic scaling to relate the horizontal temperature gradient to the stratification using the thermodynamic equation,

$$\frac{\mathrm{D}b}{\mathrm{D}t} + wN^2 = 0, \tag{14.131}$$

with implied scaling

$$\Delta b = \frac{W_E N^2 L}{U} = \frac{N^2 \beta H L}{f_0}, \tag{14.132}$$

where the second equality uses (14.129). Using (14.130) and (14.132) gives

$$\boxed{H = \left(\frac{W_E f^3}{\beta^2 N^2}\right)^{1/3}.} \tag{14.133}$$

Potential vorticity interpretation

The estimate (14.133) can be obtained and interpreted more directly: *the wind-driven circulation penetrates as far as it can alter the potential vorticity q from its planetary value βy.* Recall that, ignoring relative vorticity,

$$q = \beta y + \frac{\partial}{\partial z}\left(\frac{f_0^2}{N^2}\frac{\partial \psi}{\partial z}\right). \tag{14.134}$$

The two terms are comparable if

$$\frac{f_0^2}{N^2 H^2} UL \approx \beta L \tag{14.135}$$

or

$$H^2 \approx \frac{f_0^2 U}{N^2 \beta}. \tag{14.136}$$

Using (14.129) to eliminate U in favour W_E recovers (14.133). Thus, for a given stratification, we have an estimate of the depth of the wind-driven circulation, or at least a scaling for depth of the vertical influence of the wind.

14.8.2 The complete solution

Armed with an estimate for the depth of the wind's influence, we can obtain a solution for the continuously stratified case analogous to that found in the two-layer case section 14.7. Our assumptions are:

(i) In the limit of small dissipation streamfunction and potential vorticity have a functional relationship with each other.

(ii) Potential vorticity is homogenized within closed isolines of q or ψ. The value of q within the homogenized pool is that of the outermost contour, which here is the value of q at the poleward edge of the barotropic gyre.

(iii) Outside of the pool region, (i.e., below the depth of the wind's influence) the streamfunction is zero, and the potential vorticity is given by the planetary value, i.e., βy.

Given these, finding a solution is not difficult. If N^2 is constant and neglecting relative vorticity, the expression for potential vorticity is

$$q = \frac{\partial^2}{\partial z^2}\left(\frac{f_0^2}{N^2}\psi\right) + \beta y \tag{14.137}$$

We nondimensionalize by writing

$$z = \left(\frac{f_0^2 U}{N^2 \beta}\right)^{1/2} \hat{z}, \quad q = \beta L \hat{q}, \quad \psi = \hat{\psi} UL, \quad y = L\hat{y}, \quad w = \frac{U^2 f_0}{N^2 H}\widehat{w}, \tag{14.138}$$

where the hatted variables are nondimensional, and the scaling for w arises from the thermodynamic equation $N^2 w \sim J(\psi, f_0 \psi_z)$. With this, (14.137) becomes

$$\hat{q} = \frac{\partial^2 \hat{\psi}}{\partial \hat{z}^2} + \hat{y}, \tag{14.139}$$

The flow is then given by solving the following equations:

$$\psi_{zz} + y = y_0, \qquad -D(x,y) < z < 0, \tag{14.140a}$$
$$\psi = 0, \qquad z \le -D(x,y), \tag{14.140b}$$

where D is the (to-be-determined) depth of the bowl, y_0 is a constant, and we have dropped the hats over the nondimensional variables. The solution in $D(x,y) < z < 0$ corresponds to the closed region of the two-layer model, and the solution $z \le -D(x,y)$ corresponds to the blocked region of zero lower-layer flow. The constant y_0 is the nondimensional value of potential vorticity within the pool region, and following our reasoning in the two-layer case this is the value of the potential vorticity at the northern boundary. Dimensionally this is βL, so that in nondimensional units $y_0 = 1$.

The lower boundary conditions on (14.140a) is that $\psi = \psi_z = 0$ at $z = -D$, because in the abyss $\psi = \partial\psi/\partial z = 0$ and we require that both ψ and $\partial\psi/\partial z$ be continuous (note that the buoyancy perturbation is proportional to $\partial\psi/\partial z$). The solution that satisfies this

$$\psi = \frac{1}{2}(z+D)^2(y_0 - y), \tag{14.141}$$

and $\psi = 0$ for $z < D$ To obtain an expression for D we first note that the nondimensional vertical velocity at $z = 0$ is given by

$$w = -J(\psi, \psi_z), \tag{14.142}$$

which, using (14.141), gives

$$w = \frac{1}{2}(z+D)^2(y_0 - y)\frac{\partial D}{\partial x}. \tag{14.143}$$

At $z = 0$ the vertical velocity is the Ekman pumping velocity and (14.143) becomes

$$D^2\frac{\partial D}{\partial x} = \frac{2w_E}{(y_0 - y)} \tag{14.144}$$

But the Ekman pumping velocity is related to the barotropic streamfunction, ψ_B, by the Sverdrup relationship, so that integrating (14.144) gives

$$\boxed{D^3 = \frac{6\psi_B}{(y_0 - y)} = -\frac{6(x_E - x)w_E}{(y_0 - y)}}. \tag{14.145}$$

where the second equality holds if w_E is not a function of x. This is a solution for the depth of moving region, the bowl in which potential vorticity is homogenized. An expression for the streamfunction is then obtained by using (14.145) in (14.141), and is found to be

$$\psi = \begin{cases} \frac{1}{2}\left[z(y_0 - y)^{1/2} + (6\psi_B)^{1/3}(y_0 - y)^{1/6}\right]^2 & -D < z < 0, \\ 0 & z < -D. \end{cases} \tag{14.146}$$

The potential vorticity corresponding to this solution is

$$q = \begin{cases} y_0 & -D < z < 0, \\ y & z < -D. \end{cases} \tag{14.147}$$

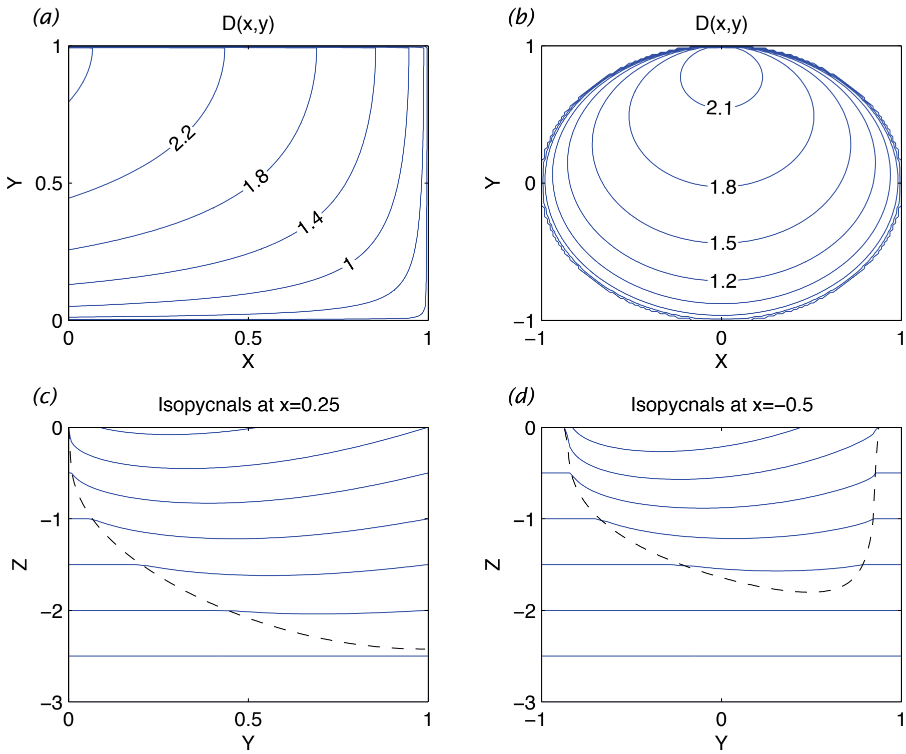


Fig. 14.17 Solutions of (14.112) for two different barotropic streamfunctions. On the left $\psi_B = (1-x)\sin\pi y$ and on the right $\psi_B = 1-(x^2+y^2)$ for $x^2+y^2<1$, zero elsewhere. The upper panels show contours of the depth of the wind-influenced region (solutions of (14.145)). The depth increases to the northwest in the left panel, and to the north in the right panel, so that in both cases the area of the bowl shrinks with depth. The lower panels are contours of $z + (\beta L/f_0)\psi_z/2$, with $\beta L/f_0 = 1/2$, obtained from (14.141) or (14.146), at $x = 0.25$ and $x = -0.5$ in the two cases. These are isopcynal surfaces, with a rather large value of $\beta L/f_0$ to exaggerate the displacement in the bowl region. The dashed lines indicate the boundary of the bowl region, outside of which the isopycnals are flat.

Solutions are illustrated in Fig. 14.17 and Fig. 14.18 for two different barotropic streamfunctions.

It is possible to extend models such as the one described above by appending a western boundary layer, and indeed the homogonization of potential vorticity depends upon the presence of such layer in allowing the flow to recirculation. However, as we saw in section 14.5.3, it is difficult for flow to leave a western boundary layer without the help of friction, and a neutrally stable, damped, stationary Rossby wave typically forms. The critical issue then is whether the presence of dissipation in the western boundary layer affects the homogenization of potential vorticity in the gyre itself. Numerical solutions with a quasi-geostrophic model do show that potential vorticity is able to homogenize under such circumstances. Homgenization in the real ocean has been observed in the Pacific and, somewhat less compellingly, in the Atlantic.[21]

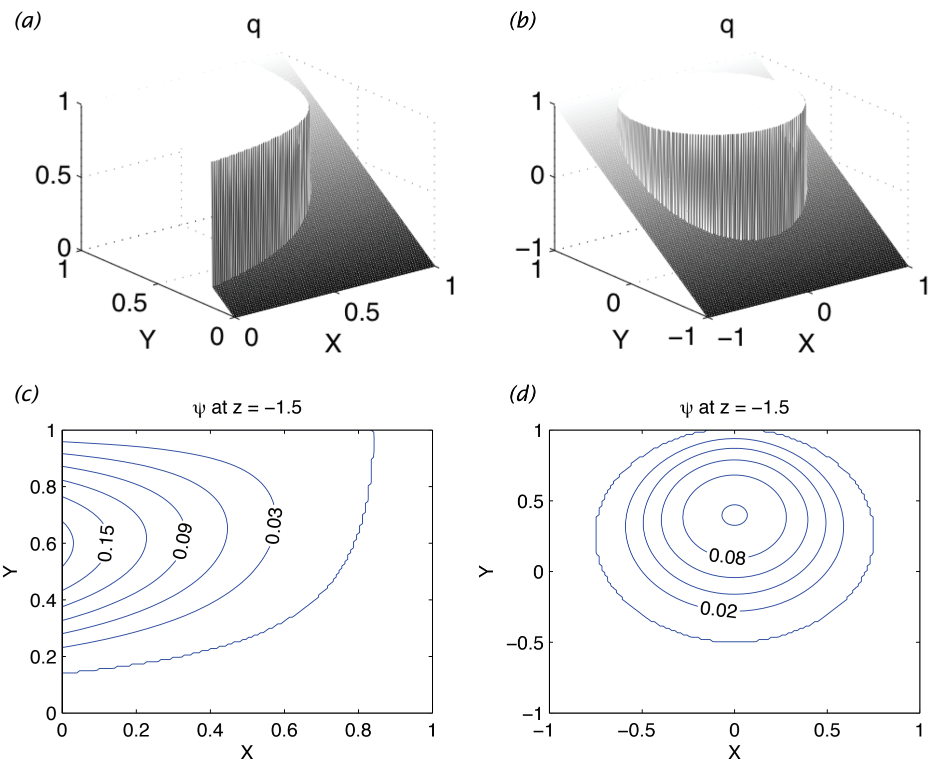


Fig. 14.18 As for Fig. 14.17, but now a perspective of potential vorticity (upper panels, obtained from (14.147)) and contours of streamfunction (lower panels, from (14.146)), at a nondimensional depth of $z = -1.5$. Within the bowls the circulation is clockwise within the bowls and the potential vorticity is uniform. Outside the bowls the fluid is stationary and the potential vorticity has the planetary value βy. The value of potential vorticity within the bowl is the planetary value at the poleward edge of the gyre, and is the same value at all depths.

Notes

1 This figure is based in part on earlier drawings by W. Schmitz and L. Talley that are largely based on observations, as well as on output from numerical models constrained to the observations at GFDL. However, it is highly schematic and not quantitative.

2 The climatological data set of Conkright et al. (2001) was assimilated into a primitive equation numerical model (MOM) by strongly relaxing the model temperature and salinity fields to those of the data set, a method arcanely known as 'robust diagnostics' or 'nudging'. Larger scale features are broadly representative of reality but smaller scale feature may be inaccurate.

3 Courtesy of R. Zhang. See also Zhang and Vallis (2006).

4 Henry Stommel (1920–1992) was one of the most creative and influential physical oceanographers of the 20'th century. Spending most of his career at Woods Hole Institute of Oceanography, his enduring contributions include the first essentially correct theory of western intensification (and so of the Gulf Stream), some of the first models

of abyssal flow and the thermohaline circulation (chapter 15), and his foundational work on the thermocline. His *forté* was in constructing elegantly simple models of complex phenomena — often models that were physically realizable in the laboratory — and at the same time testing and encouraging the testing of the models against observations. This chapter might have been entitled 'Variations on a theme of Stommel'.

5 The asymptotic solution to this boundary value problem was obtained by Wasow (1944), a few years prior to Stommel's work, and further investigated by Levinson (1950). However, it seems unlikely these two investigators were motivated by the oceanographic problem.

6 Harald Sverdrup (1888–1957) was a Norwegian meteorologist/oceanographer who is most famous for the balance that now bears his name, but he also played a leadership role in scientific policy and was the director of Scripps Institution of Oceanography from 1936–1948. The Sverdrup unit is also named for him. Originally defined as a measure of volume transport, with $1\,\mathrm{Sv} \equiv 10^6\,\mathrm{m^3\,s^{-1}}$, it is more generally thought of as a mass transport with $1\,\mathrm{Sv} \equiv 10^9\,\mathrm{kg\,s^{-1}}$, in which case it can also be used as a measure of transport in the atmosphere. The Hadley Cell, for example, has an average transport of about 100 Sv (Fig. 11.3 and Fig. 12.21).

7 Leetmaa et al. (1977) and Wunsch and Roemmich (1985) offer somewhat different views on the matter.

8 After Munk (1950).

9 Courtesy of R. Zhang. See also Zhang and Vallis (2006).

10 Hendershott (1987), Veronis (1966a).

11 Veronis (1966b) investigated nonlinear effects in the Stommel and Munk problems. I am grateful to B. Fox-Kemper for providing the solutions shown in Fig. 14.9, and for many comments on this chapter. See also Fox-Kemper and Pedlosky (2003).

12 After Charney (1955).

13 A more general discussion is given by Greenspan (1962). Il'in and Kamenkovich (1964) and Ierley and Ruehr (1986) show numerically that the friction must in fact be sufficiently strong for steady boundary-layer solutions to exist. See also Fox-Kemper (2004).

14 After Fofonoff (1954).

15 Hughes and de Cuevas (2001).

16 Welander (1968).

17 Figure kindly provided by Laura Jackson.

18 From Jackson et al. (2006).

19 Sarkisyan and Ivanov (1971).

20 This section and the next draw from Rhines and Young (1982b). Young and Rhines (1982) also considered the problem of a western boundary layer.

21 See Rhines and Young (1982a) for a numerical example of PV homogenization (numerical simulation by W. Holland). See Keffer (1985), Talley (1988) and Lozier et al. (1996) for some observed PV fields. See Holland et al. (1984) for some of both.

Further Reading

Abarbanel H. D. I. and Young, W. R., Eds., 1987. *General Circulation of the Ocean.*
This book contains several useful review articles on the oceanic general circulation. Particularly relevant to this chapter is the article by Hendershott on homogeneous single-layer models.

Pedlosky, J., 1996. *Ocean Circulation Theory.*
As well as covering the theory of wind-driven gyres, this book discusses cross-gyre flows and equatorial dynamics.

Problems

14.1 ♦ Obtain the exact analytic solution to the Stommel problem (14.6) in a rectangular basin, and not by way of a boundary-layer approximation. You may use a simple, single-gyre, zonal wind-stress.

14.2 ♦ Obtain the exact analytic solution to the Munk problem (14.42) in a rectangular basin, and not by way of a boundary-layer approximation. You may use a simple, single-gyre, zonal wind-stress.

14.3 Obtain a boundary-layer solution of the Munk problem, (14.44), but with 'free-slip' boundary conditions instead of no-slip boundary conditions. Thus, instead of setting the normal derivative of ψ to be zero at the boundaries ($\partial\psi/\partial n = 0$), set the relative vorticity at the boundary to be zero, and thus $\partial^2\psi/\partial n^2 = 0$.

14.4 ♦ Obtain a solution of either the Stommel or Munk problems in a triangular domain (or a circular domain).

14.5 What is the formal asymptotic accuracy of derivatives in the boundary-layer solution to the Stommel problem? Show that the nonlinear term $J(\psi_s, \nabla^2\psi_S)$, where ψ_S is the streamfunction, is estimated to $\mathcal{O}(1/\delta_W^3)$ accuracy, where δ_W is the width of the western boundary layer.

14.6 ♦ (a) Using regular perturbation theory explicitly obtain the nonlinear corrections to the Stommel solutions for a single gyre flow, and show that (14.63) is an appropriate approximation of this. Discuss the properties of the solution, and in particular, show where are the errors largest. Optional: Repeat this calculation using the exact solution to the Stommel problem, and compare the solutions to those obtained using the boundary-layer approximation. [This problem may be made easier by using algebraic manipulation software.]

♦ (b) Suggest a better approach to this problem that is still semi-analytic or that at least gives some insight into the nonlinear solution.

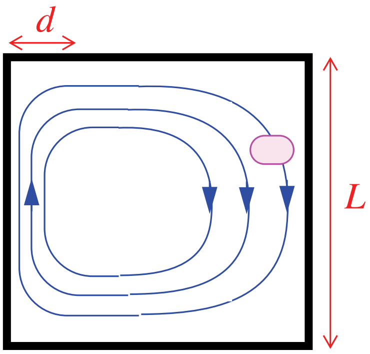
14.7 ♦ Do problem 14.6, but with lateral viscosity instead of bottom drag, as in the Munk problem.

14.8 Suppose that nonlinearity were expected to be of leading order importance in (14.58). Show that this suggests that the nondimensional version of the equation should be:

$$J(\widehat{\psi}, \widehat{\zeta}) + \gamma \frac{\partial \widehat{\psi}}{\partial \widehat{x}} = \epsilon_S' \nabla^2 \widehat{\psi} + \mathrm{curl}_z \widehat{\boldsymbol{\tau}}_T + \epsilon_M' \nabla^2 \widehat{\zeta} \tag{P14.1}$$

where $\gamma = \beta L^2/U = \beta/(|\tau|L^5)^{1/2}$ and $\epsilon_S' = r(L/|\tau|)^{1/2}$. Friction becomes less important as τ increases. Is there a limit in which friction is nowhere important?

14.9 Consider the two-dimensional, non-divergent flow in a box of side L, as pictured. The equatorward flow, of magnitude U, is weak and spread over a lateral scale L, whereas the poleward flow is concentrated in a layer of width $d \ll L$. Suppose the initial condition is a spot of tracer, as shown.



Estimate how long it takes to homogenize the tracer in terms of U, L, κ and d. *Hint:* Consider the advection-diffusion equation,

$$\frac{\partial \phi}{\partial t} + \boldsymbol{u} \cdot \nabla \phi = \kappa \nabla^2 \phi. \tag{P14.2}$$

and estimate advective and diffusive times in the western boundary layer and the interior.

14.10 For a two-layer wind-driven quasi-geostrophic model, obtain an expression for the minimum strength of the wind that leads to closed contours of $\overline{q} = \beta y + F\overline{\psi}$. In particular, for the solution illustrated in Fig. 14.15, at what values of A do closed contours first form?
(*Answer:* $A = \pi^{-1}$.)

14.11 Show by differentiating (14.146) that potential vorticity is uniform in the pool region. What is its value?

Paradise Lost.
John Milton, c. 1667.

CHAPTER

FIFTEEN

The Buoyancy Driven Circulation

OUR GOAL in this chapter and the next is to gain a rudimentary understanding of the three-dimensional dynamics and structure of the ocean circulation. In this chapter we focus on the *meridional overturning* and the associated *abyssal* circulation of the ocean, treating them as if they were solely driven by buoyancy forces, and in chapter 16 we look at the combined effects of wind and buoyancy forcing.

The meridional overturning circulation (MOC) is so-called because it is associated with sinking at high latitudes and upwelling elsewhere, although as we shall see even such a seemingly simple matter as this is not wholly settled. The circulation is also sometimes known as the 'thermohaline' circulation, reflecting a belief that it is driven by gradients in temperature and salinity, but because other mechanisms are also important that name is not appropriate as a general descriptor. In fact, the theory explaining the MOC is not in as satisfactory a state as it is for the quasi-horizontal wind-driven circulation discussed in chapter 14. In the theory of the wind-driven circulation, a rational series of approximations from the governing Navier-Stokes equations leads to a sequence of simple models (e.g., the homogeneous models of the wind-driven circulation, layered quasi-geostrophic models, etc.) whose foundations are thus reasonably secure, whose shortcomings are understood, and whose behaviour can be fairly completely analyzed. Attempts to proceed in a similar fashion with overturning circulation have been less successful; the reason is that the approximations required in order that a tractable conceptual model be constructed are unavoidably severe and, from a fluid-dynamicist's perspective, unjustifiable. Thus, although the large-scale overturning flow is well-described by the planetary geostrophic equations whose complexity is similar to that of the quasi-geostrophic equations, there is no rational simplification of these that leads to a model that is both as simple and in-

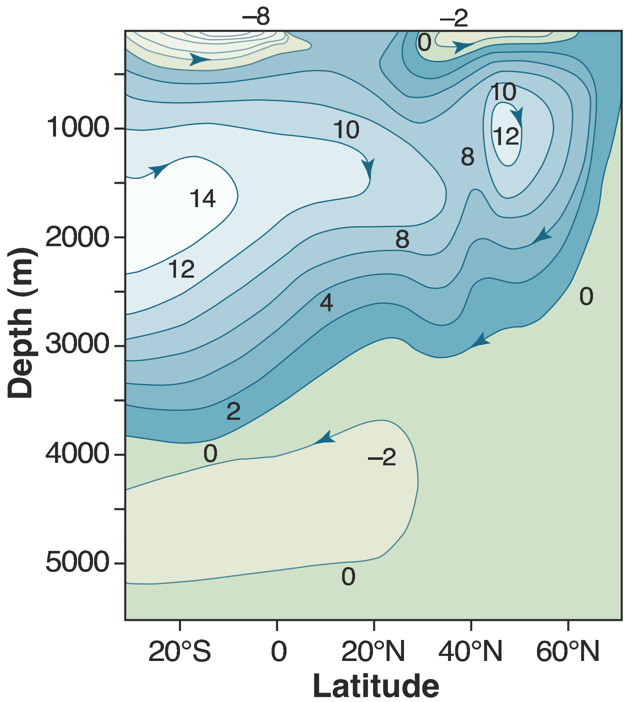


Figure 15.1 The mean meridional overturning circulation in the North Atlantic, obtained with a combination of observations and a model. The contours are the streamfunction of the zonally averaged meridional flow. The units are Sverdrups, and the circulation is mostly clockwise, with sinking at high latitudes.[2]

formative as the homogeneous Stommel model of the quasi-horizontal, wind-driven circulation. Nevertheless, progress *has* been made, through the use of both numerical models and simpler, conceptual, models, and in this chapter we concentrate on the foundations underlying these.

15.1 A BRIEF OBSERVATIONAL OVERVIEW

That there *is* a deep circulation has been known for a long time, largely from observations of tracers such as temperature, salinity and constituents such as dissolved oxygen and silica.[1] We can also take advantage of numerical models that are able to assimilate hydrographic and other observations and produce an estimate of the overturning circulation that is consistent with both the observations and the equations of motion, as illustrated in Fig. 15.1.

Associated with the overturning circulation is a stratification that has a quite distinctive structure, as illustrated in Fig. 15.2. Most of the stratification is evidently concentrated in the upper kilometre or so of the ocean, with a relatively (although not completely) unstratified abyss full of dense water that has originated from high latitudes — the isopycnals shown in Fig. 15.2 all outcrop (i.e., intersect the surface) in the North Atlantic subpolar gyre and/or in the Antarctic Circumpolar Current (ACC). The region of high stratification near the surface is known as the *thermocline* (‘thermo’ for temperature, ‘cline’ for changing); it is effectively synonymous with the *pycnocline* (for changing density), although the latter could exist without the former if there were a *halocline*, in which salinity changed rapidly. What is the cause of this circulation and the associated stratification?

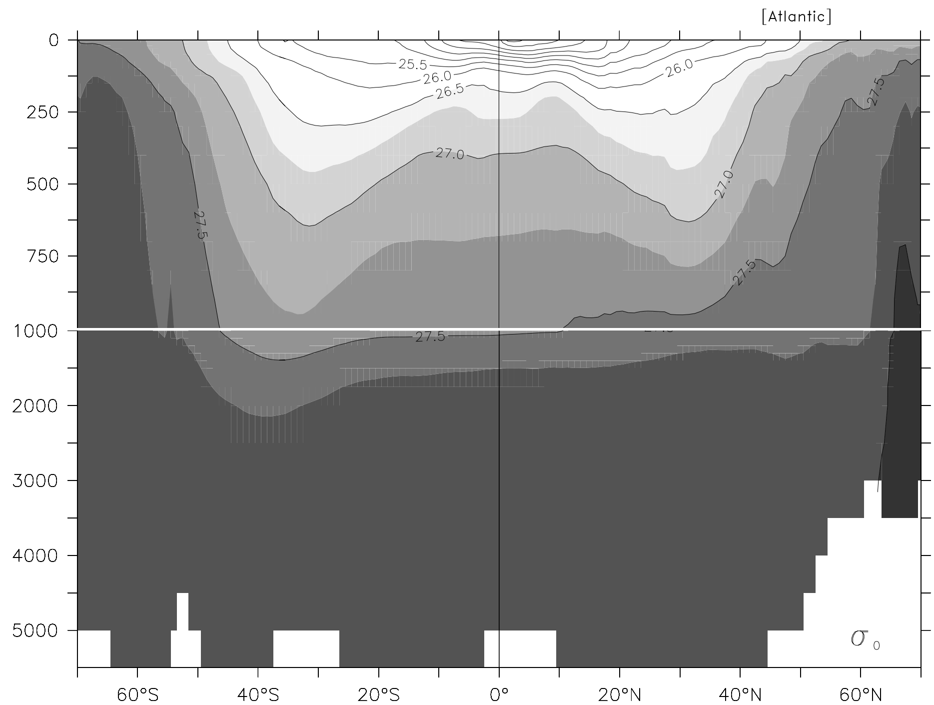


Fig. 15.2 The zonally averaged potential density (σ_θ) in the Atlantic ocean, as a function of depth (m) and latitude. Note the break in the vertical scale at 1000 m. The region of rapid change of density (and temperature) is concentrated in the upper kilometer, in the *main thermocline,* below which the ocean has a much more uniform density.[3]

15.2 †SIDEWAYS CONVECTION

Perhaps the simplest and most obvious fluid dynamical model of the overturning circulation is that of *sideways* or *horizontal convection.* The physical situation is sketched in Fig. 15.3. A fluid (two- or three-dimensional) is held in a container that is insulated on all its sides and bottom, but its upper surface is non-uniformly heated and cooled. In the purest fluid dynamical problem the heat enters the fluid solely by conduction at the upper surface, and one may suppose that here the temperature is imposed. Thus, for a simple Boussinesq fluid the equations of motion are

$$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + \boldsymbol{f} \times \boldsymbol{v} = -\nabla\phi + b\mathbf{k} + \nu\nabla^2\boldsymbol{v}, \tag{15.1a}$$

$$\frac{\mathrm{D}b}{\mathrm{D}t} = \kappa\nabla^2 b, \tag{15.1b}$$

$$\nabla\cdot\boldsymbol{v} = 0, \tag{15.1c}$$

with boundary conditions

$$b(x,y,0,t) = f(x,y), \tag{15.2a}$$

where $f(x,y)$ is a specified field, and $\partial_n b = 0$ on the other boundaries, meaning that the derivative normal to the boundary, and so the buoyancy flux, is zero. An

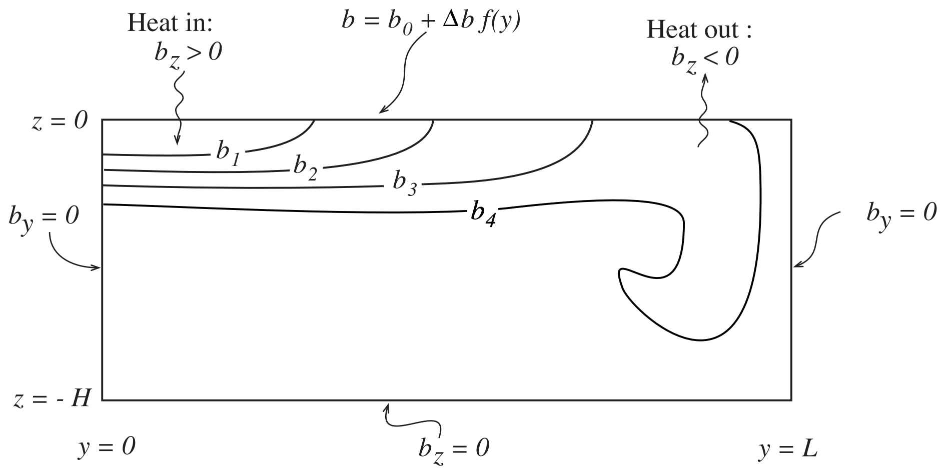


Fig. 15.3 A schema of 'sideways convection'. The y-axis represents latitude and the z-axis represents depth. The fluid is differentially heated and cooled along its top surface, whereas all the other walls are insulating. The result is, typically, a small region of convective instability and sinking near the coldest boundary, with generally upwards motion elsewhere.[4]

alternative upper boundary condition is to impose a flux condition whereby

$$\kappa \frac{\partial}{\partial z} b(x, y, 0, t) = g(x, y), \tag{15.2b}$$

where $g(x, y)$ is given. The oceanographic relevance of (15.1) and (15.2) should be clear: the ocean is heated *and* cooled from above, and although the thermal forcing in the real ocean may differ in detail (being in part a radiative flux, and in part a sensible and latent heat transfer from the atmosphere), (15.2) is a useful idealization. In some numerical models of the ocean, the heat input at the top is parameterized by way of a relaxation to some specified temperature. This is a form of flux condition in which

$$\text{Flux} = \kappa \frac{\partial b}{\partial z} = C(b^* - b), \tag{15.3}$$

and C is an empirical constant.[5] Although this may be a little more realistic than (15.2a) it will not affect the arguments below.

15.2.1 Two-dimensional convection

We may usefully restrict attention to the two-dimensional problem, in latitude and height. The two-dimensional flow may then loosely be thought of as representing the large-scale, statistically steady, zonally averaged flow of the ocean. The zonally-averaged zonal flow is then small, and concomitantly so is the Coriolis force. The incompressibility of the flow then allows one to define a streamfunction such that

$$v = -\frac{\partial \psi}{\partial z}, \quad w = \frac{\partial \psi}{\partial y}, \qquad \zeta = \nabla_x^2 \psi = \left(\frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right) \tag{15.4}$$

where ζ is the vorticity in the meridional plane. We will neglect the subscript x on the Laplacian operator where there is no ambiguity. Taking the curl of Boussinesq equations of motion (15.1) then gives

$$\frac{\partial \nabla^2 \psi}{\partial t} + J(\psi, \nabla^2 \psi) = \frac{\partial b}{\partial y} + \nu \nabla^4 \psi \tag{15.5a}$$

$$\frac{\partial b}{\partial t} + J(\psi, b) = \kappa \nabla^2 b \tag{15.5b}$$

where $J(a,b) \equiv (\partial_y a)(\partial_z b) - (\partial_z a)(\partial_y b)$.

Nondimensionalization and scaling

We non-dimensionalize (15.5) by formally setting

$$b = \Delta b\, \hat{b}, \qquad \psi = \Psi \hat{\psi}, \qquad y = L\hat{y}, \qquad z = H\hat{z}, \qquad t = \frac{LH}{\Psi}\hat{t}, \tag{15.6}$$

where the hatted variables are non-dimensional, Δb is the temperature difference across the surface, L is the horizontal size of the domain, and Ψ, and ultimately the vertical scale H, are to be determined. Substituting (15.6) into (15.5) gives

$$\frac{\partial \hat{\nabla}^2 \hat{\psi}}{\partial \hat{t}} + \hat{J}(\hat{\psi}, \nabla^2 \hat{\psi}) = \frac{H^3 \Delta b}{\Psi^2} \frac{\partial \hat{b}}{\partial \hat{y}} + \frac{\nu L}{\Psi H} \hat{\nabla}^4 \hat{\psi}, \tag{15.7a}$$

$$\frac{\partial \hat{b}}{\partial \hat{t}} + \hat{J}(\hat{\psi}, \hat{b}) = \frac{\kappa L}{\Psi H} \hat{\nabla}^2 \hat{b}, \tag{15.7b}$$

where $\hat{\nabla}^2 = (H/L)^2 \partial^2/\partial \hat{y}^2 + \partial^2/\partial \hat{z}^2$ and the Jacobian operator is similarly non-dimensional. If we now use (15.7b) to choose Ψ as

$$\Psi = \frac{\kappa L}{H}, \tag{15.8}$$

so that $t = H^2 \hat{t}/\kappa$, then (15.7) becomes

$$\frac{\partial \hat{\nabla}^2 \hat{\psi}}{\partial \hat{t}} + \hat{J}(\hat{\psi}, \hat{\nabla}^2 \hat{\psi}) = Ra\sigma\alpha^5 \frac{\partial \hat{b}}{\partial \hat{y}} + \sigma \hat{\nabla}^4 \hat{\psi}, \tag{15.9}$$

$$\frac{\partial \hat{b}}{\partial \hat{t}} + \hat{J}(\hat{\psi}, \hat{b}) = \hat{\nabla}^2 \hat{b}, \tag{15.10}$$

and the non-dimensional parameters that govern the behaviour of the system are

$$Ra = \left(\frac{\Delta b L^3}{\nu \kappa} \right), \qquad \text{(the Rayleigh number)}, \tag{15.11a}$$

$$\sigma = \frac{\nu}{\kappa}, \qquad \text{(the Prandtl number)}, \tag{15.11b}$$

$$\alpha = \frac{H}{L}, \qquad \text{(the aspect ratio)}. \tag{15.11c}$$

The Rayleigh number is a measure of the strength of the buoyancy forcing relative to the viscous term, and in the ocean it will be very large indeed, perhaps $\sim 10^{24}$ if molecular values are used. (Sometimes H is used instead of L in the Rayleigh number definition; we use L here because it is an external parameter.)

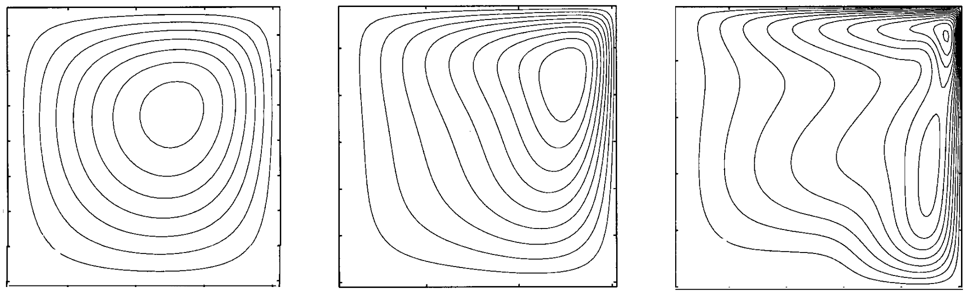


Fig. 15.4 The streamfunction in a numerical simulation of two-dimensional sideways convection.[7] The circulation is clockwise, and the imposed temperature at the top linearly decreases from left to right, and the other walls are insulating. From left to right the Rayleigh numbers are 10^4, 10^6 and 10^8, and the contour interval is 1, 4 and 10 in arbitrary units. The Prandtl number is 10.

For steady non-turbulent flows, and also perhaps for statistically steady flows, then we can demand that the buoyancy term in (15.9) is $\mathcal{O}(1)$. If it is smaller then the flow is not buoyancy driven, and if it is larger there is nothing to balance it. Our demand can be satisified only if the vertical scale of the motion appropriately adjusts and, for $\sigma = \mathcal{O}(1)$, this suggests the scalings:[6]

$$H = L\sigma^{-1/5}Ra^{-1/5} = \left(\frac{\kappa^2 L^2}{\Delta b}\right)^{1/5}, \qquad \Psi = Ra^{1/5}\sigma^{-4/5}\nu = (\kappa^3 L^3 \Delta b)^{1/5}. \quad \text{(15.12a,b)}$$

The vertical scale H arises as a consequence of the analysis, and the vertical size of the domain plays no direct role. [For $\sigma \gg 1$ we might expect the nonlinear terms to be small and if the buoyancy term balances the viscous term in (15.9) the right-hand sides of (15.12) are multiplied by $\sigma^{1/5}$ and $\sigma^{-1/5}$. For seawater, $\sigma \approx 7$ using the molecular values of κ and ν. If small scale turbulence exists, then the eddy viscosity will likely be similar to the eddy diffusivity and $\sigma \approx 1$.] Numerical experiments (Fig. 15.4 and Fig. 15.5) do provide some support for this scaling, and a few simple and robust points that have relevance to the real ocean emerge, namely:

- ⋆ Most of the box fills up with the densest available fluid, with a boundary layer in temperature near the surface required in order to satisfy the top boundary condition. The boundary gets thinner with decreasing diffusivity, consistent with (15.12). This is a diffusive prototype of the oceanic thermocline.
- ⋆ The horizontal scale of the overturning circulation is large, being at or near the scale of the box.
- ⋆ The downwelling regions (the regions of convection) are of smaller horizontal scale than the upwelling regions, especially as the Rayleigh number increases.

Let us now try to explain some of the features in a simple and heuristic way.

15.2.2 Phenomenology of the overturning circulation

No water can be denser (or, more accurately, have a greater potential density) than the densest water at the surface, and if the subsurface water is a little lighter than this the

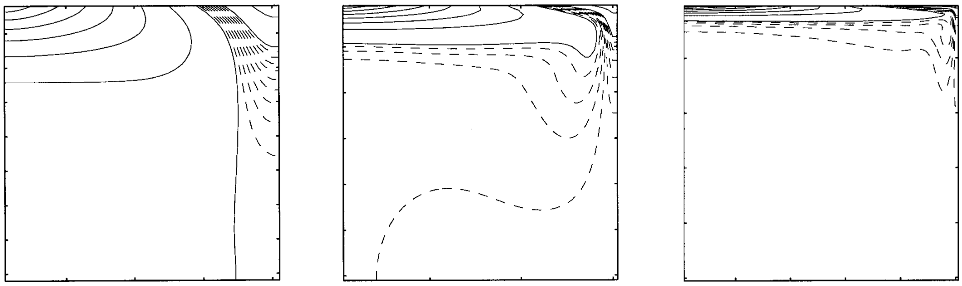


Fig. 15.5 The temperature or buoyancy field corresponding to the streamfunction fields shown in Fig. 15.4. Note an increasingly sharp gradient (a thermocline) near the top as the Rayleigh number increases, and that the bulk of the domain is filled with the densest available fluid.

surface water will be convectively unstable and sink in a plume.[8] The plume slowly entrains the warmer water that surrounds it, and then spreads horizontally when it reaches the bottom or when its density becomes similar to that of its surroundings. The presence of water denser than its surroundings creates a pressure gradient, and the ensuing flow will displace any adjacent lighter fluid, and so the domain fills with the densest available fluid. This process is a continuous one: the plumes take cold water into the interior, where the water slowly warms by diffusion, and the source of cold water at the surface is continuously replenished. If diffusion is small, the end result is that the potential density of the fluid in the interior will be slightly less than that of the densest fluid formed at the surface. (Because diffusion can act only to reduce extrema, no fluid in the interior can be colder than the coldest fluid formed at the surface.) However, the value at the surface is given by the boundary condition $b(x, y, z = 0) = f(x, y)$. Thus, the interior cannot fill all the way to the surface with this cold water and there must be a boundary layer connecting the cold, dense interior with the surface; its thickness δ is given by the height scale of (15.12a), that is $\delta \sim H = (\kappa^2 L^2/\Delta b)^{1/5}$. Such a strong boundary layer will not necessarily be manifest in the velocity field, however, because the no-normal flow boundary condition on the velocity field is satisfied by setting $\psi = 0$ as a boundary condition to the elliptic problem $\nabla^2\psi = \zeta$, where ζ is the prognostic variable in (15.5a), and this boundary condition has a global effect on the velocity field.

Why is the horizontal scale of the circulation large? The circulation transfers heat meridionally, and it is far more efficient to do this by a single overturning cell than by a multitude of small cells, so although we cannot entirely eliminate the possibility that some instability will produce such small scales of motion, it seems likely the horizontal scale of the mean circulation will be determined by the domain scale. Indeed, at low Rayleigh number we can explicitly calculate an approximate analytic solution for the flow, demonstrating this (problem 15.7). For higher Rayleigh number perturbation approaches fail and we must resort to numerical solutions; these (e.g., Fig. 15.4), do show the circulation dominated by a single overturning circulation rather than many small convective cells. We cannot rigorously prove that this will be the case for still higher Rayleigh numbers, but a general energetic argument that shows that the flow cannot in fact break up into a succession of ever smaller cells in a turbulent cascade is given in the next section.

Why is the downwelling region narrower than the upwelling? At high Rayleigh number, the extreme efficiency of convection may be responsible. Suppose that the basin is initially filled with water of an intermediate temperature, and that surface boundary conditions of a temperature decreasing linearly from low latitudes to high latitudes are imposed. The deep water will be convectively unstable, and convection at high latitudes (where the surface is coldest) will occur quickly, filling the abyss with dense water as described in the paragraphs above. After this initial adjustment, the deep, dense water at lower latitudes will slowly warmed by diffusion, but at the same time surface forcing will maintain a cold high latitude surface, thus leading to high latitude convection. A steady state or statistically steady state is eventually reached with the deep water having a slightly higher potential density than the surface water at the highest latitudes, and so with continual convection — but convection that is perforce restricted to the highest latitudes. However, it remains to put this heuristic argument on more rigorous or quantitative terms.

Finally, it is important to realise that *even for large diffusion and viscosity there is no stationary solution:* as soon as we impose a temperature gradient at the top the fluid begins to circulate, a manifestation of the dictum that a baroclinic fluid is a moving fluid, encountered in section 4.2.

15.3 THE MAINTENANCE OF SIDEWAYS CONVECTION

In many convection problems the fluid is heated from below, becomes buoyant and rises, and is cooled at the top. In contrast, in sideways convection the heating and cooling occur at the same level, and the conditions under which a circulation can be maintained are by no means clear; making them clearer is our purpose here. The energetic derivations of this section are but an extension of section 2.4.3, but now with a starring role for friction, diffusion and the boundary conditions, and the reader may wish to review that section first. To begin, we write down the equations of motion:

$$\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{f} + 2\boldsymbol{\omega}) \times \boldsymbol{v} = -\nabla B + b\mathbf{k} + \nu\nabla^2\boldsymbol{v}, \tag{15.13a}$$

$$\frac{\mathrm{D}b}{\mathrm{D}t} = \frac{\partial b}{\partial t} + \nabla \cdot (b\boldsymbol{v}) = \dot{Q} = J + \kappa\nabla^2 b, \tag{15.13b}$$

$$\nabla \cdot \boldsymbol{v} = 0, \tag{15.13c}$$

where B is the Bernoulli function and $\dot{Q}$ ($= \dot{b}$) is the total rate of heating (including diffusion, and absorbing constant factors such as heat capacity into its definition) with J its non-diffusive component. The fluid occupies a finite volume, and in a steady state $\left\langle \dot{Q} \right\rangle = 0$, where the angle brackets denote a volume and time integration.

15.3.1 The energy budget

To obtain an energy budget we follow the procedure of section 2.4.3. First take the dot product of (15.13a) with $\boldsymbol{v}$ to give

$$\frac{1}{2}\frac{\partial \boldsymbol{v}^2}{\partial t} = -\nabla \cdot (\boldsymbol{v}B) + wb + \nu\boldsymbol{v} \cdot \nabla^2\boldsymbol{v}. \tag{15.14}$$

Integrating over a domain bounded by stress-free rigid walls gives the kinetic energy equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\langle \frac{1}{2}\boldsymbol{v}^2 \right\rangle = \langle wb \rangle - \varepsilon, \tag{15.15}$$

where angle brackets denote (for the moment) just a volume integration and $\varepsilon = -\nu\left\langle \boldsymbol{v}\cdot\nabla^2\boldsymbol{v}\right\rangle = \nu\left\langle \boldsymbol{\omega}^2\right\rangle$ is the total dissipation of kinetic energy, a positive definite quantity. Thus, in a statistically steady state in which the left-hand side vanishes after time-averaging, the dissipation of kinetic energy is maintained by the buoyancy flux; that is, by a release of potential energy with light fluid ascending and dense fluid descending.

We obtain a potential energy budget by using (15.13b) to write

$$\frac{\mathrm{D}bz}{\mathrm{D}t} = z\frac{\mathrm{D}b}{\mathrm{D}t} + b\frac{\mathrm{D}z}{\mathrm{D}t} = z\dot{Q} + bw, \tag{15.16}$$

and integrating this over the domain gives the potential energy equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle bz \rangle = \left\langle z\dot{Q}\right\rangle + \langle bw \rangle. \tag{15.17}$$

Subtracting (15.17) from (15.15) gives the energy equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\langle \frac{1}{2}\boldsymbol{v}^2 - bz \right\rangle = -\left\langle z\dot{Q}\right\rangle - \varepsilon. \tag{15.18}$$

15.3.2 Conditions for maintaining a thermally-driven circulation

In a statistically steady state the left-hand side of (15.18) vanishes and the kinetic energy dissipation is balanced by the buoyancy source terms; that is

$$\boxed{\left\langle z\dot{Q}\right\rangle = -\varepsilon < 0}. \tag{15.19}$$

The right-hand side is negative definite, and to balance this the heating must be negatively correlated with height. (Recall that $\langle \dot{Q} \rangle = 0$, and the origin of the z-coordinate is then immaterial.) Thus, *in order to maintain a circulation in which kinetic energy is dissipated, the heating (including diffusive heating) must occur, on average, at lower levels than the cooling.* This result is related to, but not quite the same as, Sandström's postulate, discussed below.

In the ocean the non-diffusive heating occurs predominantly at the surface, except for the negligible effects of hydrothermal vents. Thus, $\langle Jz \rangle \approx 0$ and a kinetic-energy-dissipating circulation can *only* be maintained, in the absence of mechanical forcing, if the diffusion is non-zero — in that case heat may be diffused from the surface to depth so effectively providing a deep heat source. In the atmosphere, the heating is mostly at the surface and cooling is mostly in in the mid-troposphere so that $\left\langle z\dot{Q}\right\rangle > 0$; thus, (15.19) is readily satisfied and the circulation is not restricted.

* *Maintaining a steady baroclinic circulation*

By a rather different method — and one closer to a suggestion of Sandström dating from 1908 — we can obtain a result that is different from but related to (15.19).[9] Using (4.39) and (15.13a), the circulation in a Boussinesq system obeys

$$C = \oint \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{r}, \qquad \frac{\mathrm{D}C}{\mathrm{D}t} = \oint b\mathbf{k} \cdot \mathrm{d}\boldsymbol{r} + \oint \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r}, \tag{15.20a,b}$$

where C is the circulation, $\mathrm{d}\boldsymbol{r}$ is a path element, and $\boldsymbol{F}$ represents the frictional terms. (The Coriolis parameter plays no role in this argument, and is set to zero; alternatively, the reader may regard the argument as being set in an inertial reference frame.) Now we may write the rate of change of circulation in the form (for example problem 2.3):

$$\frac{\mathrm{D}C}{\mathrm{D}t} = \oint \left(\frac{\partial \boldsymbol{v}}{\partial t} + \boldsymbol{v} \cdot \nabla \boldsymbol{v} \right) \cdot \mathrm{d}\boldsymbol{r} = \oint \left(\frac{\partial \boldsymbol{v}}{\partial t} + \boldsymbol{\omega} \times \boldsymbol{v} \right) \cdot \mathrm{d}\boldsymbol{r}. \tag{15.21}$$

Let us assume the flow is steady, so that $\partial \boldsymbol{v}/\partial t$ vanishes. Let us further choose the path of integration to be a streamline, which since the flow is steady is also a parcel trajectory. The second term on the right-most expression of (15.21) then also vanishes and (15.20b) becomes

$$\oint b \, \mathrm{d}z = -\oint \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = -\oint \frac{\boldsymbol{F}}{|\boldsymbol{v}|} \cdot \boldsymbol{v} \, \mathrm{d}r, \tag{15.22}$$

where the last equality follows because the path is everywhere parallel to the velocity. Let us now assume that the friction retards the flow, and that $\oint \boldsymbol{F} \cdot \boldsymbol{v}/|\boldsymbol{v}| \, \mathrm{d}r < 0$. (One form of friction that has this property is linear drag, $\boldsymbol{F} = -C\boldsymbol{v}$ where C is a constant. The property is similar to, but not the same as, the property that the friction dissipates kinetic energy over the circuit.) Making this assumption, if we integrate the term on the left-hand side by parts we obtain:

$$\boxed{\oint z \, \mathrm{d}b < 0}\,. \tag{15.23}$$

Now, because the integration circuit in (15.23) is a fluid trajectory, the change in buoyancy $\mathrm{d}b$ is proportional to the heating of a fluid element as it travels the circuit (in the notation of (15.13b), $\mathrm{d}b = \dot{Q} \, \mathrm{d}t$, where the heating includes diffusive effects). Thus, the inequality implies that the net heating must be negatively correlated with height: that is, *the heating must occur, on average, at a lower level than the cooling in order that a steady circulation may be maintained against the retarding effects of friction.* (We might call this statement 'Sandström's postulate'.)

A similar result can be obtained for a compressible fluid. From equations (4.45)–(4.47) on page 175 we write the baroclinic circulation theorem as

$$\frac{\mathrm{D}C}{\mathrm{D}t} = \oint p \, \mathrm{d}\alpha + \oint \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \oint T \, \mathrm{d}\eta + \oint \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r}, \tag{15.24}$$

where η is the specific entropy. Then, by precisely the same arguments as led to (15.23), we are led to the requirements that

$$\oint T \, \mathrm{d}\eta > 0 \qquad \text{or equivalently} \qquad \oint p \, \mathrm{d}\alpha > 0. \tag{15.25a,b}$$

Eq. (15.25a) means that parcels must gain entropy at high temperatures and lose entropy at low temperatures; similarly, from (15.23b), a parcel must expand ($d\alpha > 0$) at high pressures and contract at low pressures.

For an ideal gas we can put these statements into a form analogous to (15.23) by noting that $d\eta = c_p(d\theta/\theta)$, where θ is potential temperature, and using the definition of potential temperature, (1.105). With these we have

$$\oint T\,d\eta = \oint c_p \frac{T}{\theta}\,d\theta = \oint c_p \left(\frac{p}{p_R}\right)^{\kappa} d\theta, \tag{15.26}$$

and (15.25a) becomes

$$\boxed{\oint c_p \left(\frac{p}{p_R}\right)^{\kappa} d\theta > 0}\,. \tag{15.27}$$

Because the path of integration is a fluid trajectory, $d\theta$ is proportional to the heating of a fluid element. Thus [and analogous to the Boussinesq result (15.23)], (15.27) implies that *the heating (the potential temperature increase) must occur at a higher pressure than the cooling in order that a steady circulation may be maintained against the retarding effects of friction.*

These results may be understood by noting that the heating must occur at a higher pressure than the cooling in order that work may be done, the work being necessary to convert potential energy into kinetic energy to maintain a circulation against friction. More informally, if the heating is below the cooling, then the heated fluid will expand and become buoyant and rise, and a steady circulation between heat source and heat sink can readily be imagined. On the other hand, if the heating is above the cooling there is no obvious pathway between source and sink.

Intuitive as these results may be, it is important to realize that conditions required to prove (15.23) and (15.27) are much more restrictive than those needed to prove (15.19). To prove the former, we must *assume* that the flow is absolutely steady, *and* that streamlines form a closed path, *and* that the friction has retarding properties. The second of these conditions is not generally satisfied in three-dimensions, even when the flow is steady. Furthermore, one cannot prove that Newtonian viscosity ($\nu\nabla^2\boldsymbol{v}$) will always act to retard the flow. On the other hand, (15.19) provides a condition for the maintenance of a statistically steady circulation, assuming only that the friction acts to dissipate kinetic energy.

15.3.3 Surface fluxes, diffusion, and non-turbulent flow

Suppose that the only heating to the fluid is via diffusion through the upper surface; that is $J = 0$ in (15.13b). Is a circulation possible? If the diffusivity is finite then heat can diffuse into the fluid, and thereby potentially provide an difference in altitude between the heating and the cooling. However, as $\kappa \to 0$ this mechanism ceases to operate and we therefore expect that the left-hand sides of (15.19) and (15.23) will go to zero, and the circulation will cease. In what follows we put this argument on a more rigorous footing: we will show that as $\kappa \to 0$ the kinetic energy dissipation also to zero, and therefore the flow is 'non-turbulent', the meaning of which will be made clear below.

Assuming a statistically steady state, integrating (15.13b) horizontally gives

$$\frac{\partial \overline{bw}}{\partial z} = \kappa \frac{\partial^2 \overline{b}}{\partial z^2}, \tag{15.28}$$

where an overbar indicates a horizontal and time average. Integrating this equation up from the bottom (where there is no flux) to a level z gives

$$\overline{wb} - \kappa \overline{b}_z = 0, \tag{15.29}$$

at every level in the fluid. The two terms on the left-hand side together comprise the total buoyancy flux through the level z, and this vanishes because there is no buoyancy input except at the surface. If we integrate this vertically we have

$$\langle wb \rangle = H^{-1}\kappa \left[\overline{b}(0) - \overline{b}(-H) \right], \tag{15.30}$$

where the angle brackets denote an average over the entire volume. In the limit $\kappa \to 0$, the integrated advective buoyancy flux will vanish, because the term $\overline{b}(0) - \overline{b}(-H)$ remains finite. (This follows because b is conserved on parcels, except for the effects of diffusion which can only act to reduce the value of extrema in the fluid. Thus, $\overline{b}(0) - \overline{b}(-H)$ can only be as large as the temperature difference at the surface, which is set by the boundary conditions.)

Now consider the kinetic energy budget. Using (15.15) and (15.30) we have in a statistically steady state

$$\varepsilon = H^{-1}\kappa \left[\overline{b}(0) - \overline{b}(-H) \right]. \tag{15.31}$$

Because, as noted above, the buoyancy difference on the right-hand side is bounded, the kinetic energy dissipation must go to zero if the thermal diffusivity goes to zero; that is, $\varepsilon \to 0$ as $\kappa \to 0$ and in particular $\varepsilon < \kappa \Delta b / H$ where Δb is the maximum buoyancy difference at the surface. We may also consider the limit $(\kappa, \nu) \to 0$ with fixed Prandtl number $\sigma \equiv \nu/\kappa$, and in this limit also the energy dissipation vanishes with κ.

Finally, let us see how the surface buoyancy is related to the buoyancy flux, for any value of κ. Multiplying (15.13b) by b and integrating over the domain gives the buoyancy variance equation,

$$\frac{1}{2}\frac{\mathrm{d}\langle b^2 \rangle}{\mathrm{d}t} = \kappa \left[\left. \overline{b \frac{\partial b}{\partial z}} \right|_{z=0} - \left\langle |\nabla b|^2 \right\rangle \right]. \tag{15.32}$$

We have assumed that the normal derivative of b vanishes on all surfaces except the top one ($z = 0$) and an overbar denotes a horizontal integral. In a statistically steady state,

$$\left. \overline{b \frac{\partial b}{\partial z}} \right|_{z=0} = \left\langle |\nabla b|^2 \right\rangle, \tag{15.33}$$

where the overbar and angle brackets now also imply a time average. The right-hand side is positive definite, and thus there must be a positive correlation between b and $\partial b / \partial z$, meaning there is a heat flux into the fluid where it is hot, and a heat flux out of the fluid where it is cold. This result holds no matter whether the upper boundary condition is a condition on b or on $\partial b / \partial z$.

Interpretation

The result encapsulated by (15.31) means that, for a fluid forced only at the surface by buoyancy forcing, as the diffusivity goes to zero so does the energy dissipation. For a fluid of finite viscosity the vorticity in the fluid must then go to zero, because $\varepsilon = \nu \langle \boldsymbol{\omega}^2 \rangle$; this in turn means that the flow cannot be baroclinic, because baroclinicity generates vorticity, even in the presence of viscosity (section 4.2). An even more interesting result follows for a fluid with small viscosity. In turbulent flow, the energy dissipation at high Reynolds number is not a function of the viscosity; if the viscosity is reduced, the cascade of energy to smaller scales merely continues to still smaller scale, generating vorticity at these smaller scales, and the energy dissipation is unaltered, remaining finite even in the limit $\nu \to 0$. In contrast, for a fluid heated and cooled only at the upper surface, the energy dissipation *tends to zero* as $\kappa \to 0$, whether or not one is in the high Reynolds number limit. This means that vorticity cannot be generated at the viscous scales by the action of a turbulent cascade, for that would lead to energy dissipation. Effectively, the result prohibits an ocean that is forced only at the surface by a buoyancy flux from having an 'eddy viscosity' that would enable the fluid to efficiently dissipate energy, and if there is no small scale motion producing an eddy viscosity there can be no eddy diffusivity either. Thus, such an ocean is *non-turbulent.* This is a rather different picture from that of the real ocean, where there is some dissipation of energy in the interior because of breaking gravity waves, and dissipation at the boundary in Ekman layers, and the eddy diffusivity is needed for there to be a non-negligible buoyancy-driven meridional overturning circulation.

Of course, thermal forcing in the ocean is in part an imposed flux, coming from radiation among other things, and this penetrates below the surface. However, this makes little real physical difference to the result, provided that this forcing remains confined to the upper ocean. If so, then for any level below this forcing we still have the result (15.29), and the final result (15.31) holds, assuming that the range of temperatures produced by the forcing is still finite.

15.3.4 The importance of mechanical forcing

The real ocean *does* have a deep circulation, so something is missing. Suppose we add a mechanical forcing, $\boldsymbol{F}$, to the right-hand side of (15.13a); this might represent wind forcing at the surface, or tides. The kinetic energy budget becomes

$$\varepsilon = \langle wb \rangle + \langle \boldsymbol{F} \cdot \boldsymbol{v} \rangle = H^{-1}\kappa[\overline{b}(0) - \overline{b}(-H)] + \langle \boldsymbol{F} \cdot \boldsymbol{v} \rangle . \tag{15.34}$$

In this case even for $\kappa = 0$ there is a source of energy and therefore turbulence (i.e., a dissipative circulation) can be maintained. Now, the results of (15.19) and (15.31) do not prohibit there being a thermal circulation, with fluid sinking at high latitudes and rising at low. However, in the absence of any mechanical forcing this circulation must be laminar as $\kappa \to 0$, even at high Rayleigh number, and the flow is not allowed to break in such a way that energy can be dissipated — a very severe constraint that most flows cannot satisfy. The solution most likely adopted by the fluid is for the flow to become confined to a very thin layer at the surface, with no abyssal motion at all, which is completely unrealistic vis-a-vis the observed ocean.

Now, turbulent motion at small scales provides a mechanism of mixing and so can effectively generate an 'eddy diffusivity' of buoyancy. *Given* such an eddy diffusivity, wind forcing is no longer necessary for there to be an overturning circulation. Therefore, it is useful to think of mechanical forcing as having two distinct effects:

(i) The wind provides a stress on the surface that may directly drive the large-scale circulation, including the overturning circulation. (An explicit example of this is discussed in section 16.5.)

(ii) Both tides and the wind provide a mechanical source of energy to the system that allows the flow to become turbulent and so provides a source for an eddy diffusivity and eddy viscosity.

In either case, we may conclude that presence of mechanical forcing is necessary for there to be an overturning circulation in the world's oceans of the kind observed. In the remainder of this chapter we ignore the effects of item (i), and suppose that the most important effect of the wind is that it provides an eddy diffusivity to the ocean that is much larger than its molecular value; this then allows large volumes of the ocean to become mixed, so allowing a buoyancy-driven overturning circulation (sometimes called a thermohaline circulation) to exist. We first consider extremely simple models of this circulation, so-called box models.

15.4 SIMPLE BOX MODELS

Even though they are far simpler than the real ocean, the fluid dynamical models of the previous section are still quite daunting. The analysis that can be performed is either very specific and of little generality, for example the construction of solutions at low Rayleigh number, or it is a very general form, being of the form of scaling or energetic arguments at high Rayleigh number. Models based on the fluid dynamical equations do not easily allow for the construction of explicit solutions in the parameter regime — high Rayleigh and Reynolds numbers — of interest. It is therefore useful to consider an extreme simplification of the overturning circulation, *box models.* These are very simple caricatures of the circulation, constructed by dividing the ocean into a very small number of boxes with simple rules determining the transport of fluid properties between them.[10] The purist may consider this section a diversion away from a consideration of the fluid dynamical properties of the ocean, but such box models have been quite fecund and an evident source of qualitative understanding, and thus find a place in our discussion.

15.4.1 A Two-Box Model

Consider two boxes as illustrated in Fig. 15.6. Each box is well-mixed and has a uniform temperature and salinity, T_1, T_2 and S_1, S_2. The boxes are connected with a capillary tube at the bottom along which the flow is viscous, obeying Stokes' Law. That is, the flow along the tube is proportional to the pressure gradient which, because the flow is hydrostatic, is proportional to the density difference between the two boxes. An overflow at the top keeps the upper surfaces of the two boxes at the same level. Thus, the circulation is given by

$$\Psi = A(\rho_1 - \rho_2), \tag{15.35}$$

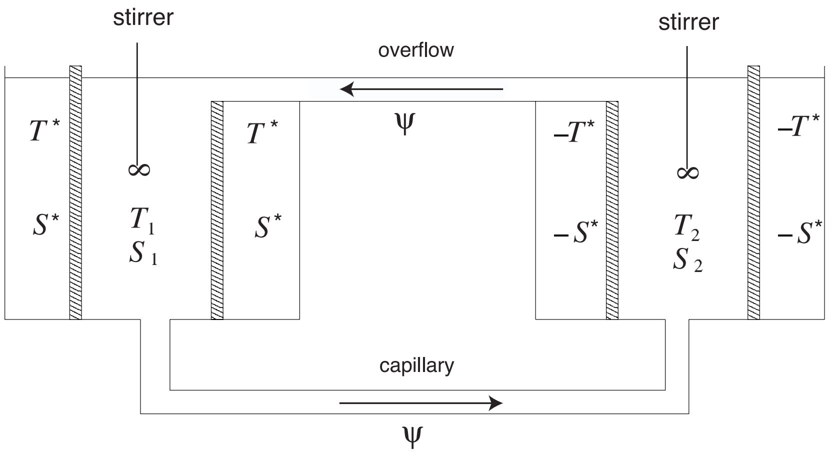


Fig. 15.6 A two-box model of relevance to the overturning circulation of the ocean. The shaded walls are porous, and each box is well-mixed by its stirrer. Temperature and salinity evolve by way of fluid exchange between the boxes via the capillary tube and the overflow, and by way of relaxation with the two infinite reservoirs at $(+T^*, +S^*)$ and $(-T^*, -S^*)$.

where Ψ is proportional to the flow in the pipe, ρ_1 and ρ_2 are the densities of the fluids in the two boxes and A is a constant. The boxes are enclosed by porous walls beyond which are reservoirs of constant temperature and salinity, and we are at liberty to choose the origin of the temperature scale such that the two reservoirs are at $+T^*$ and $-T^*$, and similarly for salinity. Thus, heat and salt are transferred into and out of the boxes as represented by simple linear laws and we have

$$\begin{aligned}\frac{\mathrm{d}T_1}{\mathrm{d}t} &= c(T^* - T_1) - 2|\Psi|(T_1 - T_2), & \frac{\mathrm{d}T_2}{\mathrm{d}t} &= c(-T^* - T_2) - 2|\Psi|(T_2 - T_1),\\ \frac{\mathrm{d}S_1}{\mathrm{d}t} &= d(S^* - S_1) - 2|\Psi|(S_1 - S_2), & \frac{\mathrm{d}S_2}{\mathrm{d}t} &= d(-S^* - S_2) - 2|\Psi|(S_2 - S_1).\end{aligned} \tag{15.36}$$

Note that the advective transfer is independent of the sign the circulation, because it occurs through both the capillary tube and the overflow.

From these equations it is apparent that the sum of the temperatures, $T_1 + T_2$ decays to zero and is uncoupled from the difference, and similarly for salinity. Defining $\widehat{T} = (T_1 - T_2)/(2T^*)$ and $\widehat{S} = (S_1 - S_2)/(2S^*)$ then gives

$$\begin{aligned}\frac{\mathrm{d}\widehat{T}}{\mathrm{d}t} &= c(1 - \widehat{T}) - 4|\Psi|\widehat{T}\\ \frac{\mathrm{d}\widehat{S}}{\mathrm{d}t} &= d(1 - \widehat{S}) - 4|\Psi|\widehat{S}\end{aligned} \tag{15.37}$$

Using a linear equation of state of the form $\rho = \rho_0(1 - \beta_T T + \beta_S S)$ (where the variables are dimensional) the circulation (15.35) becomes

$$\Psi = 2\rho_0 T^* \beta_T A \left(-\widehat{T} + \frac{\beta_S S^*}{\beta_T T^*}\widehat{S}\right). \tag{15.38}$$

Finally, nondimensionalizing time using $\tau = ct$, the equations of motion become

$$\frac{d\hat{T}}{d\tau} = (1 - \hat{T}) - |\Phi|\hat{T}, \tag{15.39a}$$

$$\frac{d\hat{S}}{d\tau} = \delta(1 - \hat{S}) - |\Phi|\hat{S}, \tag{15.39b}$$

$$\Phi = -\gamma(\hat{T} - \mu\hat{S}), \tag{15.39c}$$

where $\Phi = 4\Phi/\gamma$ and the three parameters that determine the behaviour of the system are

$$\gamma = \frac{8\rho_0 T^* \beta_T A}{c}, \qquad \delta = \frac{d}{c}, \qquad \mu = \frac{\beta_S S^*}{\beta_T T^*}. \tag{15.40}$$

The parameter γ measures the overall strength of the forcing in determining the strength of the circulation, and is the ratio of a relaxation to an advective timescale. The parameter δ is the ratio of the reciprocal time constants of temperature and salinity relaxation, and μ is a measure of the ratio of the effect of the salinity and temperature forcings on the density. Salinity transfer will normally be much slower than heat transfer so that $\delta \ll 1$, whereas if salinity and temperature are both to play a role in the dynamics we need $\mu = \mathcal{O}(1)$. We also might expect both advection and relaxation to be important if $\gamma = \mathcal{O}(1)$, and this will depend on the properties of the capillary tube.

Interpretation

The above model describes a potentially real system, one that might be constructed in the laboratory, and one with relevance to aspects of the ocean circulation. One box then represents the entire high-latitude ocean and the other the entire low-latitude ocean, and the capillary tube and the overflow carry the overturning circulation between them. The reservoirs at $\pm T^*$ and $\pm S^*$ represent the atmosphere. Typically, we would choose the low latitudes to be both heated and salted (the latter because of the low rainfall and high evaporation in the subtropics) and the high latitudes to be cooled and freshened by rainfall. Thus, T^* and S^* have the same sign, and they force the circulation in opposite directions.

It is a common fluid-dynamical experience that the behaviour of a highly-truncated system has little resemblance to that of the corresponding continuous system, and so we expect the model to be only be a cartoon of the ocean circulation. For example, we have restricted the circulation to be of basin scale, and the parameterization of the intensity of the overturning circulation by (15.39c) must be regarded with caution, because it represents a frictionally controlled flow rather than a nearly inviscid geostrophic flow. Nevertheless, observations and numerical simulations do indicate that the overturning circulation does have a relatively simple vertical and horizontal structure: the circulation in the North Atlantic is similar to that of a single cell, for example, indicating that an appropriate low-order model may be useful.

One might also question the oceanic appropriateness of the linear relaxation terms. For temperature, the bulk aerodynamic formulae often used to parameterize air-sea transport do have a similar form, but the freshening of sea-water by rainfall is more akin to an imposed (negative) flux of salinity, and evaporation is a function of temperature. An alternative might be to impose an effective salt flux so that

$$\frac{\mathrm{d}}{\mathrm{d}t}(S_1 - S_2) = 2E - 2|\Psi|(S_1 - S_2) \tag{15.41}$$

where E is an imposed, constant, effective rate of salt exchange with the atmosphere. After nondimensionalization, using E/c to non-dimensionalize salt, (15.39b) is replaced by

$$\frac{\mathrm{d}S}{\mathrm{d}\tau} = 1 - |\Phi|S. \tag{15.42}$$

Another aspect of the model that is oceanographically questionable is that the model assumes that the water masses can be mixed below the surface. Thus, when water enters one box from the other it immediately mixes with its surroundings. Without the stirrer this would not occur and the equations of box model would not represent a real system. In the real ocean, most of the mixing of water masses seems to happen near the surface (in the mixed layer) and near lateral boundaries or regions of steep topography. Elsewhere in the ocean mixing is quite small, and likely far from sufficient to mix a large volume of water in the advective or relaxation times of the box model. We will defer consideration of this and continue with an analysis of the model.

Solutions

Perhaps the most interesting aspect of the set (15.39) is that it exhibits *multiple equilibria;* that is, there are multiple steady solutions with the same parameters. Equilibria occur when the time-derivatives vanish, and the circulation then satisfies

$$\Phi = \phi(\Phi) \equiv \gamma\left(\frac{-1}{1+|\Phi|} + \frac{\mu}{1+|\Phi|/\delta}\right). \tag{15.43}$$

A graphical solution of this is obtained as the intercept of the right-hand side with the left-hand side, the latter being a straight line through the origin at an angle of 45°, and this is plotted in Fig. 15.7.

Evidently, for a range of parameters three solutions are possible, whereas for others only one solution exists. Although a fairly complete analysis of the nature of the steady solutions is possible, it is instructive to consider the special case with $\gamma \gg 1$ and $\delta \ll 1$. This corresponds to the situation in which the advective timescale is shorter than the diffusive one and temperature relaxation is much faster than salt relaxation. Two of the solutions are then close to the origin, with $\Phi \ll 1$ and satisfying

$$\Phi = \phi(\Phi) \equiv \gamma\left(-1 + \frac{\mu\delta}{\delta+|\Phi|}\right). \tag{15.44}$$

giving for small $|\Phi|$

$$\Phi \approx \pm[\delta(\mu - 1)]. \tag{15.45}$$

Only the positive solution will be stable, and this solution is driven by the density gradient in salinity; the high salinity of box 1 outweighs (so to speak) the compensating fact that it is also warmer, leading to a flow along the capillary tube from box 1 to box 2. Solving for temperature and salinity we find that $T \approx 1$ (i.e., it is close to its relaxation value and hardly altered by advection), and $S \approx 2 - \mu$ or μ, for $\mu < 2$ and

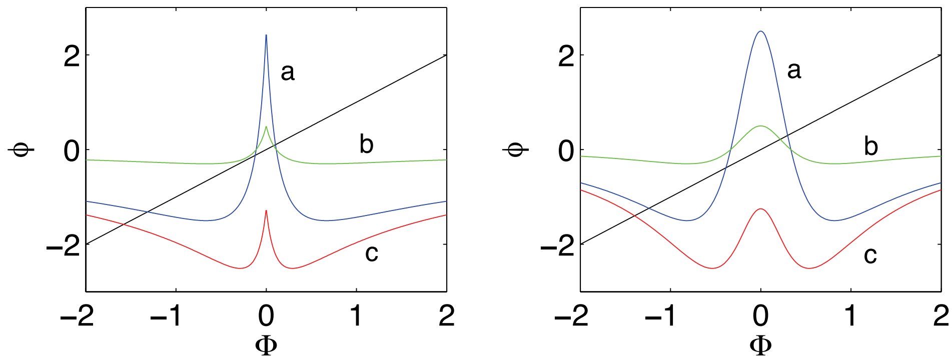


Fig. 15.7 Left panel: Graphical solution of the two-box model. The straight line has unit slope and passes through the origin, and the curved lines plot the function $\phi(\Phi)$ as given by the right-hand side of (15.43). The intercepts of the two are solutions to the equation. The parameters for the three curves are: a, $\gamma = 5$, $\delta = 1/6$, $\mu = 0.1.5$; b, $\gamma = 1$, $\delta = 1/6$, $\mu = 1.5$; c, $\gamma = 5$, $\delta = 1/6$, $\mu = 0.75$. Right panel: same except with Φ^2 in place of $|\Phi|$ on rhs of (15.43).

$\mu > 2$ respectively. The saline contribution to density can therefore be larger than that of temperature, and indeed is so for this solution.

The other solution has a circulation far from the origin, and the balance in (15.43) is between the left-hand side and the first term on the right. In the limiting case we find

$$\Phi \approx -\sqrt{\gamma}. \tag{15.46}$$

This solution has a density gradient dominated by the temperature effect: the temperature difference is $T \approx 1/\sqrt{\gamma}$ whereas the salinity difference is $S \approx \delta/\sqrt{\gamma}$, and thus its effect on density is much smaller.

15.4.2 * More boxes

More boxes can be added in a variety of ways and, now forgoing an easy relevance to a laboratory apparatus, one such is illustrated in Fig. 15.8. The three boxes represent the mid- and high-latitude Northern Hemisphere, the mid- and high-latitude Southern Hemisphere, and the equatorial regions. Each of the three boxes can exchange fluid with its neighbour, and each is also in contact with a reservoir and subject to a relaxation to a fixed value of temperature and salinity, (T_s^*, S_s^*), (T_e^*, S_e^*), (T_n^*, S_n^*). (A variation on this theme allows direct communication between the two poleward boxes.) Then, with obvious notation, we infer the equations of motion:

$$\frac{\mathrm{d}T_s}{\mathrm{d}t} = c(T_s^* - T_s) - 2|\Psi_s|(T_s - T_e), \qquad \frac{\mathrm{d}T_n}{\mathrm{d}t} = c(T_n^* - T_n) - 2|\Psi_n|(T_n - T_e),$$
$$\frac{\mathrm{d}T_e}{\mathrm{d}t} = c(T_e^* - T_e) - 2|\Psi_s|(T_e - T_s) - 2|\Psi_n|(T_e - T_n), \tag{15.47a}$$

$$\frac{\mathrm{d}S_s}{\mathrm{d}t} = d(S_s^* - S_s) - 2|\Psi_s|(S_s - S_e), \qquad \frac{\mathrm{d}S_n}{\mathrm{d}t} = d(S_n^* - S_s) - 2|\Psi_n|(S_n - S_e),$$

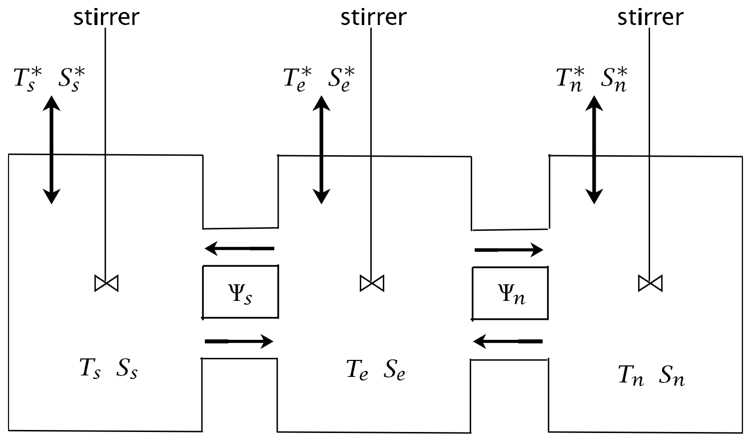


Fig. 15.8 A three-box model. Each box has a constant value of temperature and salinity within it, each exchanges fluid with its neighbour, and in each the temperature and salinity are relaxed back to fixed atmospheric values.

$$\frac{\mathrm{d}S_e}{\mathrm{d}t} = d(S_e^* - S_e) - 2|\Psi_s|(S_e - S_s) - 2|\Psi_n|(S_e - S_n), \tag{15.47b}$$

with flow rates given by the density differences.

$$\Psi_s = A\rho_0[-\beta_T(T_s - T_e) + \beta_S(S_s - S_e)], \quad \Psi_n = A\rho_0[-\beta_T(T_n - T_e) + \beta_S(S_n - S_e)]. \tag{15.48}$$

These equations may be nondimensionalized and reduced to four prognostic equations for the quantities $T_e - T_n$, $T_e - T_s$, $S_e - S_n$, $S_e - S_s$. Not surprisingly, multiple equilibria can again be found. One interesting aspect is that stable asymmetric solutions arise with symmetric forcing ($T_s^* = T_n^*$, $S_s^* = S_n^*$). These effectively have a pole-to-pole circulation, illustrated in the upper row of Fig. 15.9. Such a circulation can be thought of as the superposition of a thermal circulation in one hemisphere and a salinity-driven circulation in the other.

The box models are useful because they are suggestive of behaviour that might occur in real fluid systems, and because they provide a means of interpreting behaviour that does occur in more complete numerical models, and perhaps in the real world. However, they are by no means good approximations of the real equations of motion and without other supporting evidence the solutions found in box models should not be regarded as representing real solutions of the fluid equations for the world's oceans.[12]

15.5 A LABORATORY MODEL OF THE ABYSSAL CIRCULATION

We now return to a more fluid dynamical description of the deep ocean circulation, and consider two simple, closely related, models that are relevant to aspects of the deep circulation. The first, which we consider in this section, is a laboratory model that, although originally envisioned as being a prototype for the deep circulation, is also illustrative of the principles of the wind-driven circulation. The second model,

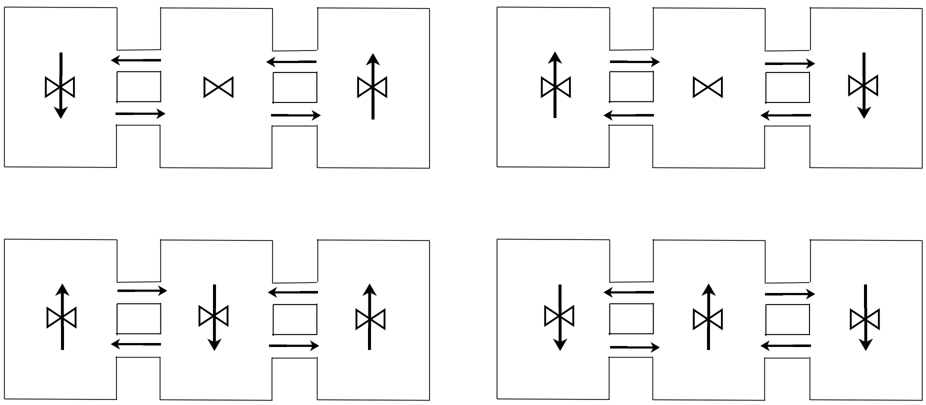


Fig. 15.9 Schematic of four solutions to the three box model with the symmetric forcing $S_s^* = S_n^*$ and $T_n^* = T_s^*$. The two solutions on the top row have an asymmetric, 'pole-to-pole', circulation whereas the solutions on the bottom row are symmetric.[11]

considered in the sections following, is explicitly a model of the deep circulation. Both models are severe idealizations that describe only limited aspects of the circulation.

15.5.1 Set-up of the laboratory model

Let us consider flow in a rotating tank, as illustrated in Fig. 15.10. The fluid is confined by vertical walls to occupy a sector, and the entire tank rotates anti-clockwise when viewed from above, like the Northern Hemisphere. When the fluid is stationary in the rotating frame, the fluid slopes up toward the outer edge of the tank and the balance of forces in the rotating frame is between a centrifugal force pointing outwards and the pressure gradient due to the sloping fluid pointing inwards. In the inertial frame of the laboratory itself, the pressure gradient pointing inwards provides a centripetal force that causes the fluid to accelerate toward the centre of the tank, resulting in a circular motion. (Recall that steady circular motion is always accompanied by an acceleration toward the centre of the circle.) This set-up, and the accompanying theory, is known as the *Stommel–Arons–Faller* model.[13] The motivation of this construct is clear, in that the sector represents an ocean basin. However, rather than driving the fluid with wind or by differential heating, we drive it with localized mass sources and sinks, for example from small pipes inserted into the tank.

15.5.2 Dynamics of flow in the tank

Let us assume that motion of the fluid in the tank is sufficiently weak that its Rossby number is small, and that it obeys the shallow water planetary geostrophic equations, namely

$$\boldsymbol{f}_0 \times \boldsymbol{u} = -g\nabla h + \Omega^2 r\,\hat{\mathbf{r}} + \boldsymbol{F}, \qquad (15.49a)$$

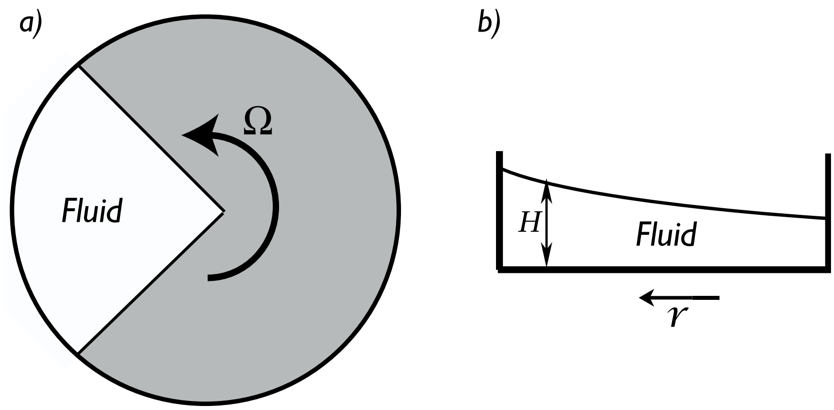


Fig. 15.10 The experimental set-up in the Stommel–Arons–Faller rotating tank experiment. (a) A plan view of the apparatus. The fluid is contained in the sector at left. (b) Side view. The free surface of the fluid slopes up with increasing radius, giving a balance (in the rotating frame) between the centrifugal force pointing outwards and the pressure force pointing inwards. Small pipes may be introduced into the fluid to provide mass sources and sinks.

$$\frac{\partial h}{\partial t} + \nabla \cdot (\boldsymbol{u}h) = S \tag{15.49b}$$

where $\hat{\mathbf{r}}$ is a unit vector in the direction of increasing r, $\boldsymbol{F}$ represents frictional terms and S represents mass sources. These two equations yield the potential vorticity equation,

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{f_0}{h}\right) = \frac{\mathrm{curl}_z\boldsymbol{F}}{h} - \frac{f_0 S}{h^2}. \tag{15.50}$$

Let us write the height field as

$$h = H(r,t) + \eta(r,\theta,t) \tag{15.51}$$

where $H(r,t)$ is the height field corresponding to the rest state of the fluid (in the rotating frame) and η the perturbation. Thus, from (15.49a)

$$0 = -g\nabla H + \Omega^2 r\hat{\mathbf{r}}, \tag{15.52}$$

which gives

$$H = \frac{\Omega^2 r^2}{2g} + \hat{H}(t), \tag{15.53}$$

where $\hat{H}$ is a measure of the overall mass of the fluid. Its rate of change is determined by the mass source

$$\frac{\mathrm{d}\hat{H}}{\mathrm{d}t} = \langle S \rangle, \tag{15.54}$$

the angle brackets indicating a domain average. The equations of motion (15.49a), and (15.49b) become

$$\boldsymbol{f}_0 \times \boldsymbol{u} = -g\nabla\eta + \boldsymbol{F}, \tag{15.55a}$$

$$\frac{\partial}{\partial t}(\eta + H) + \nabla \cdot [\boldsymbol{u}(\eta + H)] = 0. \tag{15.55b}$$

Eq. (15.55a) tells us that, away from frictional regions, the velocity is in geostrophic balance with the pressure field due to the perturbation height η.

Let us now suppose $|\eta| \ll H$, which holds if the mass source is small and gentle enough. Then (15.55b) may be written

$$\frac{\partial H}{\partial t} + \nabla \cdot (\boldsymbol{u}H) = 0. \tag{15.56}$$

In this approximation, the potential vorticity equation (15.50) becomes, away from friction and mass sources,

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{f_0}{H}\right) = 0 \qquad \text{or} \qquad \frac{\mathrm{D}H}{\mathrm{D}t} = 0. \tag{15.57a,b}$$

where the second equation follows because f_0 is a constant. (This equation also follows directly from (15.56), because the velocity is geostrophic and divergence-free where friction is absent; however, it is better thought of as a potential vorticity equation, not a mass conservation equation.) Eq. (15.57b) means that fluid columns change position in order to keep the same value of H. Further, because H only varies with r, (15.57b) becomes

$$\frac{\partial H}{\partial t} + v_r \frac{\partial H}{\partial r} = 0, \tag{15.58}$$

which, using (15.53) and (15.54), gives

$$\boxed{v_r = -\frac{g}{\Omega^2 r}\langle S \rangle}. \tag{15.59}$$

This is a remarkable result, for it implies that, if $\langle S \rangle$ is positive the flow if is *toward* the apex of the dish except at the location of the mass sources and in frictional boundary layers, *no matter where the mass source is actually located.* The explanation of this counter-intuitive result is simple enough: if $\langle S \rangle > 0$ the overall height of the fluid is increasing with time. Thus, in order that a given material column of fluid may keep its height fixed, it must move toward the apex of the dish. The full velocity field may be obtained, away from the frictional regions, using the divergence-free nature of the velocity:

$$\nabla \cdot \boldsymbol{u} = \frac{1}{r}\left(\frac{\partial (r v_r)}{\partial r} + \frac{\partial v_\theta}{\partial \theta}\right) = 0. \tag{15.60}$$

Then, using (15.59), $\partial v_\theta / \partial \theta = 0$ except at a source or sink, or in a frictional boundary layer. Assuming there is only one frictional boundary layer, $v_\theta = 0$ except at those latitudes (i.e., values of r) that contain a mass source or sink.

Suppose then, that we introduce a localized mass source somewhere in the domain, and a localized mass sink of equal strength somewhere else. According to our heuristic theory, there is no flow in the interior of the domain that can provide a passage from the mass source to the sink. Thus, there must be *boundary layers* in which frictional effects are important, and which set themselves up in such a way to satisfy mass conservation. But mass conservation alone is insufficient to determine where

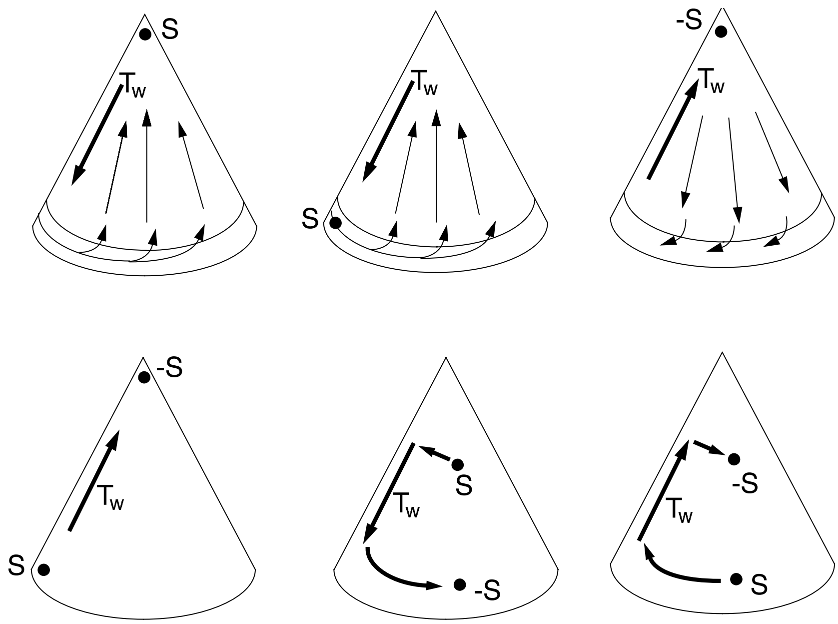


Fig. 15.11 Idealized examples of the flow in the rotating sector experiments, with various locations of a source (S) or sink ($-S$) of mass

the boundary layers might be — for this we need some vorticity dynamics. Now, away from the mass source, but including friction, the potential vorticity equation is

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{f_0}{H}\right) = \frac{\mathrm{curl}_z \boldsymbol{F}}{H}, \tag{15.61}$$

and the free surface of the water slopes downward toward the apex, as illustrated in Fig. 15.10. Now, suppose that there are mass source and a sink of equal maginudes, with the source further from the apex than the sink, as in the panel at the bottom right of Fig. 15.11. The flow from source to sink must be along either the left of right boundary of the container. This flow is toward smaller values of H, and therefore the left-hand side of (15.61) is positive (just as for poleward flow in the ocean on a sphere or β-plane). To balance this the friction in the boundary current must import a positive vorticity to the flow (i.e., $\mathrm{curl}_z \boldsymbol{F} > 0$), implying a *western* boundary layer, i.e., a boundary layer on the left of the container, for then the flow itself then has an anticyclonic (clockwise) sense and friction will normally oppose this. For example, if $\boldsymbol{F} = -\lambda \boldsymbol{u}$ the right hand side of (15.61) is $-(\lambda/H)\mathrm{curl}_z \boldsymbol{u}$ and this is positive if the flow is clockwise. A little more thought will reveal that a western boundary layer is general feature of the flow, and is not dependent on the placement of mass sources or sinks. If there is only a net source of mass, as for example in the upper left example of Fig. 15.11, then the interior mass flow will be toward the apex, the flow in the boundary layer away from the apex, but again requiring a western boundary layer to achieve a balance in the potential vorticity equation. It is clear that this flow is in some ways analogous to flow on the β-plane, and that in particular:

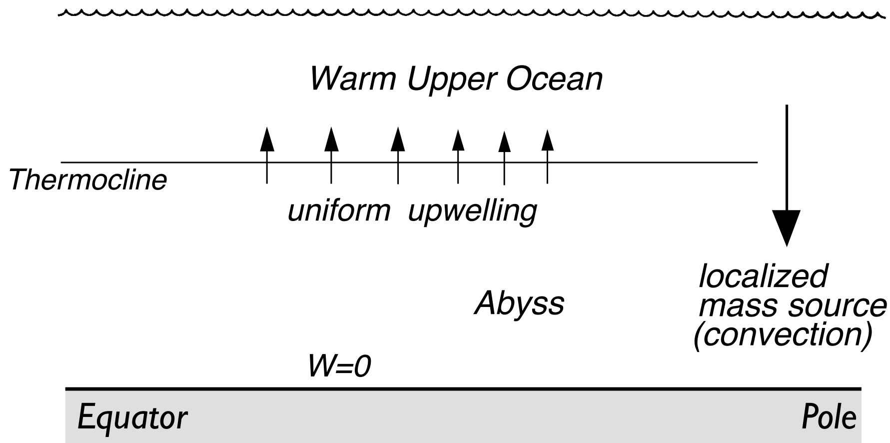


Fig. 15.12 The structure of simple Stommel–Arons ocean model of the abyssal circulation. Convection at high latitudes provides a localized mass-source to the lower layer, and upwelling through the thermocline provides a more uniform mass sink.

(i) The r-dependence of the height field provides a background potential vorticity gradient, analogous to the β-effect.
(ii) The time-dependence of H is analogous to a wind curl, for it is this that ultimately drives the fluid motion.

The analogies are drawn out explicitly in the shaded box on the next page; the box also includes a column for abyssal flow in the ocean, discussed in the next two sections.

15.6 A MODEL FOR OCEANIC ABYSSAL FLOW

We will now extend the reasoning applied to the rotating tank to the rotating sphere, and so construct a model — the *Stommel–Arons model* — of the abyssal flow in the ocean.[14] The basic idea is simple: we will model the deep ocean as a single layer of homogeneous fluid in which there is a localized injection of mass at high latitudes, representing convection (Fig. 15.12). However, unlike the rotating dish, mass is extracted from this layer by upwelling into the warmer waters above it, keeping the average thickness of the abyssal layer constant. We assume that this upwelling is nearly uniform, that the ocean is flat-bottomed, and that a passive western boundary current may be invoked to satisfy mass conservation, and which does not affect the interior flow. Obviously, these assumptions are very severe and the model can at best be a conceptual model of the real ocean. Given that, we will work in Cartesian coordinates on the β-plane, and use the planetary geostrophic approximation. Our treatment in this section is physically based but quite heuristic; in section 15.7 we are a little more mathematical and a little more formal.

The momentum and mass continuity equations are

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi \quad \text{and} \quad \nabla \cdot \boldsymbol{u} = -\frac{\partial w}{\partial z}, \tag{15.62}$$

Analogies Between a Rotating Dish, Wind-driven and Abyssal flows

Consider homogeneous models of (i) rotating dish, (ii) wind-driven flow on the β-plane, and (iii) abyssal flow on the the β-plane. We model all with a single layer of homogeneous fluid satisfying the planetary geostrophic equations. In (i) the mass source, $\langle S\rangle$, is localized and the total depth of the fluid layer changes with time; fluid columns move to keep their depth constant. In (ii) there is no mass source and depth of the fluid layer is constant; the fluid motion is determined by the wind stress curl, $\mathrm{curl}_z\boldsymbol{\tau}$, and by β. In (iii) the fluid source (convection) is localized at high latitudes and exactly balanced by a mass loss, S_u, due to upwelling everywhere else, so that layer depth is constant and S_u is uniform and negative nearly everywhere. The equations below then apply away from frictional boundary layers and localized mass sources:

	(i) Rotating dish	*(ii) Wind-driven flow*	*(iii) Abyssal flow*
PV Conservation	$\dfrac{\mathrm{D}}{\mathrm{D}t}\left(\dfrac{f_0}{H}\right) = 0$	$\dfrac{\mathrm{D}}{\mathrm{D}t}\left(\dfrac{f}{H_0}\right) = \dfrac{1}{H_0}\mathrm{curl}_z\boldsymbol{\tau}$	$\dfrac{\mathrm{D}}{\mathrm{D}t}\left(\dfrac{f}{h}\right) = -\dfrac{fS_u}{h^2}$
This leads to	$v_r\dfrac{\partial H}{\partial r} = -\dfrac{\partial H}{\partial t}$	$\dfrac{v}{H_0}\dfrac{\partial f}{\partial y} = \dfrac{1}{H_0}\mathrm{curl}_z\boldsymbol{\tau}$	$\dfrac{v}{h}\dfrac{\partial f}{\partial y} = -\dfrac{fS_u}{h^2}$
and	$v_r = -\dfrac{g}{\Omega^2 r}\langle S\rangle$	$v = \dfrac{1}{\beta}\mathrm{curl}_z\boldsymbol{\tau}$	$v = -\dfrac{fS_u h}{\beta}$
	$\langle S\rangle$ is localized mass source	$\mathrm{curl}_z\boldsymbol{\tau}$ is wind-stress curl	S_u is upwelling mass loss
Meridional mass flow away from boundaries is thus determined by:	Sign (and not location) of localized mass source, $\langle S\rangle$.	Sign of wind-stress curl, $\mathrm{curl}_z\boldsymbol{\tau}$.	Upwelling and sign of f, so polewards if $S_u < 0$ (upwelling).

which on elimination of ϕ yield the now-familiar balance,

$$\beta v = f \frac{\partial w}{\partial z}. \tag{15.63}$$

Except in the localized regions of convection, the vertical velocity is, by assumption, positive and uniform at the top of the lower layer, and zero at the bottom. Thus (15.63) becomes

$$v = \frac{f}{\beta} \frac{w_0}{H}. \tag{15.64}$$

where w_0 is the uniform upwelling velocity and H the layer thickness. Thus, the flow is *polewards* everywhere (including the Southern hemisphere), vanishing at the equator.

15.6.1 Completing the solution

Since $v = f^{-1}(\partial \phi / \partial x)$, the pressure is given by

$$\phi = \int_{x_0}^{x} \left(\frac{f^2 w_0}{\beta H} \right) \mathrm{d}x' \tag{15.65}$$

where x_0 is a constant of integration, to be determined by the boundary conditions. Because there no flow into the Eastern boundary, x_e we set ϕ = constant at $x = x_e$, and because this is a one-layer model we are at liberty to set that constant equal to zero. Thus,

$$\phi(x) = -\int_{x}^{x_e} \left(\frac{f^2 w_0}{\beta H} \right) \mathrm{d}x' = -\frac{f^2}{\beta H} w_0 (x_e - x). \tag{15.66}$$

The zonal velocity follows using geostrophic balance,

$$u = \frac{1}{f} \frac{\partial \phi}{\partial y} = \frac{2}{H} w_0 (x_e - x), \tag{15.67}$$

where we have also used $\partial f / \partial y = \beta$ and $\partial \beta / \partial y = 0$. Thus the velocity is eastward in the interior, and independent of f and latitude, provided x_e is not a function of y.

Using (15.64) and (15.67) we can confirm mass conservation is indeed satisfied:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = -\frac{2 w_o}{H} + \frac{w_0}{H} + \frac{w_0}{H} = 0 \tag{15.68}$$

15.6.2 Application to the ocean

Let us consider a rectangular ocean with a mass source at the northern boundary, balanced by uniform upwelling (see figures 15.13 and 15.14). Since the interior flow will be northwards, we anticipate a southwards flowing western boundary current to balance mass. Conservation of mass in the area poleward of the latitude y demands that

$$S_0 + T_i(y) = -T_w(y) + U(y) \tag{15.69}$$

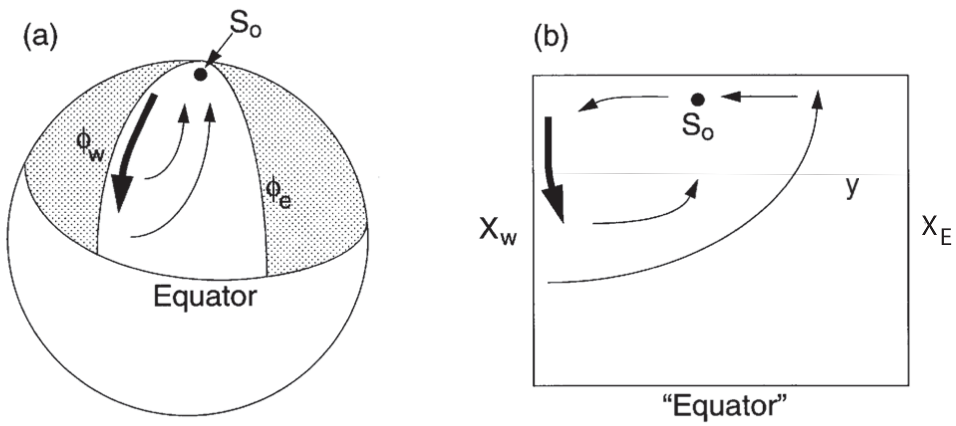


Fig. 15.13 Abyssal circulation in a spherical sector (left) and in a corresponding Cartesian rectangle (right).

where S_0 is the strength of the source, T_w the transport in the western boundary current (positive if polewards), T_i the (polewards) transport in the interior, and U is the integrated loss due to upwelling polewards of y. Then, using (15.64),

$$T_i = \int_{x_w}^{x_e} vH \, dx = \int_{x_w}^{x_e} \frac{f w_o}{\beta} \, dx = \frac{f}{\beta} w(x_e - x_w). \tag{15.70}$$

The upwelling loss is given by

$$U = \int_{x_w}^{x_e} \int_{y}^{y_n} w \, dx = w_0(x_e - x_w)(y_n - y) \tag{15.71}$$

Assuming the source term is known, then using (15.69) we obtain the strength of the western boundary current,

$$-T_w(y) = S_0 + T_i - U = S_0 + \frac{f}{\beta} w(x_e - x_w) - w_0(x_e - x_w)(y_n - y). \tag{15.72}$$

To close the problem we note that over the entire basin mass must be balanced, which gives a relationship between w and S_0,

$$S_0 = w_0 \Delta x \Delta y, \tag{15.73}$$

where $\Delta x = x_e - x_w$ and $\Delta y = y_n - y_s$. where y_s is the southern boundary of the domain. Using (15.73), (15.72) becomes

$$\begin{aligned} -T_w(y) &= -w_0 \left(\Delta x (y_n - y) - \frac{f}{\beta} \Delta x - \Delta x \Delta y \right) \\ &= w_0 \Delta x \left(y - y_s + \frac{f}{\beta} \right). \end{aligned} \tag{15.74}$$

With no loss of generality we will take $y_s = 0$ and $f = f_0 + \beta y$. Then (15.74) becomes

$$-T_w(y) = w_0 \Delta x \, (2y + f_0/\beta) \tag{15.75}$$

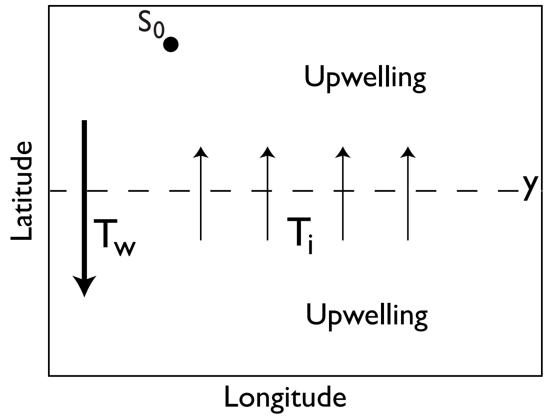


Figure 15.14 Mass budget in an idealized abyssal ocean. Polewards of some latitude y, the mass source (S_0) plus the polewards mass flux across y (T_i) are equal to sum of the southwards mass flux in the western boundary current (T_w) and the integrated loss due to upwelling (U) polewards of y. See (15.69).

or, using $S_0 = w_0 \Delta x y_n$,

$$- T_w(y) = \frac{S_0}{y_n}\left(2y - \frac{f_0}{\beta}\right). \tag{15.76}$$

With a slight loss of generality (but consistent with the spirit of the planetary geostrophic approximation) we take $f_0 = 0$, which is equivalent to supposing that the equatorial boundary of the domain is at the equator, and finally obtain

$$\boxed{T_w(y) = -2S_0 \frac{y}{y_n}.} \tag{15.77}$$

At the northern boundary this becomes

$$T_w(y) = -2S_0, \tag{15.78}$$

which means that the flow southwards from the source is twice the strength of the source itself. We also see that:

(i) The western boundary current is equatorwards everywhere;

(ii) At the northern boundary the equatorwards transport in the western boundary current is equal to *twice* the strength of the source;

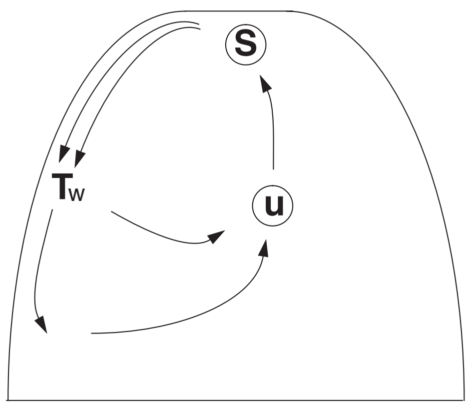


Figure 15.15 Schematic of a Stommel–Arons circulation in a single sector. The transport of Western boundary current is greater than that provided by the source at the apex, illustrating the property of *recirculation.* The transport in the western boundary current T_w decreases in intensity equatorwards, as it loses mass to the polewards interior flow, and thence to upwelling. The integrated sink, due to upwelling, U, exactly matches the strength of the source, S.

(iii) The northwards mass flux at the northern boundary is equal to the strength of the source itself.

We may check this last point directly: from(15.70)

$$T_I(y_n) = \frac{\beta y_n}{\beta} w_0 \Delta x = S. \tag{15.79}$$

The fact that convergence at the 'pole' balances T_w and S_0 does not of course depend on the particular choice we made for f and y_s.

The flow pattern evidently has the property of *recirculation* (see Fig. 15.15): this is one of the most important properties of the solution, and one that is likely to transcend all the limitations inherent in the model. This single-hemisphere model may be thought of as a crude model for aspects of the abyssal circulation in the North Atlantic, in which convection at high latitudes near Greenland is at least partially associated with the abyssal circulation. In the North Pacific there is, in contrast, little if any deep convection to act as a mass source. Rather, the deep circulation is driven by mass sources in the opposite hemisphere, and we now consider a simple model of this.

15.6.3 A two hemisphere model

Our treatment now is even more obviously heuristic, since our domain crosses the equator yet we continue to use the planetary geostrophic equations, invalid at the equator. We also persist with Cartesian geometry, even for these global-scale flows. In our defense, we note that the value of the solutions lies in their qualitative structure, not in their quantitative predictions. Let us consider a situation with a source in the Southern Hemisphere but none in the Northern Hemisphere. For later convenience we take the Southern Hemisphere source to be of strength $2S_0$, and we suppose the two hemispheres have equal area. As before, the upwelling is uniform, so that to satisfy global mass balance

$$S_0 = w_0 \Delta x \Delta y \tag{15.80}$$

where $\Delta x \Delta y$ is the area of each hemisphere. Then, given w_0, the zonally integrated polewards interior flow in each hemisphere, away from the equator, follows from Sverdrup balance,

$$T_i(y) = \frac{f}{\beta} w_0 (x_e - x_w) = S_0 \frac{y}{y_p} \tag{15.81}$$

where y_p is either y_n (the northern boundary) or y_s. The western boundary current is assumed to 'take up the slack,' that is to be able to adjust its strength to satisfy mass conservation. Thus, since $T_i(y_n) = S_0$, where S_0 is half the strength of the source in the *southern* hemisphere, it is plain that there must be a southwards flowing western boundary current near the northern end of the northern hemisphere, even in the absence of any deep water formation there!

In the northern hemisphere, the total loss due to upwelling polewards of a latitude y is given by

$$U(y) = w_0 \Delta x |y_n - y| \tag{15.82}$$

The strength of the western boundary current is then given by

$$T_w(y) = U - T_i = w_0 \Delta x (y_n - y) - \frac{f}{\beta} w_0 \Delta x = w_0 \Delta x (y_n - 2y) \tag{15.83}$$

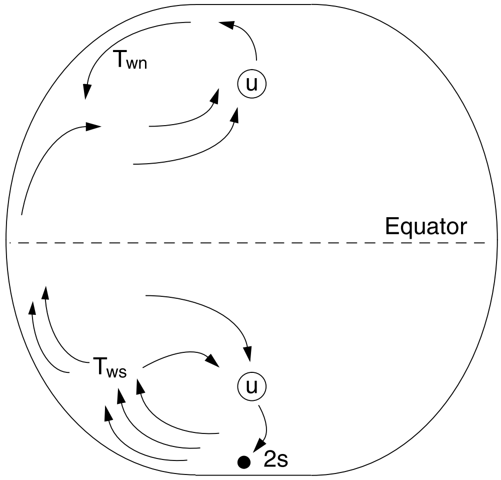


Figure 15.16 Schematic of a Stommel–Arons circulation in a two-hemisphere basin. There is only one mass source, and this is in the Southern hemisphere and for convenience it has a strength of 2. Although there is no source in the Northern Hemisphere, there is still a western boundary current and a recirculation. The integrated sinks due to upwelling exactly match the strength of the source.

using $f = \beta y$. Thus, at $y - y_n$,

$$T_w(y_n) = -w_0 \Delta x y = -S. \tag{15.84}$$

The boundary current changes sign halfway between equator and pole, at $y = y_n/2$. [In spherical coordinates, the analogous latitude turns out to be at $\theta = \sin^{-1}(1/2)$.] The solution is illustrated schematically in Fig. 15.16. We can (rather fancifully) imagine this to represent the abyssal circulation in the Pacific Ocean, with no source of deep water at high northern latitudes.

15.7 * A SHALLOW WATER MODEL OF THE ABYSSAL FLOW

We can obtain a more complete solution of the flow that explicitly includes the western boundary current by constructing a shallow water Stommel–Arons type model of the abyssal circulation.[15] However, the essential dynamics is the same as that of the previous section. The model is similar to that illustrated in Fig. 15.12 and comprises a single moving layer of homogeneous fluid lying underneath a lighter, stationary layer. The lower fluid is forced by a mass source at its polewards end that represents convection, and by a uniform mass sink everywhere else that represents diffusive upwelling into the upper layer. The mass source and sink are specified — that is, they are not functions of the flow — and are equal and opposite, so that there is no net mass source. The motion of the lower layer is then governed by the planetary geostrophic reduced-gravity shallow water equations (sections 3.2 and 5.2), to wit:

$$-fv = -g'\frac{\partial h}{\partial x} - ru, \quad fu = -g'\frac{\partial h}{\partial y} - rv, \quad \frac{\partial h}{\partial t} + \nabla \cdot (\boldsymbol{u}h) = S. \tag{15.85a,b,c}$$

Here h is the thickness of the lower layer (below the thermocline of Fig. 15.12), g' is the reduced gravity between the two layers, and r is a constant frictional coefficient; we will assume this is small, and in particular that $r \ll f$. The mass source term S on the right-hand side of the mass continuity equation represents both upwelling and a localized convective source at the polewards end of the domain. We will suppose the upwelling is uniform and that, when integrated over area, it exactly balances the convective mass source. Thus, we write $S = S_0 + S_u$ where S_u is uniform and negative, S_0 is the localized convective source, and $\int_A S \, dA = 0$.

15.7.1 Potential vorticity and polewards interior flow

Straightforward manipulation of (15.85) gives the potential vorticity equation

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{f}{h}\right) = -\frac{r}{h}\mathrm{curl}_z \boldsymbol{u} - \frac{fS_u}{h^2}. \tag{15.86}$$

Away from boundaries the first term on the right-hand side is negligible, and so, away from the convective source and in a steady state (15.86) becomes

$$\frac{\beta v}{h} + \frac{f}{h^2}\boldsymbol{u}\cdot\nabla h = -\frac{fS_u}{h^2}. \tag{15.87}$$

However, the second term on the left-hand side is small if friction is small, for then the flow is nearly in geostrophic balance and, using (15.85a) with $r = 0$, it follows that

$$f\boldsymbol{u}\cdot\nabla h = -g'\frac{\partial h}{\partial y}\frac{\partial h}{\partial x} + g'\frac{\partial h}{\partial x}\frac{\partial h}{\partial y} = 0. \tag{15.88}$$

The potential vorticity equation is then just

$$\beta v = -fS_u/h, \tag{15.89}$$

and because $S_u < 0$ the flow is polewards, regardless of the location of the convective mass source. From the perspective of the continuously stratified equations, the corresponding potential vorticity equation is just

$$\beta v = f\frac{\partial w}{\partial z} \tag{15.90}$$

Upwelling from the abyss into the upper ocean corresponds to a positive value of the stretching term $f\partial w/\partial z$, and again the interior flow is polewards.

15.7.2 The solution

It is convenient to deal with mass transports rather than velocities and we define $\boldsymbol{U} \equiv U\mathbf{i} + V\mathbf{j} \equiv uh\mathbf{i} + vh\mathbf{j}$. Away from the convective source equations of motion, (15.85), become, in steady state,

$$-fV = -\frac{\partial \Phi}{\partial x} - rU, \qquad fU = -\frac{\partial \Phi}{\partial y} - rV, \tag{15.91a}$$

$$\nabla\cdot\boldsymbol{U} = S_u, \tag{15.91b}$$

where $\Phi = g'h^2/2$. For small friction ($r \ll f$) we may write (15.91a) as

$$V = \frac{1}{f}\frac{\partial \Phi}{\partial x} - \frac{r}{f^2}\frac{\partial \Phi}{\partial y}, \qquad U = -\frac{1}{f}\frac{\partial \Phi}{\partial y} - \frac{r}{f^2}\frac{\partial \Phi}{\partial x}, \tag{15.92a,b}$$

which, after differentiation and use of (15.91b), combine to give

$$\frac{\beta}{f}\frac{\partial \Phi}{\partial x} = -fS_u - f\nabla \cdot \left(\frac{r}{f^2}\nabla \Phi\right). \tag{15.93}$$

In the interior where the effects of friction are small (15.93) becomes $\beta V = -fS$, so recovering (15.89) and imlying polewards interior flow. However, by mass conservation, the flow cannot be polewards at all longitudes, for reasons similar to those articulated in section 14.1.1 we expect there to be a frictional western boundary current. We thus let $\Phi = \Phi_I + \Phi_b$, where Φ_I is the interior field and Φ_b the boundary layer correction. The interior field obeys $\partial \Phi_I/\partial x = -(f^2/\beta)S$, and is therefore given by, for constant S,

$$\Phi_I(x, y) = \frac{f^2}{\beta} S_u(x_e - x) + \Phi_e \tag{15.94}$$

where Φ_e is the value of Φ_I at the eastern boundary, x_e. The value of Φ_e does not affect the solution and may be set to zero.

In the western boundary layer the the dominant balance in (15.93) is

$$\beta \frac{\partial \Phi_b}{\partial x} = -r\frac{\partial^2 \Phi_b}{\partial x^2}, \tag{15.95}$$

with solution

$$\Phi_b = P(y)\exp(-\beta x/r), \tag{15.96}$$

where $P(y)$ is to be determined by mass conservation: the southwards mass flux at a latitude y in the western boundary current must be equal to the sum of the polewards mass flux in the interior at that latitude plus the total mass lost to upwelling equatorwards of y (Fig. 15.14). These fluxes are:

$$\text{Upwelling flux} = \int_0^{x_e}\int_0^{y} S_u \,\mathrm{d}x\,\mathrm{d}y = S_u x_e y, \tag{15.97}$$

using the definition of S;

$$\text{Interior flux} = \int_0^{x_e} V\,\mathrm{d}x = \frac{1}{f}\int_0^{x_e}\frac{\partial \Phi_I}{\partial x}\,\mathrm{d}x = \frac{1}{f}[\Phi_e - \Phi_I(0, y)] = -\frac{f}{\beta}Sx_e, \tag{15.98}$$

using (15.94); and

$$\text{Boundary flux} = \int_0^{\infty}\frac{1}{f}\frac{\partial \Phi_b}{\partial x}\,\mathrm{d}x = P(y), \tag{15.99}$$

using (15.92a), neglecting the very small term $r\partial\Phi/\partial y$, and (15.96). By mass conservation we have

$$P(y) = S_u x_e y + \frac{f}{\beta}S_u x_e, \tag{15.100}$$

and if $f = \beta y$ and the domain goes from $y = 0$ to $y = y_n$ this becomes

$$P(y) = 2S_u x_e y = -2S_0\frac{y}{y_n}, \tag{15.101}$$

where the second equality follows because the integrated upwelling balances the mass source. Consistent with our earlier heuristic treatment, the western boundary current

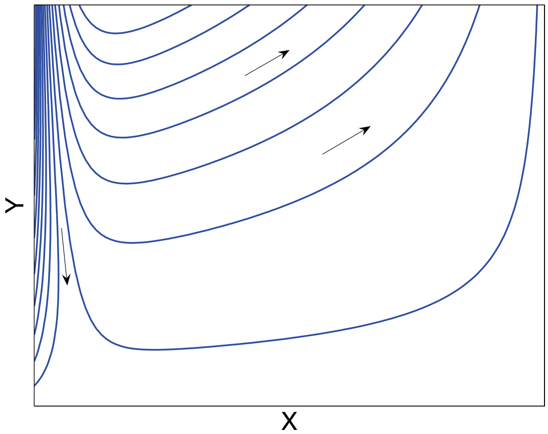


Figure 15.17 The pressure field Φ for the shallow-water Stommel–Arons model, as given by (15.102) with $r/\beta = 0.04x_e$, $f = \beta y$, and $y = 0$ at the equatorwards edge of the domain. The arrows indicate the flow direction, with the western boundary current diminishing in intensity as it moves equatorwards. The convective mass source is, implicitly, just polewards of the domain. (Note that the pressure field is not exactly a streamline.)

(i) is equatorwards, away from the convective mass source;
(ii) diminishes in intensity as it moves equatorwards, as it feeds the interior;
(iii) has a maximum mass flux of twice that of the convective source; that is, the flow recirculates.

Using (15.94) and (15.101) the complete (western boundary layer plus interior) solution is thus given by

$$\Phi = \Phi_I + \Phi_b = \frac{f^2}{\beta} S_u (x_e - x) + 2 S_u x_e y e^{-\beta x / r}, \tag{15.102}$$

and this is plotted in Fig. 15.17. This solution does not, of course, include the flow in the neighbourhood of the convective source itself, nor does it satisfy no-normal flow conditions at the polewards edge of the domain.

15.8 SCALING FOR THE BUOYANCY-DRIVEN CIRCULATION

Thus far, we have taken the strength of the upwelling as a given. In reality, this is a consequence of the presence of diapycnal diffusion, because in its absence the flow is along isopycnals. In section 15.2 we estimated the upwelling for a non-rotating model; we now do the same for a fluid obeying the steady planetary geostrophic equations, namely

$$\boldsymbol{v} \cdot \nabla b = \kappa \nabla^2 b, \qquad \nabla \cdot \boldsymbol{u} + \frac{\partial w}{\partial z} = 0 \tag{15.103a,b}$$

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla \phi, \qquad b = \frac{\partial \phi}{\partial z}. \tag{15.104a,b}$$

On the first line we have the thermodynamic equation and the mass continuity equation, and on the second line we have the momentum equations, that is geostrophic and hydrostatic balance, respectively. We use the continuously stratified equations rather than the corresponding shallow water equations because the former allow for a straightforward representation of diapycnal diffusion.

The momentum and mass continuity equations combine to give the linear geostrophic vorticity equation and the thermal wind equation and associated scales as follows:

$$\beta v = f\frac{\partial w}{\partial z} \quad \rightarrow \quad \beta U = \frac{f_0 W}{\delta}, \tag{15.105a}$$

$$f\frac{\partial \boldsymbol{u}}{\partial z} = \mathbf{k}\times\nabla b \quad \rightarrow \quad f\frac{U}{\delta} = \frac{\Delta b}{L}. \tag{15.105b}$$

We use uppercase to denote scaling variables, except that the vertical scale is denoted δ and the scaling for buoyancy variations is denoted Δb. We will take Δb is given, and equal to the buoyancy difference at the surface that is ultimately driving the motion. We also assume that the horizontal scales are isotropic, with $U = V$ and $X = Y = L$. The thermodynamic relation gives a relationship between W, κ and δ if we assume a broad upwelling region with a balance between upwards advection and diffusion:

$$w\frac{\partial b}{\partial z} = \kappa\frac{\partial^2 b}{\partial z^2} \quad \rightarrow \quad W = \frac{\kappa}{\delta}. \tag{15.106}$$

Eliminating U from (15.105) gives

$$W = \frac{\delta^2 \beta\,\Delta b}{f^2 L}, \tag{15.107}$$

and using this with (15.106) gives scalings for the vertical velocity and δ:

$$\boxed{W = \kappa^{2/3}\left(\frac{\beta\,\Delta b}{f^2 L}\right)^{1/3}, \qquad \delta = \kappa^{1/3}\left(\frac{f^2 L}{\beta\,\Delta b}\right)^{1/3}}. \tag{15.108a,b}$$

These scalings mean that, in so far as (15.104) describes the flow, the upwelling strength, and the circulation more generally, are dependent on a finite value of the diffusivity and scale as the 2/3 power of that diffusivity. The upwelling water is cold but the water at the surface is warm, and (15.108b) is a measure of the depth of the transition region — that is, the thickness of the thermocline, a topic we return to in more detail in the next chapter. We will make one important point now though: In order for the scales given in (15.108) to be at all representative of those observed in the real ocean, we must use an eddy diffusivity for κ. Using $f = 10^{-4}\,\mathrm{s^{-1}}$, $\beta = 10^{-11}\,\mathrm{m^{-1}\,s^{-1}}$, $L = 5\times 10^6\,\mathrm{m}$, $g = 10\,\mathrm{m\,s^{-2}}$, $\kappa = 10^{-5}\,\mathrm{m^2\,s^{-1}}$, $\Delta b = -g\Delta\rho/\rho_0 = g\beta_T\Delta T$ and $\Delta T = 10\,\mathrm{K}$ we find the not unreasonable values of $\delta \approx 150\,\mathrm{m}$ and $W = 10^{-7}\,\mathrm{m\,s^{-1}}$, albeit δ is rather smaller than the thickness of the observed thermocline. However, if we take the molecular value of $\kappa \approx 10^{-7}\,\mathrm{m^2}$ the values of W and δ are unrealistically small. Evidently, if the deep circulation of the ocean is buoyancy driven, it must take advantage of turbulence that enhances the small scale mixing and produces an eddy diffusivity.

15.8.1 Summary remarks on the Stommel–Arons model

If we were given the location and strength of the sources of deep water in the real ocean, the Stommel–Arons model could give us a global solution for the abyssal circulation. The solution for the Atlantic, for example, resembles a superposition of Fig. 15.15 and Fig. 15.16 (with deep water sources in the Weddell Sea and near Greenland),

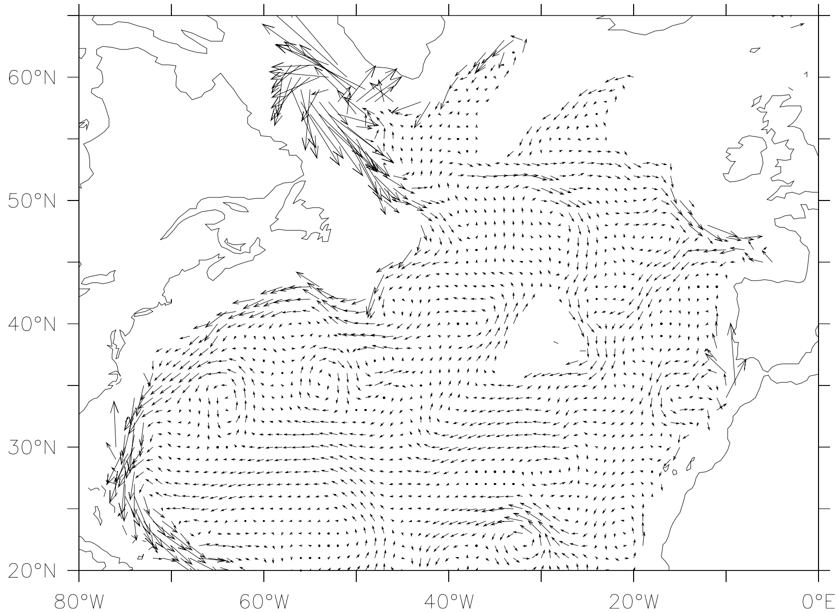


Fig. 15.18 The ocean currents at a depth of 2500 m in the North Atlantic, obtained using a combination of observations and model (as in Fig. 14.2). Note the southwards flowing *deep western boundary current.*

and that for the Pacific resembles Fig. 15.16 (with a deep water source emanating from the Antarctic Circumpolar Current). Perhaps the greatest success of the model is that it introduces the notions of deep western boundary currents and recirculation — enduring concepts of the deep circulation that remain with us today. For example, the North Atlantic does have a well-defined deep western boundary current running south along the Eastern seaboard of Canada and the United States, as seen in Fig. 15.18.[16]

However, in other important aspects the model is found to be in error, in particular it is found that there is little upwelling through the main thermocline — much of the water formed by deep convection in the North Atlantic in fact upwells in the Southern Hemisphere.[17] Are there fundamental problems with the model, or just discrepancies in details that might be corrected with a slight reformulation? To help answer that we summarize the assumptions and corresponding predictions of the model, and distinguish the essential aspects from what is merely convenient.

(i) A foundational assumption is that of linear geostrophic vorticity balance in the ocean abyss, represented by $\beta v \approx f \partial w/\partial z$, or its shallow water analog.

— The effects of mesoscale eddies are thereby neglected. As discussed in chapter 9, in their mature phase mesoscale eddies seek to barotropize the flow, and so create deep eddying motion that might dominate the deep flow.

(ii) A second important assumption is that of uniform upwelling, across isopycnals, into the upper ocean, and that $w = 0$ at the ocean bottom. When combined with (i) this gives rise to polewards interior flow, and by mass conservation a deep western boundary current. The upwelling is a consequence of a finite diffusion,

which in turn leads to deep convection as in the model of sideways convection of section 15.2.

— The uniform-upwelling assumption might be partially relaxed, while remaining within the Stommel–Arons framework, by supposing (for example) that the upwelling occurs near boundaries, or intermittently, with corresponding detailed changes to the interior flow.

— If bottom topography is important, then $w \neq 0$ at the ocean bottom. This effect may be most important if mesoscale eddies are present, for then in an attempt to maintain its value of potential vorticity the abyssal flow will have a tendency to meander nearly inviscidly along contours of constant topography. In the presence of a mid-ocean ridge, some of the deep western boundary current might travel meridionally along the eastern edge of the ridge instead of along the coast.

— The deep water may not upwell across isopycnals at at all, but may move along isopycnals that intersect the surface (or are connected to the surface by convection). If so, then in the presence of mechanical forcing a deep circulation could be maintained even in the absence of a diapycnal diffusivity. The circulation might then be qualitatively different from the Stommel–Arons model, although a linear vorticity balance might still hold, with deep western boundary currents. This is discussed in section 16.5.

Note that, even if Stommel–Arons picture were to be essentially correct, we should not consider the deep flow is being driven by deep convection at the source regions. It is a *convenience* to specify the strength of these regions for the calculations but, just as in the models of sideways convection considered in section 15.2, the overall strength of the circulation (insofar as it is thermally driven) is a function of the size of the diffusivity and the meridional buoyancy gradient at the surface.

Notes

1 See, for example, Warren (1981) who provides a review and historical background and Schmitz (1995) who surveys the observations and provides an interpretation of the deep global circulation.

2 From Wunsch (2002).

3 Figure kindly prepared by N. Stefan-Fuckar, using the data of Conkright et al. (2001).

4 Adapted from Paparella and Young (2002).

5 As in Haney (1971). The value of C, which is not necessarily related to that of κ, is often taken to be such that the heat flux is of order $30\,\mathrm{W\,m^{-2}\,K^{-1}}$, but it is not a universal constant.

6 Rossby (1965).

7 Adapted from Rossby (1998).

8 Ocean convection is reviewed by Marshall and Schott (1999).

9 Sandström (1908, 1916). Sandström's discussion was rather qualitative and generally thermodynamic in nature, and with friction playing only an implicit role. Since then various related statements with varying degrees of generality and preciseness have

been given (see e.g., Dutton 1986, Huang 1999, Paparella and Young 2002). Section 15.3.3 follows Paparella and Young.

10 The original box model is due to Stommel (1961), and many studies with variations around this have followed. Rooth (1982) developed the idea of a buoyancy-driven pole-to-pole overturning circulation, and Welander (1986) discussed, among other things, the role of boundary conditions on temperature and salinity at the ocean surface. Thual and McWilliams (1992) systematically explored how box models compare with two-dimensional fluid models of sideways convection, Quon and Ghil (1992) explored how multiple equilibria arise in related fluid models, and Dewar and Huang (1995) discussed the problem of flow in loops. Cessi and Young (1992) tried to derive simple models systematically from the equations of motion, obtaining various nonlinear amplitude equations. Our discussion is of just a fraction of all this; see also Whitehead (1995), Cessi (2001) and Dijkstra (2002) for reviews and more discussion.

11 Adapted from Welander (1986) and Dijkstra (2002).

12 Having said this, Bryan (1986), Manabe and Stouffer (1988) and Marotzke (1989) did find evidence of multiple equilibria in various three-dimensional numerical models, motivated in part by the solutions of box models.

13 After Stommel et al. (1958).

14 Following Stommel and Arons (1960).

15 Following Cessi (2001).

16 A global Stommel–Arons-like solution was presented by Stommel (1958). The discovery of deep western boundary currents, by Swallow and Worthington (1961), was motivated by the theoretical model. Using neutrally-buoyant floats underneath the Gulf Stream they found a robust equatorwards-flowing undercurrent with typical speeds of 9–$18\,\mathrm{m\,s^{-1}}$. Relevant observations of the deep circulation are summarized by Hogg (2001).

17 For example, Toggweiler and Samuels (1995).

Problems

15.1 *(a)* Obtain an expression analogous to (15.19) for the anelastic equations.

(b) ♦ Obtain, if possible, an expression analogous to (15.19) for a compressible gas (which you may suppose to be ideal, if needed). Interpret the result.

15.2 *(a)* Consider an ocean with a flat surface, forced only by radiation. Suppose that the net radiation at the surface falls from about +100 W m^{-2} at the equator to -100 W m^{-2} at the pole, and that associated heating $\dot{Q}$ varies in vertical with an e-folding scale of 10 meters with a volume integral that is zero. Is such a forcing sufficient to account for the observed dissipation of kinetic energy in the ocean, which is about 1 kW km^{-3}?

(b) Now let a wind blow over the ocean, providing an average stress of 0.05 N m^{-2}. Estimate the kinetic energy dissipation that this will lead to.

(c) Estimate the rate of heating (in degrees per day) associated with a kinetic energy dissipation rate of 1 kW km^{-3}?

15.3 ♦ Consider a model of sideways convection in which the boundary condition at the top is the relaxation, or Haney, condition $b_z = A(b^*(y) - b)$ where A (a constant) and $b^*(y)$ are given, b is the buoyancy, proportional to the temperature, the other boundaries are insulating, and the flow is statistically steady.

(a) Is it still the case that on average the fluid is heated (i.e., there is a heat flux into

the fluid through the upper surface) where it is already warm? If so, how may this be reconciled with the intuition that if the ocean surface is anomalously warm it will cool by way of a heat flux from the ocean to the atmosphere?

(b) Show that if $A < 0$ a statistically steady state cannot be reached.

(c) Suppose that b^* varies monotonically with latitude. Show that if κ is non-zero, the average surface temperature gradient will be less than that of b^*.

15.4 Write a code (e.g., in Matlab, or Fortran) that will time step the Stommel two box model, (15.39). Integrate the model to equilibrium for a variety of initial conditions and parameters. Verify some of the solutions against the graphical/analytic solutions.

15.5 Consider a variation of the Stommel box model in which the equation of motion (15.39) is replaced by

$$\frac{\mathrm{d}T}{\mathrm{d}\tau} = (1 - T) - \Phi^2 T, \quad \frac{\mathrm{d}S}{\mathrm{d}\tau} = \delta(1 - S) - \Phi^2 S, \quad \Phi = -\gamma(T - \mu S). \tag{P15.1}$$

The interbox flow equations now depends on the density difference squared, so allowing continuous equations. Show that multiple equilibria are possible, and that one is thermally driven and one salinity driven. Obtain approximate expressions for the temperature and salinity for these equilibria, in the limit of large γ and small δ if you wish. If δ is large, what changes about these solutions?

15.6 *(a)* In the Stommel two-box problem, show physically that when three solutions are present, the middle one is generally unstable. One way to do this is to suppose that the system is perturbed slightly from that equilibrium, and argue qualitatively that the forces on the system will then take it farther from that state. By use of similar arguments, show that the other solutions are generally stable.

(b) ♦ Alternatively, linearize the equations about the equilibrium points and show that small perturbations will grow if the solution is that of the middle equilibrium state, but will be damped in the other two cases.

15.7 Define $\mathcal{R} \equiv Ra\alpha^5$ and expand

$$\widehat{\psi} = \mathcal{R}\widehat{\psi}_1 + \mathcal{R}^2\widehat{\psi}_2 + \mathcal{O}(\mathcal{R}^3), \qquad \widehat{b} = \widehat{b}_0 + \mathcal{R}\widehat{b}_1 + \mathcal{O}(\mathcal{R}^2). \tag{P15.2}$$

For small aspect ratio flow show that (15.9) gives at zeroth order

$$\frac{\partial^2 \widehat{b}_0}{\partial \hat{z}^2} = 0, \tag{P15.3}$$

giving a solution that satisfies the boundary conditions at both top and bottom. Show that at next order

$$\frac{\partial^4 \widehat{\psi}_1}{\partial \hat{z}^4} = -\frac{\partial f}{\partial \hat{y}}, \qquad J(\widehat{\psi}_1, f) = \frac{\partial^2 \widehat{b}_1}{\partial \hat{z}^2}. \tag{15.4a,b}$$

Show that the solutions are of the general form $\widehat{\psi}_1(\hat{y}, \hat{z}) = (\partial f/\partial \hat{y})A(\hat{z})$ and $b_1(\hat{y}, \hat{z}) = (\partial f/\partial \hat{y})^2 B(\hat{z})$, where A and B are functions only of $\hat{z}$. Thus deduce that the horizontal form of the solution is determined by the surface forcing, provided there are no meridional walls that force $\widehat{\psi}$ to zero where $\partial f/\partial \hat{y} \neq 0$. (From notes by W. R. Young.)

15.8 ♦ Obtain the explicit solution to (P15.4) at lowest order, and at one higher order, and show that the sinking region is narrower than the upwelling region.

15.9 ♦ Suppose that seawater were transparent to solar radiation, but nearly opaque to infra-red radiation, but that its other properties were the same as real seawater. Estimate the strength of the oceanic meridional overturning circulation, stating clearly any assumptions you make. You may neglect the effects of wind. Does the addition of wind make a difference to your solution?

No fairer destiny [has] any physical theory, than that it should of itself point out the way to the introduction of a more comprehensive theory, in which it lives on as a limiting case.

Albert Einstein, *Relativity, the Special and the General Theory*, 1916.

CHAPTER

SIXTEEN

The Wind and Buoyancy Driven Circulation

IN THIS CHAPTER we try to understand the combined effect of wind and buoyancy forcing in setting the three-dimensional structure of the ocean. There are three main topics we will consider:

(i) The *main thermocline*, the region in the upper 1 km or so of the ocean where density and temperature change most rapidly.

(ii) The 'wind-driven' overturning circulation. That is, we look into whether and how the ocean might maintain a deep overturning circulation that owes its existence to the effects of wind at the surface, and that persists even as the diapycnal diffusivity in the ocean interior goes to zero.

(iii) The circulation of the flow in a channel, as a model of the Antarctic Circumpolar Current (ACC).

16.1 THE MAIN THERMOCLINE: AN INTRODUCTION

In the previous chapter we saw that a fluid that is differentially heated from above will develop both an overturning circulation and a region near the surface where the temperature changes rapidly. We now examime this in more detail, and to begin we consider the circulation in a closed, single hemispheric basin, and again suppose that there is a net surface heating at low latitudes and a net cooling at high latitudes that maintains a meridional temperature gradient at the surface. We also presume, *ab initio,* that there is a single overturning cell, with water rising at low latitudes before returning to polar regions, illustrated schematically in Fig. 16.1. The real stratification is illustrated in Fig. 16.2.

At lower latitudes the surface water is warmer than the cold water in the abyss.

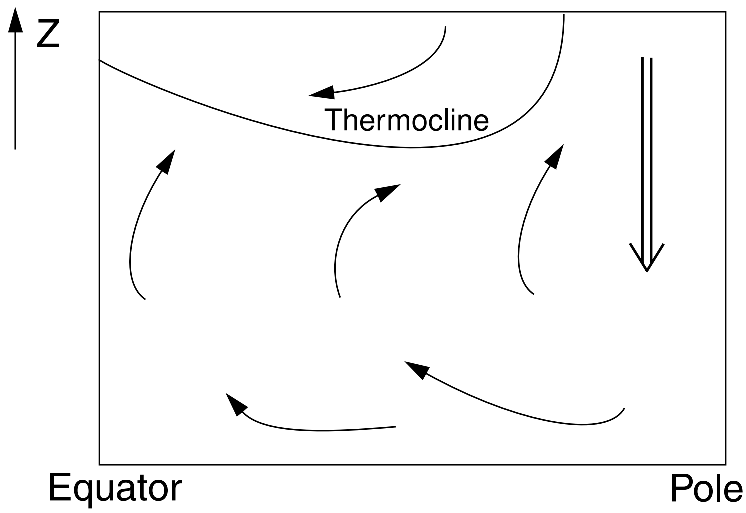


Fig. 16.1 Cartoon of a single-celled meridional overturning circulation. Sinking is concentrated at high latitudes and upwelling spread out over lower latitudes. The thermocline is the boundary between the cold abyssal waters, with polar origins, and the warmer near-surface subtropical water. Wind forcing in the subtropical gyre may also mechanically push the warm water down, deepening the thermocline.

Thus there must be a vertical temperature gradient everywhere except possibly at the highest latitudes where the cold dense water sinks. This temperature gradient is called the *thermocline.* In purely bouyancy-driven flows the thickness of the thermocline is determined by way of an advective-diffusive balance, and proportional to (some power of) the thermal diffusivity, and we first consider a simple model of this.

16.1.1 A simple kinematic model

The fact that cold water with polar origins upwells into a region of warmer water suggests that we consider the simple one-dimensional advective-diffusive balance,

$$w\frac{\partial T}{\partial z} = \kappa\frac{\partial^2 T}{\partial z^2}. \tag{16.1}$$

where w is the vertical velocity, κ is a diffusivity and T is temperature. In mid-latitudes, where this might hold, w is positive and the equation represents a balance between the upwelling of cold water and the downward diffusion of heat. If w and κ are given constants, and if T is specified at the top ($T = T_T$ at $z = 0$) and if $\partial T/\partial z = 0$ at great depth ($z = -\infty$) then the temperature falls exponentially away the surface according to

$$T = (T_T - T_B)e^{wz/\kappa} + T_B. \tag{16.2}$$

where T_B is a constant. This expression cannot be used to estimate how the thermocline depth scales with either w or κ, because the magnitude of the overturning circulation depends on κ (section 15.8). However, it is reasonable to see if the observed ocean is broadly consistent with this expression. The diffusivity κ can be

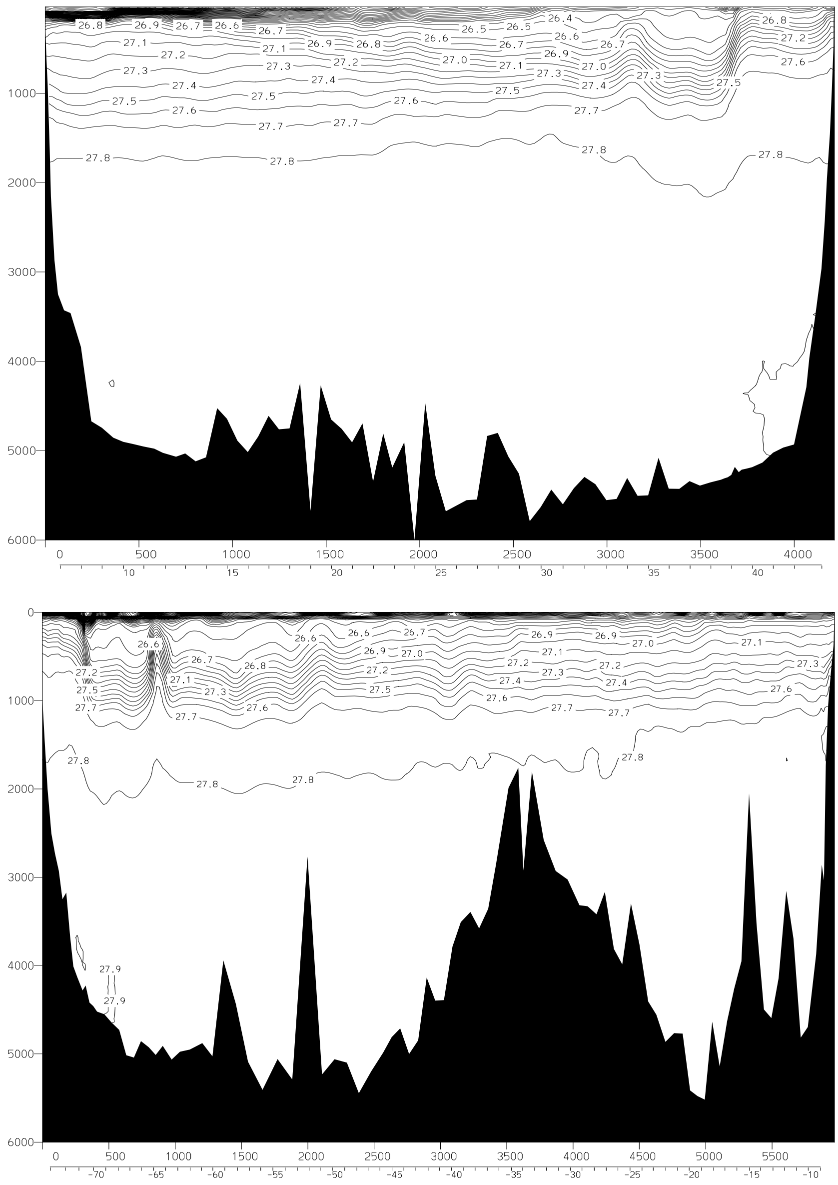


Fig. 16.2 Sections of potential density (σ_θ) in the North Atlantic: Upper panel: meridional section at 53°W, from 5°N to 45°N, across the subtropical gyre. Lower panel: zonal section at 36°N, from about 75°W to 10°W. A front is associated with the western boundary current and its departure from the coast near 40°N. In the upper northwestern region of subtropical thermocline there is a region of low stratification known as MODE water: isopycnals above this outcrop in the subtropical gyre and are 'ventilated'; isopycnals below the MODE water outcrop in the subpolar gyre or ACC.[1]

measured; it is an eddy diffusivity, maintained by small-scale turbulence, and measurements produce values that range between $10^{-5}\,\mathrm{m^2\,s^{-1}}$ in the main thermocline to $10^{-4}\,\mathrm{m^2\,s^{-1}}$ in abyssal regions over rough topography and in and near continental margins, with still higher values locally.[2] The vertical velocity is too small to be measured directly, but various estimates based on deep water production suggest a value of about $10^{-7}\,\mathrm{m\,s^{-1}}$. Using this and the smaller value of κ in (16.2) gives an e-folding vertical scale, κ/w, of just 100 m, beneath which the stratification is predicted to very small (i.e., nearly uniform potential density). Using the larger value of κ increases the vertical scale to 1000 m, which is probably closer to the observed value for the total thickness of the thermocline (look at Fig. 15.2), but using such a large value of κ in the main thermocline is probably not supported by the observations. Similarly, the deep stratification of the ocean is rather larger than that given (16.1), except with values of diffusivity on the large side of those observed.[3] Thus, there are two conclusions to be drawn:

(i) The observed thickness of the thermocline is somewhat larger than what one might infer from observed values of the diffusivity and overturning circulation.

(ii) The observed deep stratification is somewhat larger than what one might infer from the advective-diffusive balance (16.1) with observed values of diffusivity and overturning circulation.

Of course the model itself, (16.1), is overly simple but these conclusions suggest that additional physical factors may play a role in thermocline dynamics. Mechanical forcing, and in particular the wind, is one such: the wind stress curl forces water to converge in the subtropical Ekman layer, thereby forcing relatively warm water to downwell and meet the upwelling colder abyssal water at some finite depth, thus deepening the thermocline from its purely diffusive value. Indeed, in so far as we can separate the two effects of wind and diffusion, we can say that the strength of the wind influences the *depth* at which the thermocline occurs, whereas the strength of the diffusivity influences the *thickness* of the thermocline. The influence of the wind on the abyssal circulation is not quite as direct, but we find in section 16.5 that it will enable both a stronger circulation, and deep stratification, to persist even in the absence of diffusion.

16.2 SCALING AND SIMPLE DYNAMICS OF THE MAIN THERMOCLINE

We now begin to consider the dynamics that produce an overturning circulation and a thermocline. The Rossby number of the large-scale circulation is small and the scale of the motion large, and the flow obeys the planetary geostrophic equations:

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi, \qquad \frac{\partial \phi}{\partial z} = b, \tag{16.3a,b}$$

$$\nabla \cdot \boldsymbol{v} = 0, \qquad \frac{\mathrm{D}b}{\mathrm{D}t} = \kappa \frac{\partial^2 b}{\partial z^2}. \tag{16.4a,b}$$

We suppose that these equations hold below an Ekman layer, so that the effects of a wind stress may be included by specifiying a vertical velocity at the top of the domain. The diffusivity, κ, is, as we noted above, an eddy diffusivity, but since its precise form and magnitude are uncertain we must proceed with due caution, and a useful practical philosophy is to try to ignore dissipation and viscosity where possible, and to invoke them only if there is no other way out. Let us therefore scale the equations in two ways, with and without diffusion; these scalings will be the central to our theory.

16.2.1 An advective scale

As usual we denote (with one or two exceptions) scaling values with capital letter and non-dimensional values with a hat, so that for example $u = U\hat{u}$ and $u = \mathcal{O}(U)$. Let us ignore the diffusive term in (16.4b) and try to construct a scaling estimate for the depth of the wind's influence.

If there is upwelling ($w > 0$) from the abyss, and Ekman downwelling ($w < 0$) at the surface, there is some depth D_a at which $w = 0$. By cross-differentiating (16.3a) we obtain $\beta v = -f\nabla_z \cdot \boldsymbol{u}$, and combining this with (16.4a) gives the familiar geostrophic vorticity equation and corresponding scaling

$$\beta v = f\frac{\partial w}{\partial z}, \qquad \rightarrow \qquad \beta V = f\frac{W}{D_a}. \tag{16.5}$$

Here, D_a is the unknown depth scale of the motion, L is the horizontal scale of the motion, which we take as the gyre or basin scale, and V is a horizontal velocity scale. (It is reasonable to suppose that $V \sim U$, where U is the zonal velocity scale, and henceforth we will denote both by U.) The appropriate vertical velocity to use is that due to Ekman pumping, W_E; we will assume (and demonstrate later) that this is much larger than the abyssal upwelling velocity, which in any case is zero by assumption at $z = -D_a$, and this leads to the Sverdrup-balance estimate

$$U = \frac{f}{\beta}\frac{W_E}{D_a}. \tag{16.6}$$

We may determine an appropriate value of U using the thermal wind relation, which from (16.3) is

$$\boldsymbol{f} \times \frac{\partial \boldsymbol{u}}{\partial z} = -\nabla b, \qquad \rightarrow \qquad \frac{U}{D_a} = \frac{1}{f}\frac{\Delta b}{L}, \tag{16.7}$$

where Δb is the magnitude of variations (i.e., the scaling value) of buoyancy in the horizontal. Assuming the vertical scales are the same in (16.6) and (16.7) then eliminating U gives

$$\boxed{D_a = W_E^{1/2}\left(\frac{f^2 L}{\beta\,\Delta b}\right)^{1/2}}. \tag{16.8}$$

[This is in fact essentially the same as the estimate (14.130).] If we relate U and W_E using mass conservation, $U/L = W_E/D_a$, instead of using Sverdrup balance, then we write L in place of f/β and (16.8) becomes $D_a = \left(W_E f L^2/\Delta b\right)^{1/2}$, which is not qualitatively different for large scales. The important aspect of these equations is that the depth of the wind-influenced region increases with the magnitude of the wind-stress (because $W_E \propto \mathrm{curl}_z \boldsymbol{\tau}$) and decreases with the meridional temperature gradient. The former dependence is reasonably intuitive, and the latter arises because as the temperature gradient increases the associated thermal wind-shear U/D_a correspondingly increases. But the horizontal transport (the product UD_a) is fixed by mass conservation; the only way that these two can remain consistent is for the vertical scale to decrease. Taking $W_E = 10^{-6}\,\mathrm{m\,s^{-1}}$, $\Delta b = g\Delta\rho/\rho_0 = g\beta_T\Delta T \sim 10^{-2}\,\mathrm{m\,s^{-2}}$, $L = 5000\,\mathrm{km}$ and $f = 10^{-4}\,\mathrm{s^{-1}}$ gives $D_a = 500\,\mathrm{m}$. Such a scaling argument cannot be expected to give more than an estimate of depth of the wind-influenced region; nevertheless, it does indicate that the wind-driven circulation is predominantly an upper-ocean phenomenon.

16.2.2 A diffusive scale

The estimate (16.8) cares nothing about the thermodynamic equation; if do we take into account thermodynamics, with nonzero diffusivity, we recover the model of section 15.8. Thus, briefly, the scaling follows from advective-diffusive balance in the thermodynamic equation, the linear geostrophic vorticity equation, and thermal wind balance:

$$w\frac{\partial b}{\partial z} = \kappa\frac{\partial^2 b}{\partial z^2}, \qquad \beta v = f\frac{\partial w}{\partial z}, \qquad f\frac{\partial \boldsymbol{u}}{\partial z} = \mathbf{k}\times\nabla b, \tag{16.9a,b,c}$$

with corresponding scales

$$\frac{W}{\delta} = \frac{\kappa}{\delta^2}, \qquad \beta U = \frac{fW}{\delta}, \qquad \frac{U}{\delta} = \frac{\Delta b}{fL}. \tag{16.10a,b,c}$$

where δ is the vertical scale. Because there is now one more equation than in the advective scaling theory we cannot take the vertical velocity as a given, else the equations are overdetermined. We therefore take it to be the abyssal upwelling velocity, which then becomes part of the *solution,* rather than being imposed. From (16.10) we obtain the diffusive vertical scale,

$$\boxed{\delta = \left(\frac{\kappa f^2 L}{\beta\,\Delta b}\right)^{1/3}}. \tag{16.11}$$

With $\kappa = 10^{-5}\,\mathrm{m^2\,s^{-2}}$ and with the other parameters taking the values given following (16.8), (16.11) gives $\delta \approx 150\,\mathrm{m}$ and, using (16.10a), $W \approx 10^{-7}\,\mathrm{m\,s^{-1}}$, an order of magnitude smaller than the Ekman pumping velocity W_E.

A modified diffusive scaling

The scaling above assumes that the length scale over which thermal wind balance holds is the gyre scale itself. In fact there is another length scale that is more appropriate, and this leads to a slightly different scaling for the thickness of the thermocline. To obtain this scaling, we first note that the depth of the subtropical thermocline is not constant: it shoals up to the east because of Sverdrup balance, and it may shoal up polewards as the curl of the wind stress falls (and is zero at the polewards edge of the gyre). Thus, referring to Fig. 16.3, the appropriate horizontal length scale $\widetilde{L}$ is given by

$$\widetilde{L} = \delta\frac{L}{D_a}. \tag{16.12}$$

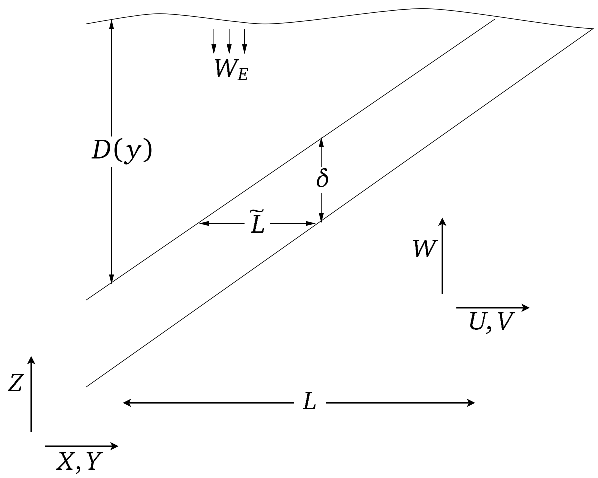


Figure 16.3 Scaling the thermocline. The diagonal lines mark the diffusive thermocline of thickness δ and depth $D(y)$. The advective scaling for $D(y)$, i.e., D_a, is given by (16.8), and the diffusive scaling for δ is given by (16.13).

This is no longer an externally imposed parameter, but must be determined as part of the solution. Using $\widetilde{L}$ instead of L as the length scale in the thermal wind equation (16.10c) gives, using (16.8), the modified diffusive scale

$$\boxed{\delta = \kappa^{1/2}\left(\frac{f^2 L}{\Delta b\,\beta D_a}\right)^{1/2} = \kappa^{1/2}\left(\frac{f^2 L}{\Delta b\,\beta W_E}\right)^{1/4}}. \tag{16.13}$$

Substituting values of the various parameters results in a thickness of about 100–200 m. The thermocline thickness now scales as $\kappa^{1/2}$. The interpretation of this scale and that of (16.11) is that the thickness of the thermocline scales as $\kappa^{1/3}$ in the absence of a wind stress, but scales as $\kappa^{1/2}$ if a wind stress is present that can provide a finite slope the the depth of the thermocline that is independent of κ, and this is confirmed by numerical simulations.[4] From (16.9a) the vertical velocity, and hence the meridional overturning circulation, no longer scales as $\kappa^{2/3}$ but as

$$W = \frac{\kappa}{\delta} \propto \kappa^{1/2}. \tag{16.14}$$

16.2.3 Summary of the physical picture

What do the vertical scales derived above represent? The wind-influenced scaling, D_a, is the depth to which the directly wind-driven circulation can be expected to penetrate. Thus, over this depth we can expect to see wind-driven gyres and associated phenomena. At greater depths lies the abyssal circulation, and this is not wind-driven in the same sense. Now, in general, the water at the base of the wind-driven layer will not have the same thermodynamic properties as the upwelling abyssal water — this being cold and dense, whereas the water in the wind-driven layer is warm and subtropical (look again at Fig. 16.1). The thickness δ characterizes the diffusive transition region between these two water masses and in the limit of very small diffusivity this becomes a *front*. One might say that D_a is the *depth* of the thermocline, while δ

is the *thickness* of the thermocline. In the diffusive region, no matter how small the diffusivity κ is in the thermodynamic equation, the diffusive term is important. Of course if the diffusion is sufficiently large, the thickness will be as large or larger than the depth, and the two regions will blur into each other, and this may indeed be the case in the real ocean. (The real world is also complicated by the effects of mesoscale eddies.) Nevertheless, these scales are a useful foundation on which to build.[5]

16.3 THE INTERNAL THERMOCLINE

We now try to go beyond simple scaling arguments and investigate in more detail the dynamics of the thermocline. In this section we consider the diffusive, or internal, thermocline and in section 16.4 we consider the advective, or ventilated, thermocline. Such an investigation requires trying to actually solve the equations of motion, but the advective term in the thermodynamic equation makes this extremely difficult; indeed this prevents us from constructing wholly analytic models, but not from constructing informative models. We begin by expressing the planetary geostrophic equations as an equation in a single unknown.

16.3.1 The M equation

The planetary geostrophic equations can be written as a single partial differential equation in a single variable, although the resulting equation is of quite high order and nonlinear. We write the equations of motion as

$$-fv = -\frac{\partial \phi}{\partial x}, \qquad fu = -\frac{\partial \phi}{\partial y}, \qquad b = \frac{\partial \phi}{\partial z}, \tag{16.15a,b,c}$$

$$\nabla \cdot \boldsymbol{v} = 0, \qquad \frac{\partial b}{\partial t} + \boldsymbol{v} \cdot \nabla b = \kappa \nabla^2 b, \tag{16.16a,b}$$

where we take $f = \beta y$. Cross differentiating the horizontal momentum equations and using (16.16a) gives the linear geostrophic vorticity relation $\beta v = f \partial w / \partial z$ which, using (16.15a) again, may be written as

$$\frac{\partial \phi}{\partial x} + \frac{\partial}{\partial z}\left(-\frac{f^2}{\beta} w\right) = 0. \tag{16.17}$$

This equation is the divergence in (x, z) of $(\phi, -f^2 w/\beta)$ and is automatically satisfied if

$$\phi = M_z \qquad \text{and} \qquad \frac{f^2 w}{\beta} = M_x. \tag{16.18a,b}$$

where the subscripts on M denote derivatives. Then straightforwardly

$$u = -\frac{\partial_y \phi}{f} = -\frac{M_{zy}}{f}, \qquad v = \frac{\partial_x \phi}{f} = \frac{M_{zx}}{f}, \qquad b = \partial_z \phi = M_{zz}. \tag{16.19a,b,c}$$

The thermodynamic equation, (16.16b) becomes

$$\frac{\partial M_{zz}}{\partial t} + \left(\frac{-M_{zy}}{f} M_{zzx} + \frac{M_{zx}}{f} M_{zzy}\right) + \frac{\beta}{f^2} M_x M_{zzz} = \kappa M_{zzzz} \tag{16.20}$$

or

$$\frac{\partial M_{zz}}{\partial t} + \frac{1}{f} J(M_z, M_{zz}) + \frac{\beta}{f^2} M_x M_{zzz} = \kappa M_{zzzz}. \tag{16.21}$$

where J is the usual horizontal Jacobian. This is the *M-equation,*[6] somewhat analogous to the potential vorticity equation in quasi-geostrophic theory in that it expresses the entire dynamics of the system in a single, nonlinear, advective-diffusive partial differential equation, although note that M_{zz} is materially conserved (in the absence of diabatic effects) by the three-dimensional flow. Because of the high differential order and nonlinearity of the system analytic solutions of (16.21) are very hard to find, and from a numerical perspective it is easier to integrate the equations in the form (16.15) and (16.16) than in the form (16.21). Nevertheless, it is possible to move forward by approximating the equation to one or two dimensions, or by *a priori* assuming a boundary-layer structure.

A one-dimensional model

Let us consider an illustrative one-dimensional model (in z) of the thermocline.[7] Merely setting all horizontal derivatives in (16.21) to zero is not very useful, for then all the advective terms on the left-hand side vanish. Rather, we look for steady solutions of the form $M = M(x, z)$, and the M-equation then becomes

$$\frac{\beta}{f^2} M_x M_{zzz} = \kappa M_{zzzz}, \tag{16.22}$$

which represents the advective-diffusive balance

$$w \frac{\partial b}{\partial z} = \kappa \frac{\partial^2 b}{\partial z^2}. \tag{16.23}$$

(We must also suppose that the value of κ varies meridionally in the same manner as does β/f^2; without this technicality M would be a function of y, violating our premise.) If the ocean surface is warm and the abyss is cold, then (16.22) represents a balance between the upward advection of cold water and the downward diffusion of warm water. The horizontal advection terms vanish because the zonal velocity, u, and the meridional bouyancy gradient, b_y, are each zero. Let us further consider the special case

$$M = (x - x_e) W(z) \tag{16.24}$$

where the domain extends from $0 \le x \le x_e$, so satisfying $M = 0$ on the eastern boundary. Equation (16.22) becomes the ordinary differential equation

$$\frac{\beta}{f^2} W W_{zzz} = \kappa W_{zzzz}, \tag{16.25}$$

where W has the dimensions of velocity squared. We nondimensionalize this by setting

$$z = H\hat{z}, \qquad \kappa = \hat{\kappa}(HW_S), \qquad W = \left(\frac{f^2 W_S}{\beta}\right) \widehat{W} \tag{16.26a,b,c}$$

where the hatted variables are nondimensional and W_S is a scaling value of the dimensional vertical velocity, w (e.g., the magnitude of the Ekman pumping velocity W_E). Eq. (16.25) becomes

$$\widehat{W} \widehat{W}_{\hat{z}\hat{z}\hat{z}} = \hat{\kappa} \widehat{W}_{\hat{z}\hat{z}\hat{z}\hat{z}}, \tag{16.27}$$

The parameter $\hat{\kappa}$ is a nondimensional measure of the strength of diffusion in the interior, and the interesting case occurs when $\hat{\kappa} \ll 1$; in the ocean, typical values are $H = 1\,\mathrm{km}$, $\kappa = 10^{-5}\,\mathrm{m\,s^{-2}}$ and $W_S = W_E = 10^{-6}\,\mathrm{m\,s^{-1}}$ so that $\hat{\kappa} \approx 10^{-2}$, which is indeed small.

The time-dependent form of (16.27), namely $\widehat{W}_{\hat{z}\hat{z}t} + \widehat{W}\widehat{W}_{\hat{z}\hat{z}\hat{z}} = \hat{\kappa}\widehat{W}_{\hat{z}\hat{z}\hat{z}\hat{z}}$, is similar to Burger's equation, $V_t + VV_z = \nu V_{zz}$, which is known to develop fronts. (The inviscid Burger's equation has the form $\mathrm{D}V/\mathrm{D}t = 0$, where the advective derivative is one-dimensional, and therefore the velocity of a given fluid parcel is preserved on the line. Suppose that the velocity of the fluid is positive but diminishes in the positive z-direction, so that a fluid parcel will catch-up with the fluid parcel in front of it — there is no pressure force to keep the fluid parcels apart. But since the velocity of a fluid parcel is fixed, then a singularity must form. In presence of viscosity, the singularity is tamed to a front.) Thus, we might similarly expect (16.27) to produce a front, but because of the extra derivatives the argument is not as straightforward and it is simplest to obtain solutions numerically.

Equation (16.27) is fourth order, so four boundary conditions are needed, two at each boundary. Appropriate ones are a prescribed buoyancy and a prescribed vertical velocity at each boundary, for example

$$\begin{aligned} &\widehat{W} = \widehat{W}_E, \quad -\widehat{W}_{\hat{z}\hat{z}} = B_0, &&\text{at top} \\ &\widehat{W} = 0, \quad\ \ -\widehat{W}_{\hat{z}\hat{z}} = 0, &&\text{at bottom} \end{aligned} \tag{16.28}$$

where $\widehat{W}_E$ is the (nondimensional) vertical velocity at the base of the top Ekman layer, which is negative for Ekman pumping in the subtropical gyre, and B_0 is a constant, proportional to the buoyancy difference across the domain. We obtain solutions numerically by Newton's method,[8] and these are shown in Fig. 16.4 and Fig. 16.5. The solutions do indeed display fronts, or boundary layers, for small diffusivity. If the wind forcing is zero (Fig. 16.5), the boundary layer is at the top of the fluid. If the wind forcing is non-zero, an internal boundary layer — a front — forms in the fluid interior with an adiabatic layer above and below. In the real ocean, where wind forcing is of course non-zero, the frontal region is known as the *internal thermocline.*

16.3.2 * Boundary layer analysis

The reasoning and the numerical solutions of the above sections all suggest that the internal thermocline has a boundary layer structure whose thickness decreases with κ. If the Ekman pumping at the top of the ocean is non-zero, the boundary layer is internal to the fluid. This suggests that we might be able to learn something about the solution by performing a boundary layer analysis, much as we did when investigating western boundary currents in section 14.1.3. The nonlinearity precludes a complete solution of the equation, but we can obtain useful information about the thickness of the boundary layer itself.

One-dimensional model

Let us now *assume* a steady two-layer structure of the form illustrated in Fig. 16.6, and that the dynamics are governed by (16.27) in a domain that extends from 0 to -1. The buoyancy thus varies rapidly only in an internal boundary layer of nondimensional

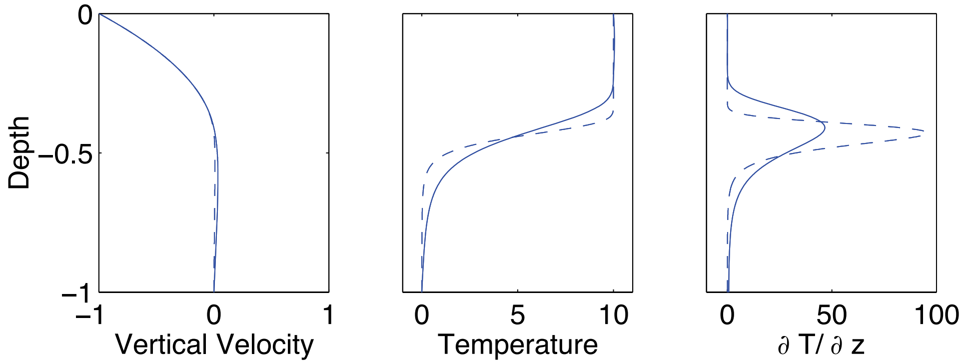


Fig. 16.4 Solution of the one-dimensional thermocline equation, (16.27), with boundary conditions (16.28), for two different values of the diffusivity: $\hat{\kappa} = 3.2 \times 10^{-3}$ (solid line) and $\hat{\kappa} = 0.4 \times 10^{-3}$ (dashed line), in the domain $0 \le \hat{z} \le -1$. 'Vertical velocity' is W, 'temperature' is $-W_{\hat{z}\hat{z}}$, and all units are the nondimensional ones of the equation itself. A negative vertical velocity, $\widehat{W}_E = -1$, is imposed at the surface (representing Ekman pumping) and $T_0 = 10$. The internal boundary layer thickness increases as $\hat{\kappa}^{1/3}$, so doubling in thickness for an eightfold increase in $\hat{\kappa}$. The upwelling velocity also increases with $\hat{\kappa}$, (as $\hat{\kappa}^{2/3}$), but this is barely noticeable on the graph because the downwelling velocity, above the internal boundary layer, is much larger and almost independent of $\hat{\kappa}$. The depth of the boundary layer increases as $\widehat{W}_E^{1/2}$, so if $\widehat{W}_E = 0$ the boundary layer is at the surface, as in Fig. 16.5.

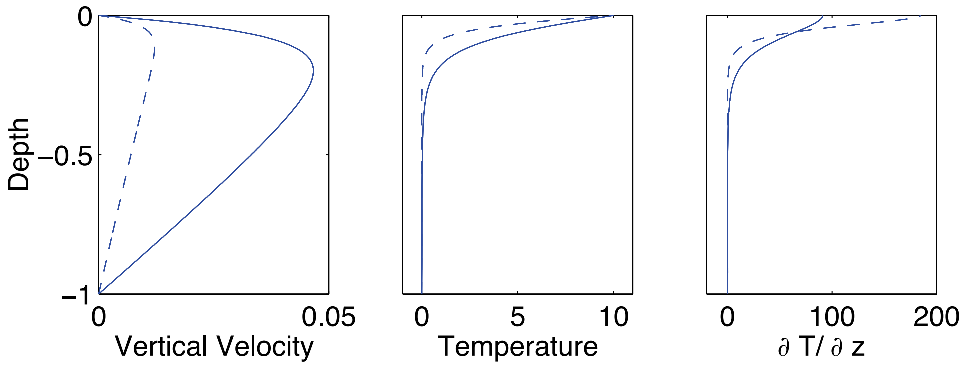


Fig. 16.5 As for Fig. 16.4, but with no imposed Ekman pumping velocity at the upper boundary ($\widehat{W}_E = 0$), again for two different values of the diffusivity: $\hat{\kappa} = 3.2 \times 10^{-3}$ (solid line) and $\hat{\kappa} = 0.4 \times 10^{-3}$ (dashed line). The boundary layer now forms at the upper surface. The boundary thickness again increases with diffusivity and, even more noticeably, so does the upwelling velocity — this scales as $\hat{\kappa}^{2/3}$, and so increases fourfold for an eightfold increase in $\hat{\kappa}$.

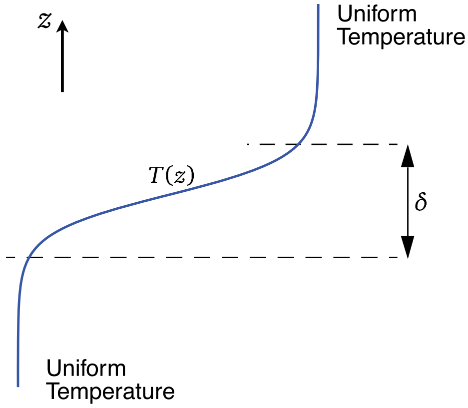


Figure 16.6 The simplified boundary-layer structure of the internal thermocline. In the limit of small diffusivity the internal thermocline forms a boundary layer, of thickness δ in the figure, in which the temperature and buoyancy change rapidly.

thickness $\widehat{\delta}$ located at $\hat{z} = -h$; above and below this the bouyancy is assumed to be only very slowly varying. Following standard boundary layer procedure, we introduce a stretched boundary layer coordinate ζ, where

$$\widehat{\delta}\zeta = \hat{z} + h. \tag{16.29}$$

That is, ζ is the distance from $\hat{z} = -h$, scaled by the boundary layer thickness $\widehat{\delta}$, and within the boundary layer ζ is an order one quantity. We also let

$$\widehat{W}(\hat{z}) = \widehat{W}_I(\hat{z}) + \widetilde{W}(\zeta) \tag{16.30}$$

where $\widehat{W}_I$ is the solution away from the boundary layer and $\widetilde{W}$ is the boundary layer correction. Because the boundary layer is presumptively thin, $\widehat{W}_I$ is effectively constant through it and, furthermore, for $\hat{z} < -h$, $\widehat{W}$ vanishes in the limit as $\kappa = 0$. We thus take $\widehat{W}_I = 0$ throughout the boundary layer. (The small diffusively-driven upwelling below the boundary layer is part of the boundary layer solution, not the interior solution.) Now, bouyancy varies rapidly in the boundary layer but it remains an order one quantity throughout. To satisfy this we explicitly scale $\widetilde{W}$ in the boundary layer by writing

$$\widetilde{W}(\zeta) = \widehat{\delta}^2 B_0 A(\zeta) \tag{16.31}$$

where B_0 is defined by (16.28) and A is an order one field. The derivatives of W are

$$\begin{aligned}\frac{\partial \widehat{W}}{\partial \hat{z}} &= \frac{1}{\widehat{\delta}}\frac{\partial \widetilde{W}}{\partial \zeta} = \widehat{\delta} B_0 \frac{\partial A}{\partial \zeta}\\ \frac{\partial^2 \widehat{W}}{\partial \hat{z}^2} &= B_0 \frac{\partial^2 A}{\partial \zeta^2},\end{aligned} \tag{16.32}$$

so that $\widehat{W}_{\hat{z}\hat{z}}$ is an order one quantity. Far from the boundary layer the solution must be able to match the external conditions on temperature and velocity, (16.28); the buoyancy condition on $W_{\hat{z}\hat{z}}$ is satisfied if

$$A_{\zeta\zeta} \to \begin{cases} 1 & \text{as } \zeta \to +\infty \\ 0 & \text{as } \zeta \to -\infty. \end{cases} \tag{16.33}$$

On vertical velocity we require that $W \to (\hat{z}/h + 1)W_E$ as $\zeta \to +\infty$, and $W \to$ constant as $\zeta \to -\infty$. The first matches the Ekman pumping velocity above the boundary layer, and the second condition produces the abyssal upwelling velocity, which as noted vanishes for $\kappa \to 0$.

Substituting (16.30) and (16.31) into (16.27) we obtain

$$B_0 A A_{\zeta\zeta\zeta} = \frac{\hat{\kappa}}{\hat{\delta}^3} A_{\zeta\zeta\zeta\zeta}. \tag{16.34}$$

Because all quantities are presumptively $\mathcal{O}(1)$, (16.34) implies that $\hat{\delta} \sim (\hat{\kappa}/B_0)^{1/3}$. We restore the dimensions of δ by using $\kappa = \hat{\kappa}(HW_S)$ and $\Delta b = B_0 L f^2 W_S/(\beta H^2)$, where Δb is the dimensional buoyancy difference across the boundary layer [note that $b = M_{zz} = (x-1)W_{zz} \sim LW_{zz} \sim LB_0 f^2 W_S/(\beta H^2)$ using (16.26)]. The dimensional boundary layer thickness, δ, is then given by

$$\delta \sim \left(\frac{\kappa f^2 L}{\Delta b\, \beta}\right)^{1/3}, \tag{16.35}$$

which is the same as the heuristic estimate (16.11). The dimensional vertical velocity scales as

$$W \sim \frac{\kappa}{\delta} \sim \kappa^{2/3}\left(\frac{\Delta b\, \beta}{\kappa k^2 L}\right)^{1/3}, \tag{16.36}$$

this being estimate of strength of the upwelling velocity at base of the thermocline and, more generally, the strength of the diffusively-driven component of meridional overturning circulation of the ocean.

Somewhat different one-dimensional thermocline models may be constructed (see problem 16.3), and these have slightly different scaling properties. However, their qualitative features transcend their detailed construction, and in particular:

- ⋆ The thickness of the internal thermocline increases with increasing diffusivity, and decreases with increasing temperature difference across it. In particular, as the diffusivity tends to zero the thickness of the internal thermocline tends to zero.
- ⋆ The strength of the upwelling velocity, and hence the strength of the meridional overturning circulation, increases with increasing diffusivity and increasing temperature difference.

* *The three-dimensional equations*

We now apply the same boundary-layer techniques to the three-dimensional M-equation.[9] The main difference is that the depth of the boundary layer is now a function of x and y, so that the stretched coordinate ζ is given by

$$\hat{\delta}\zeta = z + h(x, y). \tag{16.37}$$

(The coordinates (x, y, z) in this subsection are nondimensional, but we omit their hats to avoid too cluttered a notation.) Just as in the one-dimensional case we rescale M in the boundary layer and write

$$M = B_0 \hat{\delta}^2 \hat{A}(x, y, \zeta), \tag{16.38}$$

where the scaling factor $\hat{\delta}^2$ again ensures that the temperature remains an order-one quantity. In the boundary layer the derivatives of M become

$$\frac{\partial M}{\partial z} = \frac{1}{\hat{\delta}}\frac{\partial A}{\partial \zeta}, \tag{16.39}$$

and

$$\frac{\partial M}{\partial x} = \hat{\delta}^2 B_0 \left(\frac{\partial A}{\partial \zeta}\frac{\partial \zeta}{\partial x} + \frac{\partial A}{\partial x}\right) = \hat{\delta}^2 B_0 \left(\frac{\partial A}{\partial \zeta}\left(\frac{1}{\hat{\delta}}\frac{\partial h}{\partial x}\right) + \frac{\partial A}{\partial x}\right) \tag{16.40}$$

Substituting these into (16.20) we obtain, omitting the time-derivative,

$$\begin{aligned} \hat{\delta}\left[\frac{1}{f}\left(A_{\zeta x}A_{\zeta\zeta y} - A_{\zeta y}A_{\zeta\zeta x}\right) + \frac{\beta}{f^2}A_x A_{\zeta\zeta\zeta}\right] &+ \frac{\beta}{f^2}h_x A_\zeta A_{\zeta\zeta\zeta} \\ + \frac{1}{f}\Big[h_x\left(A_{\zeta\zeta}A_{\zeta\zeta y} - A_{\zeta y}A_{\zeta\zeta\zeta}\right) &+ h_y\left(A_{\zeta x}A_{\zeta\zeta\zeta} - A_{\zeta\zeta}A_{\zeta\zeta x}\right)\Big] \\ &= \frac{\kappa}{B_0\hat{\delta}^2}A_{\zeta\zeta\zeta\zeta}, \end{aligned} \tag{16.41}$$

where the subscripts on A and h denote derivatives. If $h_x = h_y = 0$, that is if the base of the thermocline is flat, then (16.41) becomes

$$\frac{1}{f}\left[A_{\zeta x}A_{\zeta\zeta y} - A_{\zeta y}A_{\zeta\zeta x}\right] + \frac{\beta}{f^2}A_x A_{\zeta\zeta\zeta} = \frac{\kappa}{B_0\hat{\delta}^3}A_{\zeta\zeta\zeta\zeta}. \tag{16.42}$$

Since all the terms in this equation are, by construction, order one, we immediately see that the nondimensional boundary layer thickness $\hat{\delta}$ scales as

$$\hat{\delta} \sim \left(\frac{\kappa}{B_0}\right)^{1/3}, \tag{16.43}$$

just as in the one-dimensional model. On the other hand, if h_x and h_y are order one quantities then the dominant balance in (16.41) is

$$\frac{1}{f}\left[h_x(A_{\zeta\zeta}A_{\zeta\zeta y} - A_{\zeta y}A_{\zeta\zeta\zeta}) + h_y(A_{\zeta x}A_{\zeta\zeta\zeta} - A_{\zeta\zeta}A_{\zeta\zeta x})\right] = \frac{\kappa}{B_0\hat{\delta}^2}A_{\zeta\zeta\zeta\zeta} \tag{16.44}$$

and

$$\hat{\delta} \sim \left(\frac{\kappa}{B_0}\right)^{1/2}, \tag{16.45}$$

confirming the heuristic scaling arguments. Thus, if the isotherm slopes are fixed independently of κ (for example by the wind stress), then as $\kappa \to 0$ an internal boundary layer will form whose thickness is proportional to $\kappa^{1/2}$. We expect this to occur at the base of the main thermocline, with purely advective dynamics being dominant in upper part of the thermocline, and determining the slope of the isotherms (i.e., the form of h_x and h_y), as in Fig. 16.3. Interestingly, the balance in the three-dimensional boundary layer equation does not in general locally correspond to $wT_z \approx \kappa T_{zz}$. Both at $\mathcal{O}(1)$ and $\mathcal{O}(\delta)$ the horizontal advective terms in (16.41) are of the same asymptotic size as the vertical advection terms. In the boundary layer the thermodynamic balance is thus $\boldsymbol{u} \cdot \nabla_z T + wT_z \approx \kappa T_{zz}$, whether the isotherms are sloping or flat. We might have anticipated this, because the vertical velocity passes through zero within the boundary layer.

What are the dynamics above the diffusive layer, presuming that it does does not extend all the way to the surface? Answering this leads us into our next topic, the 'ventilated thermocline.'

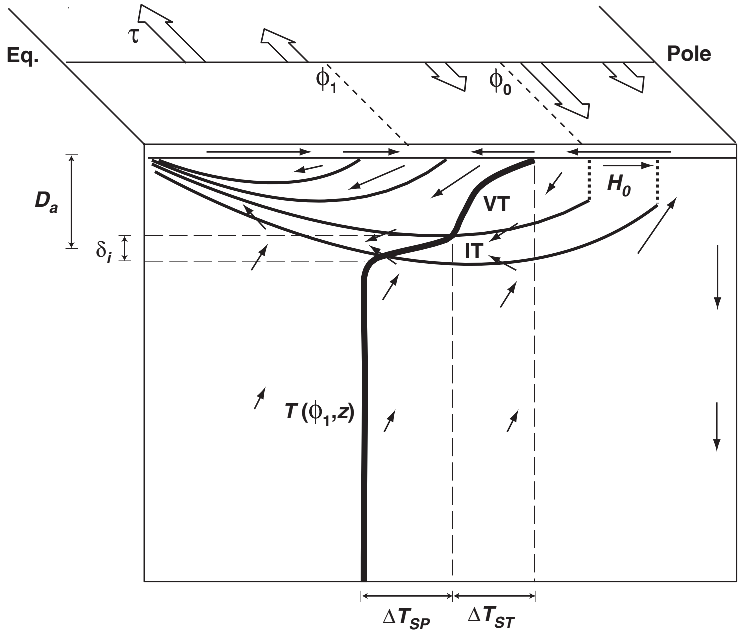


Fig. 16.7 Schema of the large-scale circulation and structure of the main thermocline, in a single-hemisphere ocean driven by wind-stress (broad arrows) and a meridional gradient of heating at the surface. The subtropical-subpolar gyre boundary is a constant latitude ϕ_0, where the wind-stress curl changes sign. 'VT' denotes the ventilated thermocline, an advective regime of thickness D_a, and 'IT' denotes the internal thermocline, a diffusive internal boundary layer of thickness δ_I. The thin arrows indicate the meridional overturning circulation and the flow in the Ekman layer near the ocean surface. The thick line is a temperature profile at latitude ϕ_1: The temperature drop across the internal thermocline is ΔT_{SP}, equal to the meridional temperature difference across the subpolar gyre; the temperature drop across the ventilated thermocline is ΔT_{ST}, the temperature difference across the subtropical gyre.[10]

16.4 THE VENTILATED THERMOCLINE

We now consider the nature of the dynamics *above* the diffusive layer, presuming that the diffusivity is sufficiently small that there is a meaningful separation of the internal boundary layer and the advective dynamics above. In this advective region there is no general reason that the temperature profile should be uniform, and we envision an essentially adiabatic region that is both wind-driven and stratified. This region of the thermocline has become known, for reasons that will become apparent, as the *ventilated thermocline.* The main thermocline is comprised of the internal thermocline plus a ventilated region, and to set our bearings it may be useful to refer now to the overall picture sketched in Fig. 16.7, and to come back to it again later on. To elucidate the structure of the ventilated thermocline we will assume:[11]

(i) The motion satisfies the ideal, steady, planetary geostrophic equations.

(ii) The surface temperature, and the vertical velocity due to Ekman pumping, are given. (These surface conditions are, in reality, influenced by the ocean's dy-

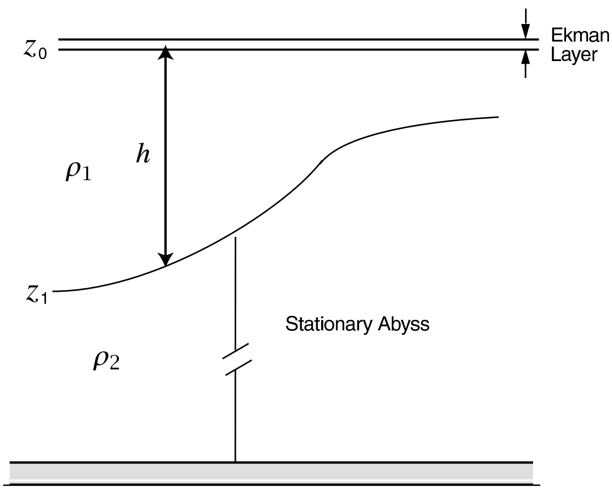


Figure 16.8 A reduced gravity, single layer model. A single moving layer lies above a deep, stationary layer of higher density. The upper surface is rigid. A thin Ekman layer may be envisioned to lie on top of the moving layer, providing a vertical velocity boundary condition.

namics, but we assume that we can calculate a solution with specified surface conditions.) At the base of the wind-influenced region we will impose $w = 0$.

(iii) Rather than use the continuously stratified equations, we will assume the solution can be adequately represented by a small number of layers, each of constant density. The abyss is represented by a single stationary layer.

(iv) We will not take into account the possible effects of a western boundary current. In that sense the model is an extension of the Sverdrup interior of homogeneous models.

The model is thus not a complete one, yet we may hope that it is revealing about the structure of the real ocean.

16.4.1 A reduced gravity, single-layer model

The simplest possible model along these lines is to suppose the ocean is composed of just two layers, as illustrated in Fig. 16.8. The upper layer of density ρ_1 is wind-driven, whereas the lower layer of density ρ_2 is assumed stationary; this is called a 'one-and-a-half-layer' model or a 'reduced gravity single-layer' model. Pertinent questions are: *(i)* How deep is the upper layer? *(ii)* What is the velocity field in it?

In the planetary geostrophic approximation, the momentum and mass conservation equations of the reduced gravity shallow water model may be written as:

$$\boldsymbol{f} \times \boldsymbol{u} = -g'\nabla h, \qquad \nabla \cdot \boldsymbol{u} = -\frac{\partial w}{\partial z}, \tag{16.46a,b}$$

where ∇ is a two dimensional operator (as it will be for the rest of this section) and $g' = g(\rho_2 - \rho_1)/\rho_0$ is the *reduced gravity*. Taking the curl of (16.46a) gives the geostrophic vorticity equation, $\beta v + f\nabla \cdot \boldsymbol{u} = 0$, and integrating this over the depth of the layer and using mass conservation gives

$$h\beta v = f(w_E - w_b), \tag{16.47}$$

where w_E is the velocity at the top of the layer, due mainly to Ekman pumping, and w_b is the vertical velocity at the layer base. If the flow is steady, w_b is zero for then

$$w_b = \boldsymbol{u} \cdot \nabla h = -\frac{g'}{f}\frac{\partial h}{\partial y}\frac{\partial h}{\partial x} + \frac{g'}{f}\frac{\partial h}{\partial x}\frac{\partial h}{\partial y} = 0. \tag{16.48}$$

Using this result and geostrophic balance, (16.47) becomes

$$\frac{g'}{f}\beta h \frac{\partial h}{\partial x} = f w_E \tag{16.49}$$

which integrates to

$$\boxed{h^2 = -2\frac{f^2}{g'\beta}\int_x^{x_e} w_E \,\mathrm{d}x' + H_e^2}, \tag{16.50}$$

where H_e is the (unknown) value of h at the eastern boundary x_e, and this is a constant to satisfy the no-normal flow condition. The value of H apart, the equation contains complete information about the solution. We note:

- ⋆ The depth of the moving layer scales as the magnitude of the wind stress (or Ekman pumping velocity) to the one-half power.
- ⋆ The horizontal solution is very similar to the simpler Sverdrup interior solution previously obtained in section 14.1.3.
- ⋆ There is no solution if w_E is positive; that is, if there is Ekman upwelling.
- ⋆ The solution depends on the unknown parameter H_e, the layer depth at the eastern boundary.[12]

16.4.2 A two-layer model

If there is a meridional temperature gradient at the surface then isopycnals *outcrop*, or intersect the surface. Thus, at some latitude (say $y = y_2$, assumed not to be a function of longitude) layer 2 passes underneath layer 1, which is of lower density (and higher temperature), as sketched in Fig. 16.9. Thus, polewards of y_2 the dynamics are just those of a single layer discussed above, whereas equatorwards of y_2 layer 2 does not feel the wind directly, and its dynamics are are governed by two principles:

(i) Sverdrup balance. This still applies to the vertically integrated motion, and thus to the sum of layer 1 and layer 2.

(ii) Conservation of potential vorticity. The motion in layer 2 is shielded from the wind forcing, and the effects of dissipation are assumed negligible. Thus, the fluid parcels in the layer will conserve their potential vorticity.

These two principles are sufficient to provide a complete description of the dynamics. We first use potential vorticity conservation to obtain an expression for the depths of each layer in terms of the total depth of the moving fluid, h, and then use Sverdrup balance to obtain h.

Use of potential vorticity conservation

Conservation of potential vorticity in the region equatorwards of y_2 is, for steady flow,

$$\boldsymbol{u}_2 \cdot \nabla q_2 = 0 \qquad (y < y_2). \tag{16.51}$$

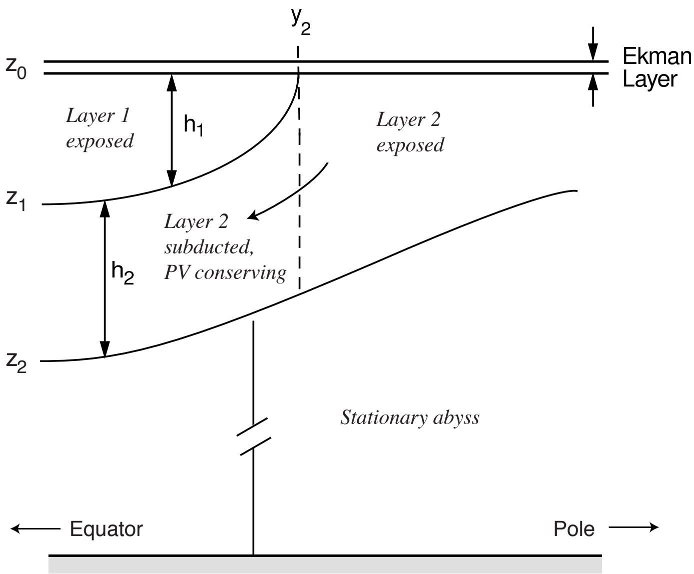


Figure 16.9 A two-layer model of the ventilated thermocline. Two moving layers lie above an infinitely deep, stationary layer of higher density. Models with more moving layers may be constructed by straightforward extension.

where $q_2 = f/h_2$. Now, the velocity field in layer 2 is given by $\boldsymbol{u}_2 = (g_2'/f)\mathbf{k} \times \nabla h$ where $h = h_1 + h_2$ is the total depth of the fluid (see the appendix to this chapter). Thus, (16.51) becomes

$$-\frac{g_2'}{f}\frac{\partial h}{\partial y}\frac{\partial}{\partial x}\left(\frac{f}{h_2}\right) + \frac{g_2'}{f}\frac{\partial h}{\partial x}\frac{\partial}{\partial y}\left(\frac{f}{h_2}\right) = \frac{g_2'}{f}J\left(\frac{f}{h_2}, h\right) = 0. \tag{16.52}$$

This is an equation relating h and h_2 and it has the general solution

$$q_2 = \frac{f}{h_2} = G_2(h) \tag{16.53}$$

where G_2 is an *arbitrary* function of its argument. However, we *know* what the potential vorticity of layer 2 is at the moment it is subducted; it is just

$$q_2(y_2) = \frac{f(y_2)}{h_2} = \frac{f_2}{h}, \tag{16.54}$$

where $f_2 \equiv f(y_2)$, and $h_2 = h$ because $h_1 = 0$. This relationship must therefore hold everywhere in layer 2, equatorwards of y_2. That is,

$$G_2(h) = \frac{f_2}{h} = \frac{f_2}{-z_2}. \tag{16.55}$$

Thus, in the subducted region,

$$\frac{f}{z_1 - z_2} = -\frac{f_2}{z_2} \tag{16.56}$$

whence

$$z_1 = \left(1 - \frac{f}{f_2}\right) z_2 = -\left(1 - \frac{f}{f_2}\right) h. \tag{16.57}$$

From this we obtain expressions for the layer depths as functions of the total depth, h, namely

$$h_2 = z_1 - z_2 = \frac{f}{f_2}h \quad \text{and} \quad h_1 = -z_1 = \left(1 - \frac{f}{f_2}\right)h. \tag{16.58}$$

Note that because potential vorticity, f/h_2, is conserved, as a fluid column moves equatorward its height must decrease.

Use of Sverdrup balance

Equations (16.58) contain the unknown total depth h, and we now use Sverdrup balance to find this and close the problem. Because the velocity at the base of layer 2 is zero, this may be written as

$$\beta(h_1 v_1 + h_2 v_2) = f w_E \tag{16.59}$$

where the velocities in each layer are given by [see (16.121)]

$$f v_1 = \frac{\partial}{\partial x}(g_2' h + g_1' h_1) \quad \text{and} \quad f v_2 = \frac{\partial}{\partial x}(g_2' h). \tag{16.60}$$

Thus the Sverdrup relationship becomes,

$$\beta h_1 \frac{\partial}{\partial x}(g_2' h + g_1' h_1) + \beta(h - h_1) g_2' \frac{\partial h}{\partial x} = f^2 w_E, \tag{16.61}$$

or

$$\frac{\partial}{\partial x}\left(g_2' h^2 + g_1' h_1^2\right) = \frac{2f^2}{\beta} w_E. \tag{16.62}$$

On integrating this gives

$$\left(h^2 + \frac{g_1'}{g_2'} h_1^2\right) = D_0^2 + C. \tag{16.63}$$

where

$$D_0^2(x, y) = -\frac{2f^2}{\beta g_2'} \int_x^{x_e} w_E(x', y)\, \mathrm{d}x' \tag{16.64}$$

which by construction vanishes at the eastern wall ($x = x_e$). The constant of integration C may be interpreted as follows. We write $C = H^2 + (g_1'/g_2')H_1^2$ where H is the (unknown) total depth of layers 1 and 2 at the eastern boundary, and H_1 is the depth of layer 1. These must both be constants in order to satisfy the no normal flow condition. However, H_1 must be zero, because at the outcrop line $h_1 = 0$. Thus, H_1 is zero at $y = y_2$, and therefore zero everywhere, and $C = H^2$.

Using (16.58) and (16.63) we obtain a closed expression for h, namely

$$\boxed{h = -z_2 = \frac{(D_0^2 + H^2)^{1/2}}{\left[1 + (g_1'/g_2')(1 - f/f_2)^2\right]^{1/2}}}. \tag{16.65}$$

Using (16.57) and (16.58) the depths in each layer, and the corresponding geostrophic velocities, can be obtained.

A typical solution is shown in Fig. 16.10. The upper layer exists only equatorwards of the outcrop latitude, $y_2 = 0.8$, and isolines of total thickness correspond to streamlines of the lower layer. We see, as expected, the overall shape of a subtropical gyre,

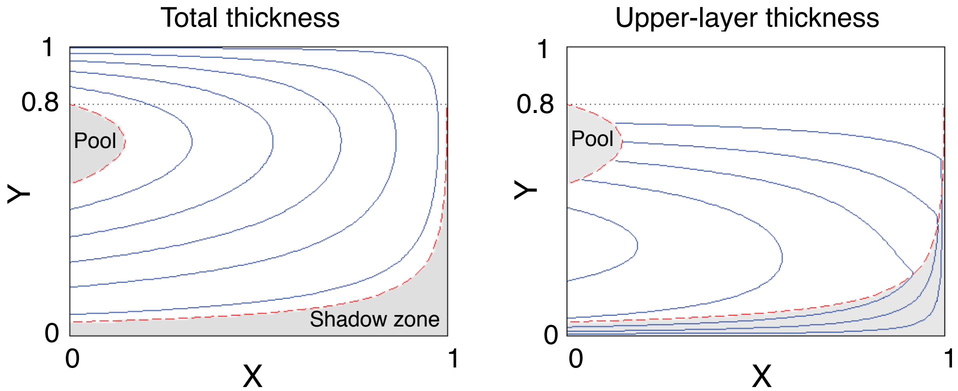


Fig. 16.10 Contour plots of total thickness and upper layer thickness in a two-layer model of the ventilated thermocline. The thickness generally increases westward, and the flow is clockwise. The shadow zone and the western pool are shaded, and no contours are drawn in the latter. The outcrop latitude, $y_2 = 0.8$, is marked with a dotted line. [Parameters are $g_1' = g_2' = 1$, $\beta = 1$, $f_0 = 0.5$, $H = 0.5$, and $w_E = -\sin(\pi y)$.]

with the circulation beng closed by an implicit western boundary current that is not part of the calculation. Two regions are shaded in the figure, the 'pool' region in the west and the 'shadow zone' in the south-east. The solutions above do not apply to these, and they require special attention.

16.4.3 The shadow zone

In the fluid interior the potential vorticity of a parcel in layer 2 is determined by tracing its trajectory back to its outcrop latitude where the potential vorticity is given. That trajectory is determined by its velocity, and this turn is determined by inverting the potential vorticity. Now, parcels subducted at y_2 sweep equatorwards and westwards, so that a parcel, labelled '$\boldsymbol{a}$' say, subducted at the eastern boundary will in general leave the eastern boundary tracing a southwestern trajectory. Consider another parcel, '$\boldsymbol{b}$' say, in the interior that lies eastward of the subducted position of $\boldsymbol{a}$, in the shaded region of Fig. 16.11. It is impossible to trace $\boldsymbol{b}$ back to the outcrop line without trajectories crossing, and this is forbidden in steady flow. Rather, it seems as if the trajectory of $\boldsymbol{b}$ would emanate from the eastern wall. What is the potential vorticity there?

At the eastern boundary the condition of no normal flow at the boundary demands that h be constant (so that $u_2 = 0$), and h_1 be constant (so that $u_1 = 0$). But if a parcel in layer 2 moves *along* the boundary potential vorticity conservation demands that f/h_2 is constant, and therefore h_2 must change, contradicting the no-normal flow requirement. Thus, the velocity at the boundary can have neither a normal component nor a tangential component, and so we cannot trace parcels in the shaded region back to the wall. Rather, in the absence of closed trajectories (for example, eddying motion), we may assume that the shaded region is stagnant, and h is constant. Of course, potential vorticity is everywhere given by f/h_2, which varies spatially, but since there

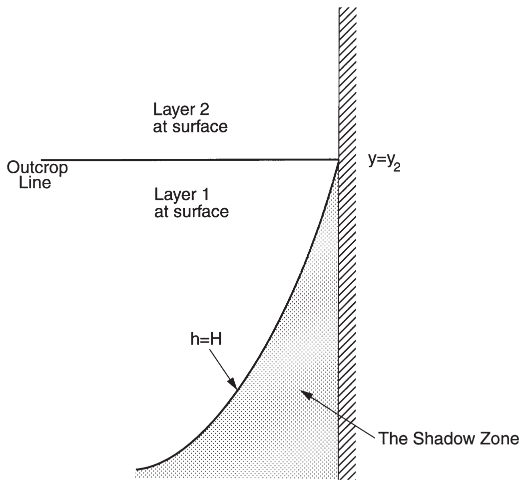


Figure 16.11 The shadow zone in the ventilated thermocline. Layer 2 outcrops at $y = y_2$. A column moving equatorwards along the eastern boundary in layer 2 is subducted at y_2. It cannot remain against the eastern wall and both preserve its potential vorticity, which implies the column shrinks, at the same time that the no-normal flow condition is satisfied, as by geostrophy this implies the layer depth is constant. Thus, the column must move westward, along the boundary of a 'shadow zone' within which is there is no motion. The streamline it follows is the isoline of constant total thickness of the two moving layers [see (16.121) or (16.123c)].

is no motion potential vorticity is still, rather trivially, conserved along trajectories. This region is aptly called the *shadow zone,* since the region falls under the shadow of the eastern boundary; an analogous region arose in the quasi-geostrophic discussion of section 14.7.

To obtain an expression for the fields within the shadow zone, first note that because h is constant, its value is equal to that on the eastern wall; that is, $h = H$. The wind forcing must then all be taken up by the upper layer, and Sverdrup balance then implies

$$\beta v_1 h_1 = f w_E, \tag{16.66}$$

and using (16.60) we obtain an expression for h_1, to wit:

$$h_1^2 = -\frac{2f^2}{\beta g_1'} \int_x^{x_e} w_E(x', y)\, \mathrm{d}x' = \frac{g_2'}{g_1'} D_0^2, \tag{16.67}$$

which is zero at the eastern wall. In the lower layer the thickness is just $h_2 = H - h_1$. The boundary of the shadow zone is given by the trajectory of a fluid parcel in layer 2 that emanates from the eastern boundary at the outcrop line where $h_1 = 0$ and $h = h_2 = H$. Since the flow is steady, the trajectory is an isoline of h. Thus, from (16.65) we have

$$h^2 = \frac{(D_0^2(x_s, y_s) + H^2)}{[1 + g_1'/g_2'\,(1 - f/f_2)^2]} = H^2. \tag{16.68}$$

where (x_s, y_s) denotes the boundary of the shadow zone. (Note that $x_s = x_e$ at $y = y_2$.) From this

$$D_0^2(x_s, y_s) = H^2 \left[\frac{g_1'}{g_2'} \left(1 - \frac{f}{f_2}\right)^2 \right] \tag{16.69}$$

which, given the wind-stress, is an equation for the shadow zone boundary x_s as a function of y.

16.4.4 † The western pool

Polewards of the outcrop latitude the fluid of layer 2 feels the wind directly and the layer thickness is determined by Sverdrup balance. Equatorwards of the outcrop latitude the properties of this layer are determined by potential vorticity conservation, with the potential vorticity being determined by the layer thickness at the outcrop. However, just as there is a region in the east where trajectories cannot be traced back to the outcrop, there is a 'pool' region in the west that is bounded by the trajectory that emerges from the western boundary at the outcrop latitude. Within the pool, trajectories cannot be traced back to the outcrop (Fig. 16.10), and one might imagine that they emerge from the western boundary current. In the context of ventilated themocline theory, there are two plausible hypotheses for determining the layer depths within this region:

(i) Within layer 2, potential vorticity is homogenized;

(ii) Because there is no source for layer 2 water, layer 2 water does not exist and the pool consists solely of ventilated, layer 1 water.

Note the analogy between these two hypotheses and those for the stratification of the troposphere summarized on page 534.

(i) Potential vorticity homogenization

The pool region is a region of recirculation, receiving water from and depositing water into the western boundary current. Thus, following the ideas described in chapter 10 and employed in section 14.7, we hypothesize that the potential vorticity within this region becomes homogenized. The value of potential vorticity within the pool is just the value of potential vorticity at its boundary, and this is given by $f_2/h_2(w)$, where $h_2(w)$ is the thickness of layer 2 at the western boundary at the outcrop latitude. This is given using (16.50) with $f = f_2$ and $g' = g'_2$, and thus the potential vorticity in the pool is given by

$$q_{pool} = \frac{f_2}{D_w^2 + H^2} \tag{16.70}$$

where $D_w^2 = -2(f_2^2/g_2'\beta)\int_{x_w}^{x_e} w_E(x', y_2)\,\mathrm{d}x'$. The thickness of layer 2 in the pool must be consistent with this, and so is given by

$$h_2 = \frac{f}{q_{pool}}. \tag{16.71}$$

The thickness of layer 1 is determined by using Sverdrup balance, (16.59), which, given h_2 and geostrophy, reduces to an equation for h_1.

(ii) The ventilated pool

> *'I came for the waters.' 'What waters?' 'I was misinformed.'*
> From *Casablanca* (1942).

The homogenization hypothesis, although plausible, depends on the assumption of downgradient diffusion of potential vorticity by eddies. Also, because there is no source of layer-2 water in the pool, we must suppose that it is ventilated by eddy pathways that meander down from the surface. An alternative hypothesis, and one

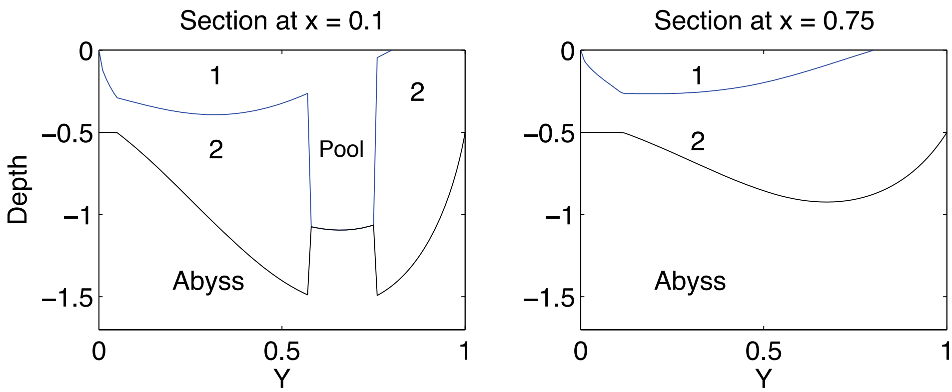


Fig. 16.12 Two north-south section of layer thickness, at different longitudes, from the same solution as Fig. 16.10 and assuming a ventilated western pool. The numbers refer to the fluid layer. The section on the left passes through the western ventilated pool region, where all the Sverdrup transport is taken up by the top layer. The region near $y = 0$ in both plots where the total depth of the thermocline is constant is the shadow zone.

that does not rely on the properties of mesoscale eddies, is to suppose that the western pool is filled with water that is directly ventilated from the surface. That is, if there is no surface source for a water mass, we simply suppose that that water mass does not exist. In the two-layer model, this means that the western pool is filled entirely with layer-1 fluid. If a nonventilated (e.g., layer 2) fluid is present initially, then we hypothesize that it is slowly expunged by the continuous downwards Ekman pumping of layer-1 water into the pool.[13]

Because layer-2 fluid is absent, layer-1 fluid extends all the way down to the stagnant abysss; it takes up all the Sverdrup transport, and this determines the depth of the ventilated pool. Thus, rounding up the usual equations, we set $h = h_1$ in (16.63) to give

$$h_1^2 = D_1^2 + g_2' H^2 \tag{16.72}$$

where

$$D_1^2(x,y) = -\frac{2f^2}{\beta g_{1a}'} \int_x^{x_e} w_E(x',y)\,\mathrm{d}x' \tag{16.73}$$

and $g_{1a}' = g_1' + g_2'$ is the reduced gravity between layer 1 and the abyss and H, as before, is the thickness of layer 2 at the eastern boundary. Because $g_{1a}' > g_2'$ this pool will generally be shallower than the total depth of moving fluid ($h_1 + h_2$) just outside, but the depth of layer-1 fluid alone will be much greater; that is, there will be a discontinuities in layer depths at the pool boundary. A section through the pool region is shown in Fig. 16.12.

Although the reader may be shocked by the appearance of discontinuities in layer depths in a fluid model, the model does provide a simple mechanism for the appearance of *mode water.* This is a distinct mass of weakly stratified, low potential vorticity water appearing in the northwest corner of the North Atlantic subtropical gyre (where it is sometimes called '18 degree water'), with analogs in the other gyres of the world's oceans, and so-called because it appears as a distinct mode in a census of water properties. The proximate mechanism for mode water formation is convection in winter, but for such convection to occur the large-scale ocean circulation must maintain a weakly stratified region, and it is the ventilated pool that enables this, and sets the formation in the context of thermocline structure. In reality, the vertical isopycnals predicted by the simple model will, of course, be highly baroclinically unstable, and the ensuing mesoscale eddies will erode the pool interface and cause the isopycnals to slump, so that the discontinuities in layer depths will be manifest only as rapid changes or fronts.

We conclude by emphasizing that both the ventilated pool theory and the potential vorticity homogenization theory rely on hypotheses that cannot be derived from the governing equations of motion without making additional physical assumptions that are neither a priori true nor obvious. For an overview of the entire main thermocline, refer to the shaded box on the facing page.

16.5 † A MODEL OF DEEP WIND-DRIVEN OVERTURNING

Our goal in this section is to construct a model that illustrates that the overturning circulation of the ocean, and a concomitant deep stratification, may have a wind-driven component that persists even as the diffusivity goes to zero.[14]

How might deep stratification be maintained in the absence of a diapycnal diffusivity? In that case, no upwelling can occur through the stratification, because that is a diabatic process. Rather, the deep water must be directly connected to surface, perhaps by a convective pathway. Let us recall two *de facto* principles that were useful in our discussion of the overturning circulation and the thermocline:

(i) A basin will, in the absence of mechanical forcing, tend to fill up with the densest available fluid.

(ii) Light fluid forced down by wind may displace the cold fluid, so producing stratification.

A completely closed ocean thus fills completely with dense polar water, except in the upper several hundred meters where the main thermocline forms, and whose thickness is the sum of wind-driven and diffusive components. However, suppose that the polewards part of the basin is not fully enclosed but is periodic, as illustrated in Fig. 16.13, with a sill across it at mid-depth, and suppose too that the surface bondary conditions are such that the surface temperature decreases monotonically polewards. A fully enclosed basin exists only beneath the level of the sill, and we may expect the densest water in the basin, formed at the polewards edge of the domain, to fill the basin only below the level of the sill, and that above this may lie warmer water with origins at lower latitudes. Furthermore, suppose that an eastward wind blows over the channel that produces a equatorial flow in the Ekman layer. Then mass conservation demands that there must exist a *subsurface* return flow, and thus a meridional overturning circulation is set up. Note the essential role of the channel in this: if the gap were closed, then the return flow could take place at the surface via a western boundary current, as in a conventional subpolar gyre, and no overturning circulation need be set up.

Thermocline Dynamics — a Summary and Overview

The model of the main thermocline that we have constucted in sections 16.1 to 16.4 is schematically illustrated in Fig. 16.7. Some of the features, and limitations, of this model are listed below.

- ⋆ The main subtropical thermocline consists of an advective upper region overlying a diffusive base.
  - The diffusive base forms forms the *internal thermocline,* and in limit of small diffusivity this is an internal boundary layer. The advective region forms the *ventilated thermocline.* The separation of the two regions may, in reality, not be sharp.
  - The relative thicknesses of these layers is a function of various parameters, notably the strength of the wind and the magnitude of the diffusivity.
- ⋆ Above the ventilated thermocline there may be a mixed layer with a seasonally varying depth. In certain regions, for example at the polewards edge of the subtropical gyre, convection may deepen the mixed layer as far as the base of the thermocline.
- ⋆ A western boundary layer is needed to close the circulation and the heat budget.
- ⋆ The single-hemisphere model assumes that the water that sinks at high latitude either upwells through the main thermoline or returns to the subpolar gyre beneath the main thermocline. In reality much of this water may cross into the other hemisphere, so that, for example, the deep water in the North Atlantic may upwell in the Antarctic Circumpolar Current.
  - In this case, the diffusion-dependent overturning circulation represents only part of the overall meridional overturning circulation.
  - Nevertheless, there would remain a diffusive internal thermocline, because there is still a boundary between the warm subtropical water and cold abyssal water.
- ⋆ Within the ventilated thermocline there are two regions — the shadow zone and the western pool — whose dynamics are not determined without additional assumptions. A plausible assumption for the western pool, namely that all the water within it is ventilated, leads to a model of mode water.
- ⋆ Mesoscale eddies may play an important role in thermocline dynamics. Numerical simulations and some observations suggest that eddies may be particularly important in the internal thermocline, and they may provide a tendency for potential vorticity to homogenize.

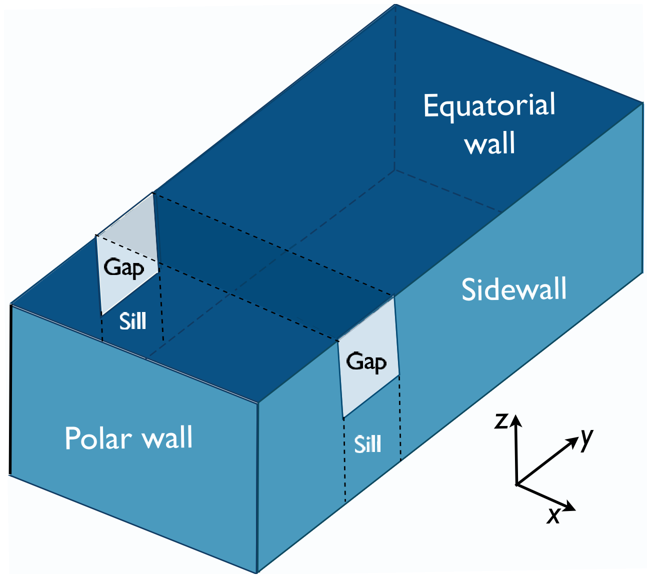


Figure 16.13 Idealized geometry of the Southern Ocean: a re-entrant channel, partially blocked by a sill, is embedded within a closed rectangular basin; thus, the channel has periodic boundary conditions, whereas elsewhere there is no normal flow. The channel is a crude model of the Antarctic Circumpolar Current, with the area over the sill analagous to the Drake Passage.

16.5.1 A single hemisphere model

We consider first a single-hemisphere basin with a periodic channel near its polewards edge. We suppose it to be in the southern hemisphere, so the channel represents the Antactic Circumpolar Current (ACC), and that the dynamics are Boussinesq and planetary-geostrophic. We will choose extremely simple forms of wind and bouyancy forcing to allow us to obtain an analytic solution, and then later discuss how the qualitative forms of these solutions might more generally apply.

Wind and buoyancy forcing

Thermodynamic forcing is imposed by fixing the surface buoyancy, b_s. (In the discussion following salinity is absent, and buoyancy is virtually equivalent to temperature.) South of the gap we suppose the buoyancy is constant, then that it linearly increases across the gap, and is constant again polewards of the gap. Thus, there is no temperature gradient across the subtropical gyre, focussing attention on the influence of the channel. Thus, referring to Fig. 16.14:

$$b_s = \begin{cases} b_1, & y_0 \le y \le y_1 \\ b_1 + \dfrac{(b_2 - b_1)(y - y_1)}{y_2 - y_1}, & y_1 \le y \le y_2 \\ b_2, & y \ge y_2. \end{cases} \tag{16.74}$$

where $b_2 > b_1$, and both are constants, and we may take $b_1 = 0$ and $y_0 = 0$.

The wind forcing is purely zonal, and it is convenient to express this in terms of the Ekman transport and associated pumping (refer to section 2.12). In the channel the Ekman transport is chosen to be (realistically) equatorwards and (less realistically) constant, a simplification that avoids complications of wind-driven upwelling in the channel. South (polewards) of the channel there is a conventional subpolar gyre, with an Ekman upwelling and an equatorwards Ekman transport that joins smoothly to

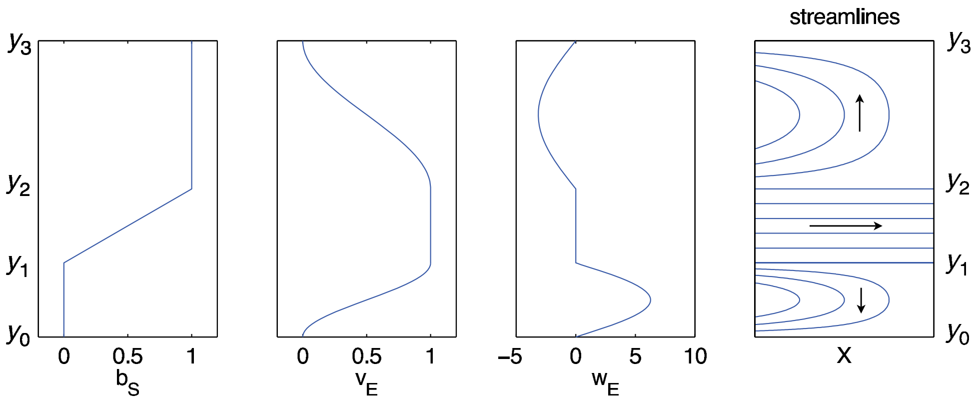


Fig. 16.14 The surface buoyancy b_s, merdional Ekman velocity v_E, vertical Ekman velocity w_E and the solution streamlines for the geostrophic horizontal flow, omitting the western boundary currents. The ordinate in all plots is latitude, with the pole at the bottom, and the four fields are given by, respectively, (16.74), (16.75a), (16.75b), and (16.76), with purely zonal flow given by (16.79) in the channel.

that of the channel. Equatorwards of the channel there is conventional subtropical gyre, with Ekman downwelling. All this may be achieved by specifying:

$$v_E = \begin{cases} \dfrac{V}{2}\left[1 - \cos\left(\dfrac{\pi y}{\Delta y_1}\right)\right] \\ V \\ \dfrac{V}{2}\left[1 + \cos\left(\dfrac{\pi(y - y_2)}{\Delta y_2}\right)\right] \end{cases} \qquad w_E = \begin{cases} W_1 \sin\left(\dfrac{\pi y}{\Delta y_1}\right) & 0 \le y < y_1 \\ 0 & y_1 \le y < y_2 \\ -W_2 \sin\left(\dfrac{\pi(y - y_2)}{\Delta y_2}\right) & y_2 \le y < y_3. \end{cases} \tag{16.75a,b}$$

where $\Delta y_1 = y_1$ and $\Delta y_2 = y_3 - y_2$. The meridional Ekman transport, v_E, is related to the Ekman pumping by $w_E = \partial v_E/\partial y$, so that $W_i = \pi V/(2\Delta y_i)$, where V is a constant that determines the magnitude of the meridional Ekman flow. If f were constant, the wind stress curl would be proportional to the w_E field above. The precise details of the forcing do not affect the qualitative form of the solution — they merely allow an analytic solution to be obtained — but there are two essential aspects to it:

(i) The surface is cold south of the channel, warm north of the channel, and there is a temperature gradient across the channel.

(ii) The Ekman flow is equatorwards within the channel, with conventional gyres to either side.

The meridional extent of the region south of the channel and the wind forcing within it are relatively unimportant, and this region could be shrunk to nearly zero.

Solution in the gyres

Below the depth of the sill the basin is fully enclosed, and therefore up to that level the basin will fill with the densest available water (much as described in section 15.2), except where it may be displaced by warmer fluid equatorwards of the gap that is pumped down below the level of the sill by the wind (Fig. 16.15). Thus, all of the

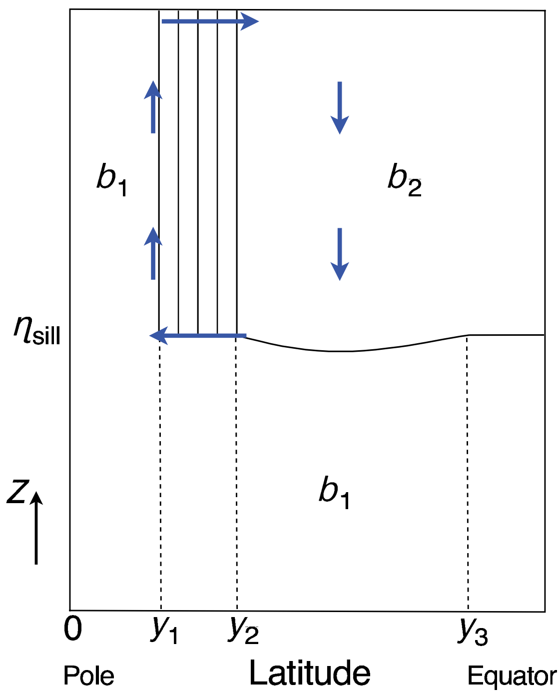


Figure 16.15 Cross section of the structure of the single-hemisphere ocean model described in section 16.5.1. The domain is zonally closed equatorwards of y_2 and polewards of y_1, with a zonally-periodic channel between latitudes y_1 and y_2 and above the sill, which has height η_{sill}. The arrows indicate the fluid flow driven by the equatorwards Ekman transport in the channel, and the solid lines are isopycnals.

domain south of the channel, and nearly everywhere below the sill the water has buoyancy b_1. Polewards of the channel, then, the fluid is barotropic and its vertically integrated horizontal circulation is given by Sverdrup balance, $\beta V = f w_E/H$, where V is the vertically integrated flow. With the wind-stress of (16.75) we get a conventional barotropic subpolar gyre (and associated western boundary current) by the same methods as we employed in chapter 14.

Above the sill, net meridional geostrophic transfer is forbidden in the channel region, because $\overline{v}_g = \partial\overline{\phi}/\partial x = 0$, where the overbar denotes a zonal average. Equatorwards of the channel the region above the sill will therefore tend to fill with the densest water available to it, and this is water with buoyancy equal to b_2 (which is the buoyancy of the water as it emerges from the channel). However because of the presence of wind forcing, the base of this layer is not flat; rather, this fluid obeys the dynamics of the reduced-gravity single-layer ventilated thermocline model discussed in section 16.4.1. In such a model the depth of the fluid on the eastern boundary is constant, and this must be specified. Here, this is given by the height of the sill, and therefore $h(x = x_e, y) = h_e = H - \eta_{sill}$, where H is the total thickness of the fluid and η_{sill} is the sill height. Then, using (16.50) and (16.75), the thickness of the moving layer equatorwards of the sill is given by, for $y_2 < y < y_3$.

$$h^2 = D^2(x, y) + h_e^2 \tag{16.76}$$

where

$$D^2 = -\frac{2f^2}{g'\beta}\int_x^{x_e} w_E \, dx' = \frac{2f^2}{g'\beta} W_2(x_e - x)\sin\left(\frac{\pi(y - y_2)}{\Delta y_2}\right) \tag{16.77}$$

where $g' = b_2 - b_1$. The solution is closed by the addition of a western boundary current. Note that because $h > h_e$, the light fluid is pushed below the level of the sill in the subtropical gyre.

Solution in the channel

In the channel, the fluid in the Ekman layer flows equatorwards, and therefore there must be a compensating polewards flow at depth. This will occur at and just below the level of the sill: it cannot be deeper, because here the basin is full of denser, b_1 fluid, and in the absence of eddying or ageostrophic flow it cannot be shallower because of the geostrophic constraint. Now, because of the temperature gradient across the channel the polewards flowing fluid is warmer than the fluid at the surface, and therefore convectively unstable. Convection ensues, the result of which is the entire column of fluid between the top of the sill and the surface mixes and takes on the temperature of the surface. Thermal wind demands that there be a zonal flow associated with this meridional temperature gradient, so this temperature distribution is advected eastwards into the interior of the channel. Because the interior is presumed to to be adiabatic, this temperature fields extends zonally throughout the channel. Thus, in steady state, the temperature *everywhere* in the channel above the level of the sill is given by:

$$b(x,y,z) = b_s(y) = b_1 + \frac{(b_2 - b_1)(y - y_1)}{y_2 - y_1}, \qquad y_1 \le y \le y_2, \ \ z > \eta_{sill.} \tag{16.78}$$

Note that convective mixing does not rely on a diapycnal diffusivity other than a molecular one: Convective plumes are generally turbulent, generating small scales in the fluid interior where mixing and entrainment may occur; failing that, the lighter fluid is displaced to the surface where it cools by way of interaction with the atmosphere. The zonal velocity within the channel is then given by thermal wind balance, so that

$$u(x,y,z) = -\frac{1}{f}\left(\frac{b_2 - b_1}{y_2 - y_1}\right)(z - \eta_{sill}). \tag{16.79}$$

(Note that $f < 0$ so that the shear is positive.)

Regarding the depth-integrated zonal momentum budget, the wind-stress at the surface is balanced by a pressure force against the sill walls. This pressure gradient arises through the meridional circulation, for the southwards return flow just below the level of the sill is associated with a zonal pressure gradient that is exactly equal, but opposite, to the stress exerted by the wind. That is to say, in the Ekman layer the wind stress is balanced by the Coriolis force on the equatorwards flow in the Ekman layer, which by mass conservation is equal and opposite to the Coriolis force on the deep polewards flow, which by geostrophy is equal to the net pressure force on the sill walls. The wind-stress plays no role in determining the zonal transport of the channel: if the wind increases the meridional overturning, and the pressure force, increase but with no change to the transport. This is a somewhat unrealistic feature of the model, for in reality the form stress induced by the flow over bottom topography, and that balances the wind stress, is a likely to be a function of the zonal transport as well as the meridional transport.

A qualitative summary

The circulation of the model may be described as follows. The entire basin polewards of the channel fills with dense, b_1, water. Below the sill this fluid extends equatorward, filling the lower part of the channel and subtropical basin, up to the level of the sill. Now, Ekman pumping in the channel forces near-surface fluid equatorwards, which warms as it goes, entering the subtropical basin with buoyancy b_2. This fluid fills the basin down to the level of the sill, where it encounters the dense, b_1, fluid. The subtropical basin is wind-driven, and it forms a subtropical gyre with a single moving layer. Its dynamics are completely determined by specifying the wind, the reduced gravity ($g' = b_2 - b_1$), and the depth of the fluid at the eastern boundary (the sill depth). Because of the requirements of mass conservation, there must be a polewards return flow at depth, and so at the level of the sill warm water flows polewards. This flow is convectively unstable (because the water is lighter than that at the surface), and so the entire column of fluid mixes and its density takes on the value at the surface. The meridional temperature gradient gives rise to an eastwards flow, and this temperature field is advected zonally, and in steady state the temperature distribution is zonally symmetric and given by (16.78). The overturning circulation within the ACC is known as the Deacon cell, and this is a crude model of it. It is considered further in section 16.6.

If we were to add some diapycnal diffusivity to this situation, the sharp boundary between the two fluid masses at the sill height would be diffused to a front of finite thickness, with some upwelling and water mass transformation occurring across the front. This diffusive loss of dense fluid would be compensated by water-mass formation at the surface, polewards of the channel, leading to a deep, diffusively-driven circulation. That is, the deep water mass of b_1 fluid would circulate: this is a crude model of the 'Antarctic Bottom Water' cell.

Suppose now that the wind were everywhere zero, and the diffusivity small but non-zero. The cold, b_1 fluid would still quickly completely fill the basin polewards of the channel, and would fill the basin equatorwards of the channel up to the level of the sill. However, with no wind to drive an overturning circulation dense b_1 water would slowly drift ageostrophically across the channel, displacing warmer water until the *entire basin* were filled with the dense fluid, except for a thin boundary layer at the top needed to satisfy the upper boundary condition. The final state would be one of no motion, and no stratification, below this boundary layer. The important conclusion to the drawn is the following: *A deep meridional circulation, and a deep stratification, can be maintained, even as the diapycnal diffusivity goes to zero, in the presence of a wind forcing and a circumpolar channel.* Of course there are a number of idealised or unrealistic aspects to this model, perhaps the most egregious being:

- ⋆ The vertical isopycnals in the channel will be highly baroclinically unstable. This will cause the isopycnals to slump and will potentially set up an eddy-induced circulation. We consider this in section 16.6.
- ⋆ There is no surface temperature gradient across the subtropical gyre. If present it would lead to the formation of a 'main' subtropical thermocline, a full treatment of which would require determining its eastern boundary conditions. This would, however, be unlikely to qualitatively affect the presence of a deep, wind-driven overturning circulation.
- ⋆ The wind stress in the model channel is chosen so that the meridional Ekman

transport is constant. (This means the wind-stress is chosen to vary in the same fashion as the Coriolis parameter, and if f were constant, the wind-stress curl would vanish.) Thus, there is no wind-driven downwelling or upwelling in the channel, and this simplifies the solution. Numerical simulations suggest that this choice does not affect the qualitative nature of the overturning circulation or temperature distribution.

16.5.2 A cross-equatorial wind-driven circulation

Let us now qualitatively and heuristically extend the above model to consider flow across the equator. The essential addition is that we suppose that the ocean basin extends to high northern latitudes, where there is, potentially, another source of cold deep water. To keep the model simple and tractable we will assume a very simple temperature structure:

$$b_s = \begin{cases} b_1, & 0 \le y \le y_1 \\ b_1 + \dfrac{(b_2 - b_1)(y - y_1)}{y_2 - y_1}, & y_1 \le y \le y_2 \\ b_2, & y_2 \le y \le y_4 \\ b_3 & y > y_4. \end{cases} \tag{16.80}$$

where the geometry is illustrated in Fig. 16.16. Given that $b_2 > b_1$, there are three cases to consider:

(i) $b_3 > b_2$. This is not oceanographically relevant to today's climate, nor does it provide another potential deep water source.

(ii) $b_3 < b_1$. The northern water is now the densest in the ocean, and would fill up the entire basin north of the channel (except near the surface in regions where some b_2 water is pushed down by the wind), and so provide no mid-depth stratification.

(iii) $b_1 < b_3 < b_2$. This is the most interesting and relevant case, and the only one we explore further.

As regards the wind, we will assume that south of the equator this is given by (16.75). North of the equator the wind forcing does not affect the qualitative nature of the overturning circulation, and may be taken to be zero.

Descriptive solution

In case (iii), the entire basin below the sill fills with b_1 water, except where wind forcing forces warmer fluid below the sill level, as before. However, unlike the earlier case, the fluid above the sill is predominantly b_3 water from high northern latitudes. This forms in high polar latitudes and fills most of the basin above the sill, from the basin boundary in the north to the channel in the south (as discussed more below). However, except at latitudes where the b_3 is formed, it does not reach the surface because of the presence of b_2 water. That water is pushed down by the wind in the southern hemisphere to some as yet undetermined depth (discussed below), the boundary between b_2 and b_3 water then forming the upper ocean thermocline.

These water masses circulate because of the wind forcing in the channel. As in the single hemisphere case, northwards flowing water emerges from the channel with

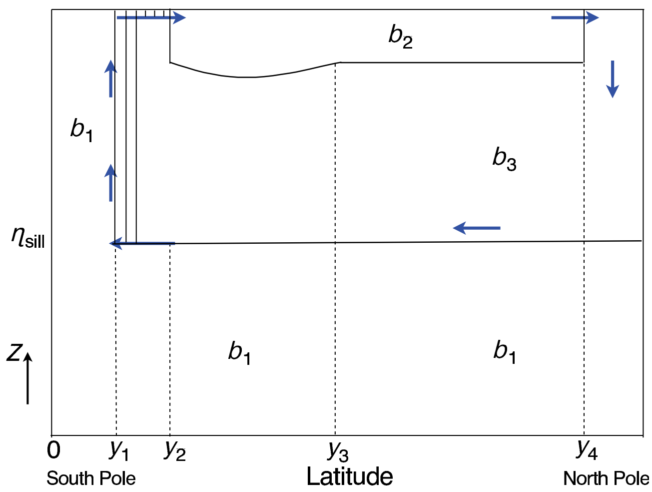


Figure 16.16 As for Fig. 16.15, but now for a two hemisphere ocean with a source of dense water, b_3, at high northern latitudes. The solid lines are isopycnals, and here the wind is zero in the Northern Hemisphere.

buoyancy b_2. This emerges into a region of Ekman downwelling, with a northwards transport carried by a western boundary current. This transport crosses the equator finally reaching the latitudes where b_3 water is formed where it sinks and returns equatorwards, again in a western bundary current. (Away from the western boundary layer there is no meridional flow in the absence of diffusion, because the flow satisfies $\beta v = f\partial w/\partial z$ and there is no upwelling.) This water then crosses the sill. However, unlike in the single hemisphere case, in the northern part of the sill this water is denser than the surface water; no convection occurs and so the b_3 water extends upwards to the surface, where it warms by contact with the atmosphere and advected equatorwards to become b_2 water. Further south the surface buoyancy in the channel is less than b_3, and the column now convectively mixes, much as in the single-hemisphere case. The solution is completed by specifying the thickness of the layer of b_2 water at the surface. Now, if the circulation is in steady state, the meridional transport between the gyres must equal that of the northwards Ekman flow at the northern edge of the circumpolar channel, and given the wind forcing, this is determined by the depth of the layer at the eastern boundary, a constant. Thus, in this model, global constraints determine the depth of the eastern boundary of the thermocline.

Suppose that the wind were everywhere zero. Then, as in the single-hemisphere case, the circulation would eventually die. Again, though, slow ageostrophic motion across the channel would first allow the entire basin, within and on both sides of the channel, to fill with the densest available water, and in the final steady state there would be no stratification (and no motion) below a thin surface layer.

Suppose, on the other hand, that a small amount of diffusion were added to the wind-forced model above. Then there would be mass exchange between the layers and, in particular, the deep cell of b_1 water would begin to diffusively circulate. In addition, the middepth cell would begin to upwell through the b_2–b_3 interface, and develop a diffusively driven circulation, in much the same way as is illustrated in Fig. 16.7.

Summary remarks

The key result of this model is that, even as the diffusivity falls and the interior of the ocean becomes more and more adiabatic, a meridional cross-hemispheric circulation can be maintained, provided that the wind across the circumpolar channel remains finite. The diabatic water mass transformations all occur at the surface or in convection: these processes require a finite diffusivity, but this diffusivity can be the molecular one because the associated mixing involves turbulence, which can generate arbitrarily small scales. (Note that the convection that occurs in the circumpolar channel reduces the potential energy of the column, and requires no mechanical input of energy.) Aside from the region of the ACC, the meridional transport will occur (in this model) in western boundary layers. Indeed, we may still expect to see a southwards flowing deep western boundary current south of y_4 and below the b_2 water in Fig. 16.16, just as in the implicitly diffusive Stommel-Arons model. In the ACC itself, the meridional transport occurs in a sub-surface current, nestled against the sill.

Although the overturning circulation in this model is 'wind-driven', the possibility that it may be cross-equatorial depends upon the thermodynamic forcing; in particular, if there is no source of dense water in the northern hemisphere, then the basin above the sill simply fills with b_2 water, as in the model of section 16.5.1, and there need be little or no inter-hemispheric flow. We emphasize, too, that our model of interhemispheric flow is quite heuristic: we have essentially *posited* that b_2 water may continuously flow across the equator, without examining the equatorial dynamics in any detail. Finally, we note that in the simple model we have discussed, the overturning circulation is equal to that of the Ekman pumping in the channel. In reality, the ACC is a complex beast, and some of this water most likely recirculates within the channel, and this leads us into our next topic.

16.6 † FLOW IN A CHANNEL, AND THE ANTARCTIC CIRCUMPOLAR CURRENT

We now take a closer look at the Antarctic Circumpolar Current (ACC) itself, with less of a focus on how the ACC connects the rest of the worlds ocean and more of a focus on its own internal dynamics. This current system, sketched in Fig. 16.17, differs from other oceanic regimes primarily in that the flow is, like that of the atmosphere, predominantly zonal and re-entrant. The two obvious influences on the circulation are the strong, eastward winds (the 'roaring forties', the 'furious fifties', and the 'screaming sixties') and the buoyancy forcing associated with the meridional gradient of atmospheric temperature and radiative effects that cause ocean cooling at high latitudes and warming at low. Providing a detailed description of the resulting flow is properly the province of numerical models, and here our goals are much more modest, namely to describe some of the basic dynamical mechanisms that determine the structure and transport of the system.[16]

16.6.1 Steady and eddying flow

Consider again the simplified geometry of the Southern Ocean as sketched in Fig. 16.13. The ocean floor is flat, except for a ridge (or 'sill') at the same longitude as the the gyre walls; this is a crude representation of the topography across the Drake Passage, that part of the ACC between the tip of South America and the Antarctic

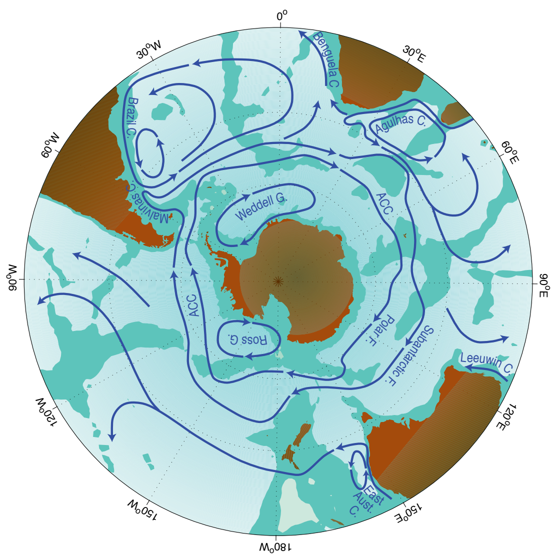


Figure 16.17 Schema of the major currents in the Southern Ocean. Shown are the South Atlantic subtropic gyre, and the two main cores of the ACC, associated with the Polar front and the sub-Antarctic front.[15]

Peninsula. In the planetary geostrophic approximation, the steady response is that of nearly vertical isopycnals in the area above the sill, as illustrated in Fig. 16.15. Below the sill a meridional flow can be supported and the isotherms spread polewards, as illustrated in numerical solutions using the primitive equations (Fig. 16.18).[17]

The stratification of the non-eddying simulation is similar to that predicted by the idealized model illustrated in Fig. 16.15. However, the steep isotherms within the

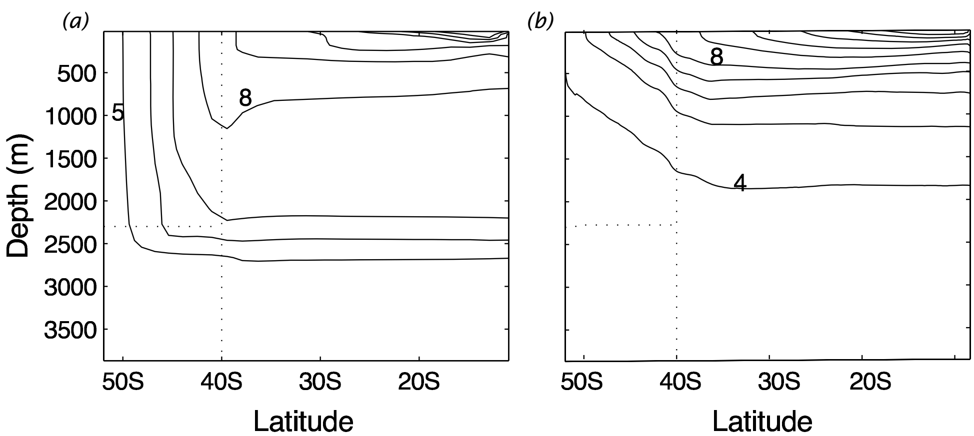


Fig. 16.18 The zonally-averaged temperature field in numerical solutions of the primitive equations in a domain similar to that of Fig. 16.13 (except that here the channel and sill are nestled against the polewards boundary). Panel (a) shows the steady solution of a diffusive model with no baroclinic eddies, and (b) shows the time-averaged solution in a higher resolution model that allows baroclinic eddies to develop. The dotted lines show the channel boundaries and the sill.[18]

channel contain a huge amount of available potential energy (APE), and the flow is highly baroclinically unstable, and if baroclinic eddies are allowed to form, the solution is dramatically different: the isotherms slump, releasing that APE and generating mesoscale eddies. An important conclusion is that *baroclinic eddies are of leading-order importance in the dynamics of the ACC.* A dynamical description of the ACC without eddies would be *qualitatively* in error, in much the same way as would a similar description of the mid-latitude troposphere (i.e., the Ferrel Cell). (However, it is less clear whether these eddies are important in the interaction of the ACC with the rest of the worlds oceans.) These eddies transfer both heat and momentum, and much of the rest of our description will focus on their effects.

16.6.2 Vertically integrated momentum balance

The momentum supplied by the strong eastward winds must somehow be removed. Presuming that lateral transfers of momentum are small the momentum must be removed by fluid contact with the solid earth at the bottom of the channel. Thus, let us first consider the vertically integrated momentum balance in a channel, without regard to how the momentum might be vertically transferred. We begin with the frictional-geostrophic balance

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi + \frac{\partial \widetilde{\boldsymbol{\tau}}}{\partial z}, \tag{16.81}$$

where $\widetilde{\boldsymbol{\tau}}$ is the kinematic stress (and henceforth we drop the tilde). Integrating over the depth of the ocean gives [c.f. (14.96)]

$$\boldsymbol{f} \times \widehat{\boldsymbol{u}} = -\nabla\widehat{\phi} - \phi_b \nabla\eta_b + \boldsymbol{\tau}_w - \tau_f, \tag{16.82}$$

where $\boldsymbol{\tau}_w$ is the stress at the surface (due mainly to the wind) and $\boldsymbol{\tau}_f$ is the frictional stress at the bottom, a hat denotes a vertical integral and ϕ_b is the pressure at $z = \eta_b$, where η_b is the z-coordinate of the bottom topography.

The x-component of this equation is just

$$-f\widehat{v} = -\frac{\partial\widehat{\phi}}{\partial x} - \phi_b\frac{\partial\eta_b}{\partial x} + \tau_w^x - \tau_f^x, \tag{16.83}$$

and on integrating around a line of latitude the term on the left-hand side vanishes by mass conservation and we are left with

$$\oint\left(-\phi_b\frac{\partial\eta_b}{\partial x} + \tau_w^x - \tau_f^x\right)\mathrm{d}x = 0. \tag{16.84}$$

The first term is the form drag, encountered in sections 3.5 and 14.6, and observations and numerical simulations indicate that it is this, rather than the frictional term τ_f^x, that predominantly balances the wind stress in the zonally and vertically integrated momentum balance.[19] We address the question of *why* this should be so in section 16.6.5.

The vorticity balance is also dominated by a balance between bottom pressure torque and wind stress curl. Taking the curl of (16.82), noting that $\nabla\cdot\widehat{\boldsymbol{u}} = 0$, gives

$$\beta\widehat{v} = -\mathbf{k}\cdot\nabla\phi_b \times \nabla\eta_b + \mathrm{curl}_z\boldsymbol{\tau}_w - \mathrm{curl}_z\boldsymbol{\tau}_f. \tag{16.85}$$

Now, on integrating over an area bounded by two latitude circles and applying Stokes's theorem the β term vanishes by mass conservation and we regain (16.84). This means that Sverdrup balance, in the usual sense of $\beta v \approx \mathrm{curl}_z \boldsymbol{\tau}_w$, cannot hold in the zonal average: the left-hand side vanishes but the right-hand side does not. The same could be said for the zonal integral of (16.85) across a gyre, but the two cases do differ: In a gyre Sverdrup balance can (in principle) hold over most of the interior, with mass balance being satisfied by the presence of an intense western boundary current. In contrast, in a channel where the dynamics are zonally homogeneous then v must be, on average, zero at *all* longitudes and form drag and/or frictional terms must balance the wind-stress curl in a given water column. Sverdrup balance is thus a less useful foundation for channel dynamics (at least zonally homogeneous ones) than it is for gyres. Of course, the real ACC is *not* zonally homogeneous, and may contain regions of polewards Sverdrup flow balanced by equatorwards flow in boundary currents along the eastern edges of sills and continents, and the extent to which Sverdrup flow is a leading-order descriptor of its dynamics is partly a matter of geography.

We cannot in general completely neglect nonconservative frictional terms, on two counts. First, they are the means whereby kinetic energy is dissipated. Second, if there is a contour of constant orographic height encircling the domain (i.e., encircling Antartica) then the form drag will vanish when integrated along it. However, the same integral of the wind stress will not vanish, and therefore must be balanced by something else. To see this explicitly, write the vertically integrated vorticity equation, (16.85), in the form

$$\beta \widehat{v} + J(\phi_b, \eta_b) = \mathrm{curl}_z \boldsymbol{\tau}_w - \mathrm{curl}_z \boldsymbol{\tau}_f. \tag{16.86}$$

If we integrate over an area bounded by a contour of constant orographic height (i.e., constant η_b) then both terms on the left-hand side vanish, and the wind-stress along that line must be balanced by friction. In the real ocean there may be no such contour that is confined to the ACC — rather, any such contour would meander through the rest of the ocean; indeed, no such confined contour exists in the idealized geometry of Fig. 16.13.

16.6.3 Form drag and baroclinic eddies

How does the momentum put in at the surface by the wind stress make its way to the bottom of the ocean where it may be removed by form drag? We saw in section 16.5.1 that one mechanism is by way of a mean meridional overturning circulation, with an upper branch in the Ferrel cell and a lower branch at the level of the sill, with no meridional flow between. However, the presence of baroclinic eddies changes things in two related ways:

(i) Eddy form drag can pass momentum vertically within the fluid

(ii) Eddies can allow a net meridional mass flux.

Momentum dynamics of layers

Let us first model the channel as a finite number of fluid layers, each of constant density and lying one on top of the other — a 'stacked shallow water' model, equivalent to a model expressed in isopycnal coordinates. The wind provides a stress on the upper layer, which sets it into motion, and this in turn, via the mechanism of form

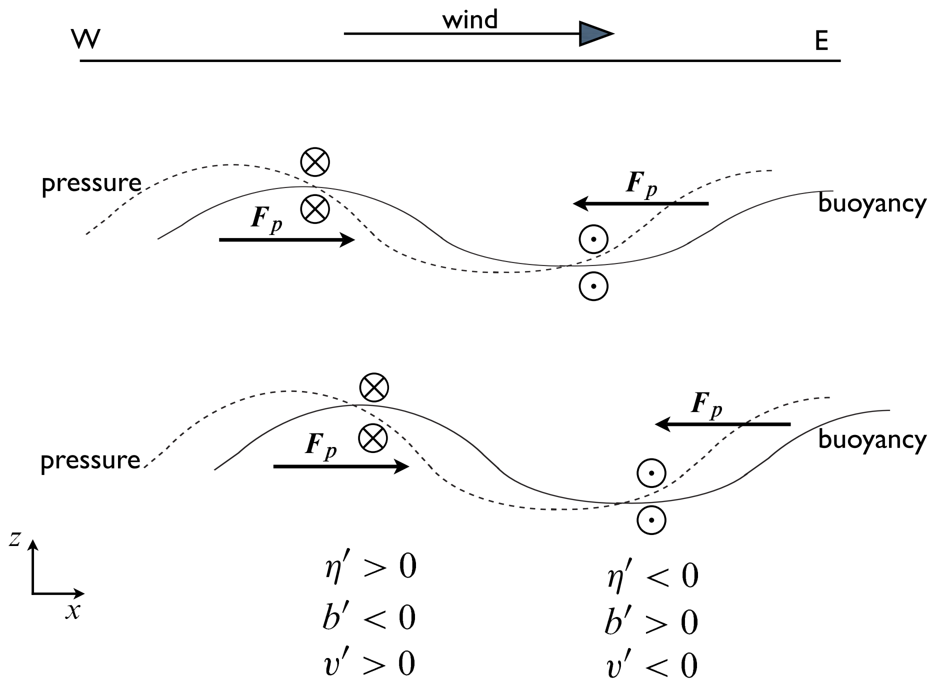


Fig. 16.19 Eddy fluxes and form drag in a Southern Hemisphere channel, viewed from the south. In this example, cold (less buoyant) water flows equatorwards and warm water polewards, so that $\overline{v'b'} < 0$. The pressure field associated with this flow (dashed lines) provides a form-drag on the successive layers, $\boldsymbol{F}_p$, shown. At the ocean bottom the westward form drag on the fluid arising through its interaction with the orography of the sea-floor is equal and opposite to that of the eastward wind-stress at the top. The mass fluxes in each layer are given by $\overline{v'h'} \approx -\partial_z(\overline{v'b'}/N^2)$. If the magnitude of buoyancy displacement increases with depth then $\overline{v'h'} < 0$.

drag, provides a stress to the layer below, and so on until the bottom is reached. The lowest layer then equilibrates via form drag with the bottom topography or via Ekman friction, and the general mechanism is illustrated in Fig. 16.19.

Recalling the results of section 3.5, the form drag at a layer interface is given by

$$\tau_i = -\overline{\eta_i \frac{\partial p_i}{\partial x}} = -\rho_0 f \overline{\eta_i v_i} \tag{16.87}$$

where p_i is the pressure and η_i the displacement at the i'th interface (i.e., between the i'th and $i+1$ layer as in Fig. 16.20), and the overbar denotes a zonal average. If we define the averaged meridional transport in each layer by

$$V_i = \int_{\eta_i}^{\eta_{i-1}} \rho_0 v \, \mathrm{d}z \tag{16.88}$$

then, neglecting the meridional momentum flux divergence (as explained in the next subsection), the time- and zonally-averaged zonal momentum balance for Boussinesq layers of fluid are:

$$-f\overline{V}_1 = \tau_w - \tau_1 = \overline{\eta_1 \frac{\partial p_1}{\partial x}} + \tau_w, \tag{16.89a}$$

$$-f\overline{V}_i = \tau^x_{i-1} - \tau_i = -\overline{\eta_{i-1}\frac{\partial p_{i-1}}{\partial x}} + \overline{\eta_i \frac{\partial p_i}{\partial x}}, \tag{16.89b}$$

$$-f\overline{V}_N = \tau_{N-1} - \tau_N = -\overline{\eta_{N-1}\frac{\partial p_{N-1}}{\partial x}} + \overline{\eta_b \frac{\partial p_b}{\partial x}} - \tau_f, \tag{16.89c}$$

where the subscripts 1, i and N refer to the top layer, an interior layer, and the bottom layer, respectively. Also, η_b is the height of bottom topography and τ_w is the zonal stress imparted by the wind which, we assume, is confined to the uppermost layer. The term τ_f represents drag at the bottom due to Ekman friction, but we have neglected any other viscous terms or friction between the layers.

The vertically integrated meridional mass transport must vanish, and thus summing over all the layers (16.89) becomes

$$0 = \tau_w - \tau_f - \tau_N \tag{16.90}$$

or, noting that $\tau_N = -\overline{\eta_b \partial p_b/\partial x}$,

$$\boxed{\tau_w = \tau_f - \overline{\eta_b \frac{\partial p_b}{\partial x}}}\,. \tag{16.91}$$

Thus, the stress imparted by the wind (τ_w) may be communicated vertically through the fluid by form drag, and ultimately balanced by the sum of the bottom form stress (τ_N) and bottom friction (τ_f).

Momentum dynamics in height coordinates

We now look at these same dynamics in height coordinates, using the quasi-geostrophic TEM formalism, and it may be helpful to review section 7.3 before proceeding. As in (7.66), we write the zonally-averaged momentum equation in the form

$$-f_0\overline{v}^* = \nabla_m \cdot \boldsymbol{\mathcal{F}} + \frac{\partial \tau}{\partial z} \tag{16.92}$$

where $\overline{v}^* = \overline{v} - \partial/\partial z(\overline{v'b'}/\overline{b}_z)$ is the residual meridional velocity, τ is the zonal component of the kinematic stress (wind induced and frictional) and $\boldsymbol{\mathcal{F}}$ is the Eliasssen-Palm flux, which satisfies

$$\nabla_m \cdot \boldsymbol{\mathcal{F}} = -\frac{\partial}{\partial y}\overline{u'v'} + \frac{\partial}{\partial z}\left(\frac{f_0}{N^2}\overline{v'b'}\right) = \overline{v'q'}. \tag{16.93}$$

The stress, τ, is typically important only in an Ekman layer at the surface and in a frictional layer at the bottom. Now, if the horizontal velocity and buoyancy perturbations are related by $v' \sim b'/N$ (see chapter 9) then the two terms comprising the potential vorticity flux scale as

$$\frac{\partial}{\partial y}\overline{u'v'} \sim \frac{v'^2}{L_e}, \qquad \frac{\partial}{\partial z}\left(f_0\frac{\overline{v'b'}}{\overline{b}_z}\right) \sim \frac{v'^2}{L_d} \tag{16.94}$$

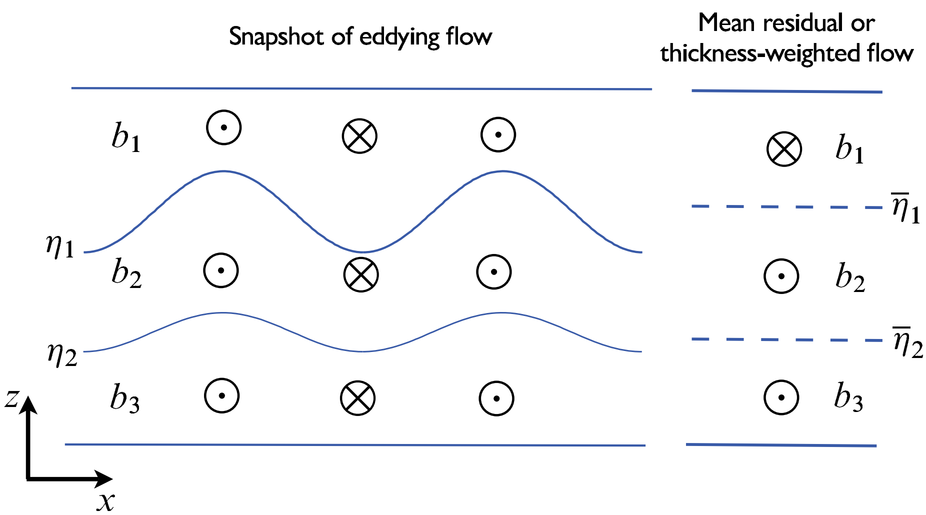


Fig. 16.20 An example of the meridional flow in an eddying channel. The eddying flow may be organized such that, even though at any given level the Eulerian meridional flow may be small, there is a net flow in a given isopycnal layer. The residual ($\overline{v}^*$) and Eulerian ($\overline{v}$) flows are related by $\overline{v}^* = \overline{v} + \overline{v'h'}/\overline{h}$; thus, the thickness-weighted average of the eddying flow on the left gives rise to the residual flow on the right, where $\overline{\eta}_i$ denotes the mean elevation of the isopycnal interface η_i.

where L_e is the scale of the eddies and L_d is the deformation radius. If the former is much larger than the latter, as we might expect in a field of developed geostrophic turbulence (and as is indeed observed in the ACC), then the potential vorticity flux is dominated by the buoyancy flux [so also justifying the neglect of the lateral momentum fluxes in (16.89)] and (16.92) becomes

$$-f_0\overline{v}^* \approx \frac{\partial \tau}{\partial z} + \frac{\partial}{\partial z}\left(f_0 \frac{\overline{v'b'}}{\overline{b}_z}\right). \tag{16.95}$$

If we integrate this equation over the depth of the channel the term on the left-hand side vanishes and we have

$$\tau_w = \tau_f - \left[f_0 \frac{\overline{v'b'}}{\overline{b}_z}\right]_{-H}^{0}, \tag{16.96}$$

where τ_w is the wind stress and τ_f is the frictional stress at the bottom. As we noted in section 7.4.3, the vertical component of the EP flux is equivalent to a form stress acting on a fluid layer, and (16.96) expresses essentially the same momentum balance as (16.91) (where it was assumed that the stress at the top arises from the wind). Thus, the EP flux expresses the passage of momentum vertically through the water column, and it is removed at the bottom through frictional stresses and/or form drag with the orography.

Mass fluxes and thermodynamics

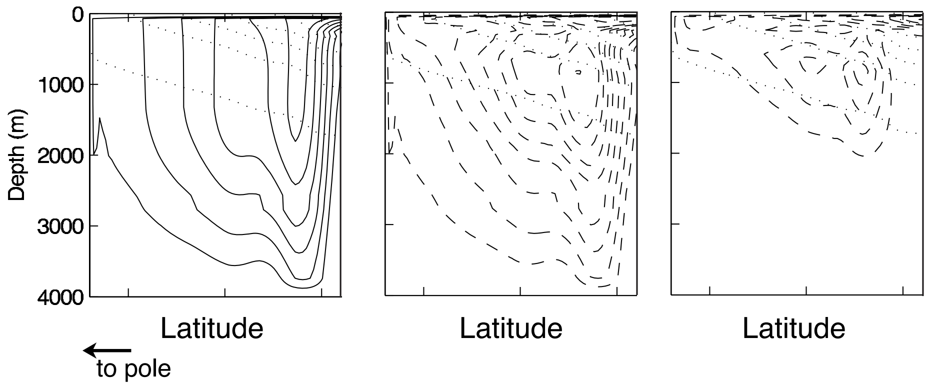


Fig. 16.21 The meridional circulation in the re-entrant channel of an idealized, eddying numerical model of the ACC (as in Fig. 16.18, but showing only the region south of 40°S). Left panel, the zonally averaged Eulerian circulation. Middle panel, the eddy induced circulation. Right panel, the residual circulation. Solid lines represent a clockwise circulation and dashed lines, anticlockwise. The faint dotted lines are the mean isopycnals. Over much of the channel the model ocean is losing buoyancy (heat) to the atmosphere and so the net, or residual, flow at the surface is polewards.

Associated with the form drag is a meridional mass flux in each layer, which in the layered model appears as V_i (a thickness flux) in each layer. The satisfaction of the momentum balance at a particular latitude goes hand-in-hand with the satisfaction of the mass balance. Above any topography the Eulerian mean momentum equation is (with quasi-geostrophic scaling and neglecting eddy momentum fluxes),

$$f_0 \overline{v}_a = \tau \tag{16.97}$$

where $\overline{v}_a$ is the ageostrophic meridional velocity and τ the zonal stress. That is, all the zonally-averaged meridional flow is ageostrophic and, in this approximation, it is non-zero only near the surface (i.e., equatorwards Ekman flow) and below the level of the topography, where it can be supported by friction and/or form drag. Even in an eddying flow, the Eulerian circulation is primarily confined to the upper Ekman layer and a frictional or topographically interupted layer at the bottom, as illustrated in Fig. 16.21. This is of course a perfectly acceptable description of the flow, and is not an artifact in any way. However, and analogously to the atmospheric Ferrel Cell (section 11.7 and 12.2.2), if the flow is unsteady this circulation does not necessarily represent the flow of water parcels, nor does it imply that water parcels cross isopycnals, as might be suggested by the leftmost panel of Fig. 16.21. The flow is better represented by the residual flow, or the thickness-weighted flow, and as sketched in Fig. 16.20 there can be a net meridional flow in a given layer (i.e., of a given water mass type) *even when the net meridional Eulerian flow at the level of mean height of the layer is zero.*

It is important nevertheless to realize that residual flow is really a thickness flux, or equivalently a buoyancy flux, and that the eddies do not carry net mass. In any given isopycnal layer the residual circulation represents a flux of mass of fluid of a particular buoyancy class, and this may be non zero if there is a correlation the thickness of the layer and the velocity (or between velocity and buoyancy) so that $\overline{v'h'} \neq 0$ and $\overline{v'b'} \neq 0$. However, in a Boussinesq fluid eddies do not themselves flux mass per se, because a mass flux is a volume flux and $\overline{v'} = 0$. Thus, for example, eddies cannot locally compensate for the meridional mass transport in the top Ekman layer. This mass flux will return in an ageostrophic bottom boundary layer or, if there is bottom topography, in a flow supported by form stress; this balance being represented by (16.91) or (16.96).

The vertically integrated mass flux must vanish, and even though one component of this — the equatorwards Ekman flow — is determined mechanically, the overall sense of the residual circulation cannot be determined by the momentum balance alone: thermodynamic effects play a role. As in (7.101), the thermodynamic equation may be written in TEM form as

$$\frac{\partial \overline{b}}{\partial t} + J(\psi^*, \overline{b}) = Q[b] \tag{16.98}$$

where $J(\psi^*, \overline{b}) = (\partial_y \psi^*)(\partial_z b) - (\partial_z \psi^*)(\partial_y b) = \overline{v}^* \partial_y b + \overline{w}^* \partial_z b$, ψ^* is the streamfunction of the residual flow and $Q[b]$ represents heating and cooling, which occurs mainly at the surface. In the ocean interior and in statistically steady state we therefore have

$$J(\psi^*, \overline{b}) = 0, \tag{16.99}$$

the general solution of which is $\psi^* = G(\overline{b})$ where G is an arbitrary function. That is, the residual flow is along isopycnals, and this is approximately satisfied by the numerical solution shown in Fig. 16.21.

At the surface, however, the flow is generally not adiabatic, because of heat exchange with the atmosphere. In the simulations shown there is a net heat flux, and consequent buoyancy loss, from the ocean to the atmosphere at the polewards edge of the domain. Heat balance is then achieved by a polewards residual flow of warmer fluid at the surface and sinking at the the highest latitudes. If there were no surface fluxes, and the flow were everywhere adiabatic, then we could expect the residual circulation to vanish. Note that the sense of the subsurface circulation determines how the form drag varies with depth; if the residual flow were zero for example, then, either from (16.89) or from (16.95), we see that the form drag must be constant with depth.

16.6.4 † An idealized adiabatic model

We finally consider a simple model of the ACC that, although very idealized, provides a starting point for understanding more complete numerical models and the system itself.[20] The simplifying assumption we make is that the flow is adiabatic everywhere; it then follows that the net overturning, as given by the residual circulation, is zero. We can see this by first noting that in a statistically steady state the flow satisfies (16.99), implying that the residual flow is along isopycnals. However, if there is a meridional buoyancy gradient at the surface there can be no surface residual flow (because this would be cross-isopycnal); it then follows that there can be no net flow along isopycnals in the interior, because if these outcrop there would be a net fluid source, and hence diapycnal flow, at the surface. This idealized limit has thus led to the 'vanishing of the Deacon Cell'. Although it is a somewhat extreme limit, Fig. 16.21 shows that in a numerical simulation the residual flow is in fact weaker than either the Eulerian or the eddy-induced flow.

The zonal momentum equation in this limit follows from (16.95), which with $\overline{v}^* = 0$ gives

$$\frac{\partial \tau}{\partial z} \approx -\frac{\partial}{\partial z}\left(f_0 \frac{\overline{v'b'}}{\overline{b}_z}\right). \tag{16.100}$$

The equivalent balance for the Eulerian flow is, using the definition of $\overline{v}^*$,

$$f_0 \overline{v} = f_0 \frac{\partial}{\partial z}\left(\frac{\overline{v'b'}}{\overline{b}_z}\right). \tag{16.101}$$

In the residual equation, (16.100), the wind stress is balanced by the divergence of the Eliassen-Palm flux, which is dominated by the contribution from the buoyancy flux, and in the ocean interior where the stress is small the meridional buoyancy flux will be constant with height. In (16.101) the Coriolis force on the equatorwards flow in the Ekman layer is balanced by a form drag. The constancy of the form drag on the right-hand side of (16.101) implies there can be little meridional flow in the interior; the equatorial flow in the top Ekman layer is thus balanced by a return flow at the bottom of the ocean involving topographic form stress or a bottom Ekman layer.

Integrating (16.100) from the surface (where $\tau = \tau_w$) to a stress-free level in the interior (where $\tau = 0$) gives

$$\tau_w = f_0 \frac{\overline{v'b'}}{\overline{b}_z}, \tag{16.102}$$

if the buoyancy flux at the surface is small. If we are now willing to parameterize the eddy fluxes in terms of the mean flow, then we can predict the stratification. Thus, let $\overline{v'b'} = -\kappa \partial \overline{b}/\partial y$, and noting that $s = -\overline{b}_y/\overline{b}_z$ is the slope of the isopycnals, we find

$$\tau_w = \kappa f_0 s = \kappa \frac{f_0^2}{\overline{b}_z} \frac{\partial \overline{u}}{\partial z}. \tag{16.103}$$

where the second equality uses thermal wind balance. Thus, given κ, we can predict the isopcynal slope [$s \sim \tau_w/(\kappa f_0)$] and, potentially, the total baroclinic transport of the ACC as a function of the wind stress. Further progress then depends on making a specific choice for κ, for example as described in chapter 10, but our reach has already exceeded our grasp. In a more realistic diabatic model, the sense of the residual circulation can be inferred if the diabatic fluxes at the surface are known, but at the same time these fluxes depend in a complicated way on both the lateral eddy fluxes and the general circulation itself.

16.6.5 Form stress and Ekman stress at the ocean bottom

Earlier, we noted that the stress at the ocean bottom is observed to be dominated by form stress, rather than Ekman friction, in the ACC. A simple scaling argument helps understand why this should be. The form stress scales like

$$\tau_{form} \sim \eta_b \frac{\partial p_b}{\partial x} \sim \eta_b V f, \tag{16.104}$$

where V is a scaling for the horizontal velocity. The frictional stress due to an Ekman layer scales like

$$\tau_{Ekman} \sim A\frac{\partial u}{\partial z} \sim \frac{AV}{d} \sim dVf \tag{16.105}$$

where A is a coefficient of viscosity and $d = \sqrt{A/f}$ is the Ekman layer thickness. The ratio of these two stresses thus scales as

$$\frac{\tau_{form}}{\tau_{Ekman}} \sim \frac{\eta_b}{d}. \tag{16.106}$$

We therefore expect the form stress to dominate the Ekman stress if the variations in topography are greater than the Ekman layer thickness, and if the flow goes *over* the topography rather than around it. In the ACC the topography is hundreds or even thousands of meters high whereas the bottom Ekman layer may be of order tens of meters, and furthermore the predominantly eastward flow must (unlike the siutation in gyre circulations) go over the topography. Thus, form stress dominates the frictional, Ekman layer, stress at the bottom of the ocean.

16.6.6 Differences between gyres and channels

In the dynamics of the ACC, the wind stress itself seems to play an important role, whereas in our discussion of gyres in chapter 14 the wind stress *curl* was dominant. What is the root of this difference?[21] Suppose that we change the wind stress, but not its curl, in a closed basin. The vertically integrated gyral flow, as given for example by the Stommel solution (14.39) or its two-gyre counterpart, does not change at all. However, the vertical structure of this flow will in general change; for example, if the wind is made uniformly more eastward, there will be a corresponding increase in the equatorward flux in the Ekman layer that must return polewards at depth (note that the western boundary current balances only the Sverdrup flow). At the same time, the added force from the wind must be balanced by an increased pressure difference between the western and eastern boundaries. This may be achieved if the the sea-surface tilts upward to the east, so producing a net (vertically integrated) polewards geostrophic flow. The subsurface isopycnal slopes may then adjust in order to reduce this flow to near zero in the abyss. The added force provided by the by the basin walls on the fluid in the basin is a kind of form drag (rather like the force provided by the sill in section 16.5), and integrated around the basin this force must be equal and opposite to the force supplied by the wind. In contrast, in a channel adding a constant wind produces a direct change in its zonal transport. This is because the wind-stress is balanced by form drag and bottom friction (with the former likely dominating), and both of these, in practice, depend on the zonal flow at the channel bottom.

APPENDIX: MISCELLANEOUS RELATIONSHIPS IN A LAYERED MODEL

Here we collect various expressions relating pressure, density and velocity in a geostrophic and Boussinesq layered model. The layers and the interfaces are numbered, increasing downwards, as in Fig. 16.22, and the bottom layer is stationary.

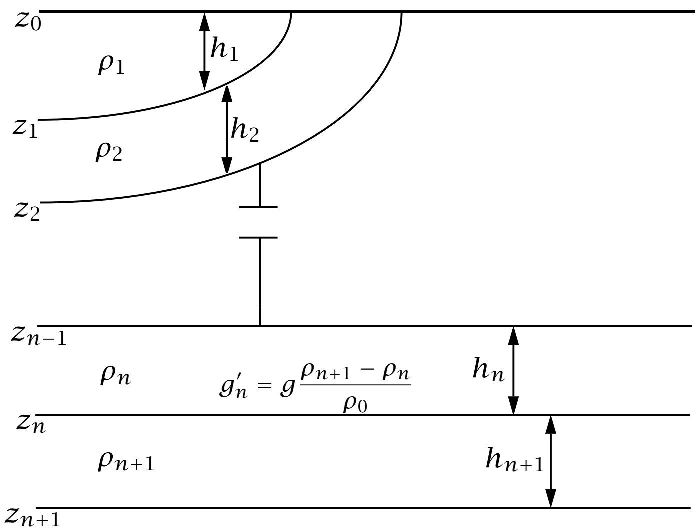


Fig. 16.22 Structure and notational conventions used for a multi-layered model.

16.A.1 Hydrostatic balance

Hydrostatic balance is $\partial p/\partial z = -\rho g$. We can integrate this to give, in layers n and $n-1$,

$$p_n = -\rho_n g z + p'_n(x, y), \qquad p_{n-1} = -\rho_{n-1} g z + p'_{n-1}(x, y). \tag{16.107}$$

Since pressure is continuous, at $z = z_{n-1}$ these two expressions are equal so that

$$-\rho_n g z_{n-1} + p'_n = p_{n-1} = -\rho_{n-1} g z_{n-1} + p'_{n-1} \tag{16.108}$$

whence

$$g'_{n-1} z_{n-1} = \frac{1}{\rho_0}(p'_n - p'_{n-1}) \tag{16.109}$$

where $g'_n = g(\rho_{n+1} - \rho_n)/\rho_0$ is the *reduced gravity*, and ρ_0 is the constant, reference, value of the density used in the Boussinesq approximation, taken to be equal to ρ_1.

16.A.2 Geostrophic and thermal wind balance

In the Boussinesq approximation, geostrophic balance is:

$$\rho_0 f \boldsymbol{u}_n = \mathbf{k} \times \nabla p_n \tag{16.110}$$

Using (16.110) with (16.109) gives

$$\boldsymbol{u}_{n+1} - \boldsymbol{u}_n = \frac{g'_n}{f} \mathbf{k} \times \nabla z_n \tag{16.111}$$

which is the appropriate form of thermal wind balance for this system. Let us suppose that at sufficient depth there is no motion, and in particular that layer $N+1$ is stationary and contains no pressure gradients. That is, $p'_{N+1} = 0$ and so, from (16.108) and (16.110)

$$p'_N = -(\rho_{N+1} - \rho_N)gz_N = -g'_N \rho_0 z_N. \tag{16.112}$$

Integrating upwards we obtain the pressure in each layer,

$$p'_n = -\rho_0 \sum_{i=n}^{i=N} g'_i z_i \tag{16.113}$$

where $n \le N$. Thus, the geostrophic velocities in each layer are given by

$$f\boldsymbol{u}_n = -\mathbf{k} \times \nabla \left(\sum_{i=n}^{i=N} g'_i z_i \right). \tag{16.114}$$

The quantity in brackets on the right-hand side is not a velocity streamfunction because it is $f\boldsymbol{u}_n$, and not $\boldsymbol{u}_n$ that is given by its curl. Nevertheless, the velocity is normal to its gradient, and therefore its isolines define streamlines.

The upper surface of the ocean is assumed to be fixed; this is the 'rigid-lid' approximation. Thus, $z_0 = 0$ and $h_1 = -z_1$. More generally, the layer thicknesses and the interfaces between the layers are related by

$$z_n = -\sum_{i=M}^{i=n} h_i \tag{16.115}$$

where M is the index of the uppermost layer, and $n \ge M$. If there is no outcropping, then $M = 1$.

The geostrophic velocity in the lowest moving layer is given by

$$f\boldsymbol{u}_N = -g'_N \mathbf{k} \times \nabla z_N = g'_N \mathbf{k} \times \nabla h. \tag{16.116}$$

This means that lines of constant depth of the lowest layer are also streamlines; the velocity moves parallel to the depth contours. The vertical velocity at the base of the lowest layer is given by, for steady flow,

$$w(z = -h) = \boldsymbol{u}_N \cdot \nabla h = \frac{1}{f} g'_N (\mathbf{k} \times \nabla h) \cdot \nabla h = 0. \tag{16.117}$$

That is, there is no vertical motion at the base of the moving layers.

16.A.3 Explicit cases

A one-layer reduced-gravity model

The perturbation pressure in the moving layer (layer 1) is

$$p'_1 = -\rho_0 g'_1 z_1 = \rho_0 g'_1 h_1. \tag{16.118}$$

The geostrophic velocity is given by

$$f\boldsymbol{u}_1 = \frac{1}{\rho_0} \mathbf{k} \times \nabla p'_1 = g'_1 \mathbf{k} \times \nabla h_1. \tag{16.119}$$

(In a single layer model, the subscripts are often omitted.)

A two-layer model

The perturbation pressure in the upper and lower moving layers are given by

$$p_1' = -\rho_0(g_2' z_2 + g_1' z_1) = \rho_0(g_2' h + g_1' h_1), \tag{16.120a}$$

$$p_2' = -\rho_0 g_2' z_2 = \rho_0 g_2'(h_1 + h_2) = \rho_0 g_2' h, \tag{16.120b}$$

[c.f. (16.112)], where $h = h_1 + h_2 = -z_2$.

The corresponding geostrophic velocities are

$$f\boldsymbol{u}_1 = \mathbf{k} \times \nabla(g_2' h + g_1' h_1), \tag{16.121a}$$

$$f\boldsymbol{u}_2 = \mathbf{k} \times \nabla(g_2' h). \tag{16.121b}$$

A three-layer model

The perturbation pressures in the three moving layers are

$$p_1 = -\rho_0[g_3' z_3 + g_2' z_2 + g_1' z_1] = \rho_0[g_3' h + g_2'(h_1 + h_2) + g_1' h_1], \tag{16.122a}$$

$$p_2 = -\rho_0[g_2' z_2 + g_3' z_3] = \rho_0[g_2'(h_1 + h_2) + g_3' h], \tag{16.122b}$$

$$p_3 = -\rho_0 g_3' z_3 = \rho_0 g_3' h, \tag{16.122c}$$

where $h = h_1 + h_2 + h_3 = -z_3$. The corresponding geostrophic velocities are:

$$f\boldsymbol{u}_1 = \mathbf{k} \times \nabla[g_3' h + g_2'(h_1 + h_2) + g_1' h_1] \tag{16.123a}$$

$$f\boldsymbol{u}_2 = \mathbf{k} \times \nabla[g_2'(h_1 + h_2) + g_3' h] \tag{16.123b}$$

$$f\boldsymbol{u}_3 = \mathbf{k} \times \nabla[g_3' h]. \tag{16.123c}$$

Notes

1 Courtesy of L. Talley.

2 See (among others) Toole et al. (1994), Polzin et al. (1997), Gregg (1998) and Ledwell et al. (1998).

3 For estimates of the strength of the overturning circulation in the ocean, and its relation to diapycnal diffusivity and the observed stratification, see Munk (1966), revisited by Munk and Wunsch (1998). See also Huang (1999) and, for a review, Wunsch and Ferrari (2004).

4 Vallis (2000).

5 The modern development of the theory of the thermocline began with two back-to-back papers in 1959 in the journal *Tellus*. Welander (1959) suggested an adiabatic model, based on the ideal-fluid thermocline equations (i.e., the planetary geostrophic equations, with no diffusion terms in the buoyancy equation), whereas Robinson and Sommel (1959) proposed a model that is intrinsically diffusive. In this model [developed further by Stommel and Webster (1963), Salmon (1990), and others] the thermocline is an internal boundary layer or front that forms at the convergence of two different homogeneous water types, warm surface fluid above and cold abyssal fluid below. Meanwhile, the adiabatic model continued its own development (see Veronis 1969), culminating in the ventilated thermocline model of Luyten et al. (1983) and

its continuous extensions (e.g., Huang 1988). Later, Welander (1971b) and Colin De Verdiere (1989) noted that the diffusion might become important below an adiabatic near-surface flow, and Samelson and Vallis (1997) eventually suggested a model in which the upper thermocline is adiabatic, as in the ventilated thermocline model, but has a diffusive base, constituting an internal boundary layer. The role of mesoscale eddies in all of this is still unclear; however, numerical simulations suggest they do play a role in determining the structure of the thermocline, at least in its diffusive base and perhaps everywhere. The likelyhood that some of the deep overturning circulation is wind-driven, and inter-hemispheric, will also affect the dynamics of the diffusive layer.

6 Welander (1971a).

7 Drawing from Salmon (1990).

8 Newton's method is an iterative way to numerically solve certain types of differential equations. The solutions here are obtained using about 1000 uniformly spaced grid points to span the domain, taking just a few seconds of computer time. Because of the boundary layer structure of the solutions employing a nonuniform grid would be even more efficient for this problem, but there is little point in designing a streamlined hat to reduce the effort of walking.

9 Following Samelson (1999b).

10 Adapted from Samelson and Vallis (1997).

11 Following Luyten et al. (1983).

12 That the eastern boundary depth is undetermined is perhaps the main incomplete aspect of the theory. It may be that the depth is determined by global thermodynamic and/or mass constraints. For example, there must be a given polewards transport across the boundary of the subtropical-subpolar gyre to balance the equatorward transport in the Ekman layer and that of the meridional overturning circulation.

13 Dewar et al. (2005). This paper also discusses the nature of discontinuities at the pool boundaries, and their treatment via shock conditions.

In the shadow zone, layer-2 fluid also has no direct surface source, so one may wonder why is this not also expunged. However, the shadow zone is not a recirculating regime and the Ekman induced displacement will be much less efficient. More directly, the eastern boundary condition $h_1(x = x_e) = 0$ precludes the vanishing of layer-2 water there, and this boundary condition is propagated westward into the interior.

14 Drawing from the numerical, conceptual and analytic models of Toggweiler and Samuels (1995, 1998), Döös and Coward (1997), Vallis (2000), Webb and Suginohara (2001), Nof (2003) and Samelson (1999a, 2004).

15 From Rintoul et al. (2001).

16 See Rintoul et al. (2001) and Olbers et al. (2004) for reviews.

17 These simulations, described in Henning and Vallis (2005), solve the primitive equations in a domain similar to Fig. 16.13. The wind forcing produces a polewards Ekman drift across the channel, as well as a subtropical gyre, and there is a meridional temperature gradient across the whole domain, so giving rise to a subtropical thermocline.

18 Adapted from Henning and Vallis (2005).

19 Munk and Palmén (1951), Gille (1997), Stevens and Ivchenko (1997).

20 Models of this ilk derive from Johnson and Bryden (1989). Straub (1993), Hallberg and

Gnanadesikan (2001), Karsten et al. (2002), Marshall and Radko (2003), Henning and Vallis (2005), and others, consider related issues and extensions.

21 See also Warren et al. (1996) and Hughes (2002).

Further Reading

Siedler, G., Church, J. and Gould, J., Eds., 2001. *Ocean Circulation and Climate.*
A comprehensive collection of review articles, with a general emphasis on observations and modelling of the large-scale circulation and its role in the climate system.

Problems

16.1 Show that the height of the upper layer in a model of the ventilated thermocline is continuous at the edge of the shadow zone, although the gradient normal to the shadow zone boundary need not be. [Hint: In the ventilated region $h_1 = (1 - f/f_2)h$, and in the shadow zone $g_1'/g_2'h_1^2 = D_0^2$. Show that these lead to a shadow zone boundary that is the same as (16.69).]

16.2 Obtain an expression for the boundary of the western pool in a two-layer model of the ventilated thermocline.

16.3 ♦ A one-dimensional equation similar to (16.27) was put forward by Stommel and Webster (1963) as a simple model of the thermocline. The equation is

$$\left(2W - z\frac{\partial W}{\partial z}\right)\frac{\partial^3 W}{\partial z^3} = \kappa\frac{\partial^4 W}{\partial z^4}. \tag{P16.1}$$

Perform a boundary layer analysis on this equation, and show that if the downwards vertical velocity at the surface $z = 0$ is non-zero then the thickness of an internal boundary layer scales as $\kappa^{1/2}$. How does the vertical velocity scale with κ in this case? Why do these differ from the corresponding scalings for (16.27)? How does the boundary layer thickness scale if it is at the surface ($z = 0$)?

16.4 Construct a one-dimensional thermocline boundary layer equation by simply ignoring the variations in x and y in A, so by substituting $A(x, y, \zeta) = B(\zeta)$ in (16.41). Show that this leads to an equation of the form

$$C\frac{\mathrm{d}B}{\mathrm{d}\zeta}\frac{\mathrm{d}^3 B}{\mathrm{d}\zeta^3} = \frac{\kappa}{\delta^2}\frac{\mathrm{d}^4 B}{\mathrm{d}\zeta^4}, \tag{P16.2}$$

where C depends on β, f, h_x and T_0. If C were independent of x and y, show that this equation is essentially the same as the boundary layer equation that emerges from (P16.1). Is C in fact constant (i.e., not a function of x) in general?

16.5 If the Ekman pumping velocity at the surface, W_E, is zero then we cannot use it to non-dimensionalize W in (16.27). In this case what would be an appropriate choice for the scaling value of W (i.e., for W_S)? Does this choice affect whether $\hat{\kappa}$ is 'small' and $\widehat{W}$ is $\mathcal{O}(1)$ in (16.27)?

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